

# **C.11 ANALIZA DINAMICĂ ȘI INGINERIA INVERSĂ A COMPORTAMENTULUI**

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# PRINCIPALELE PĂRȚI ALE PREZENTĂRII

## C.11 Detonarea malware și ingineria inversă a comportamentului:

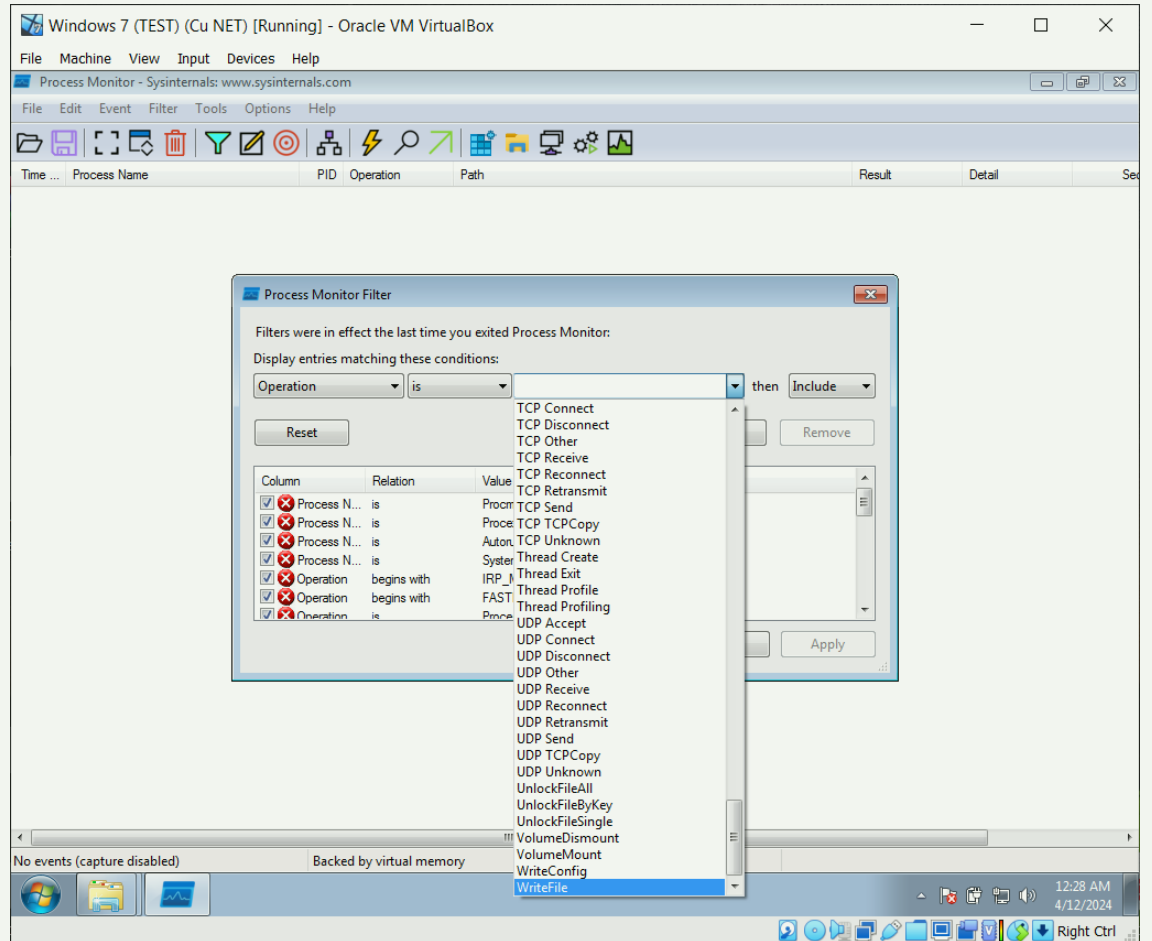
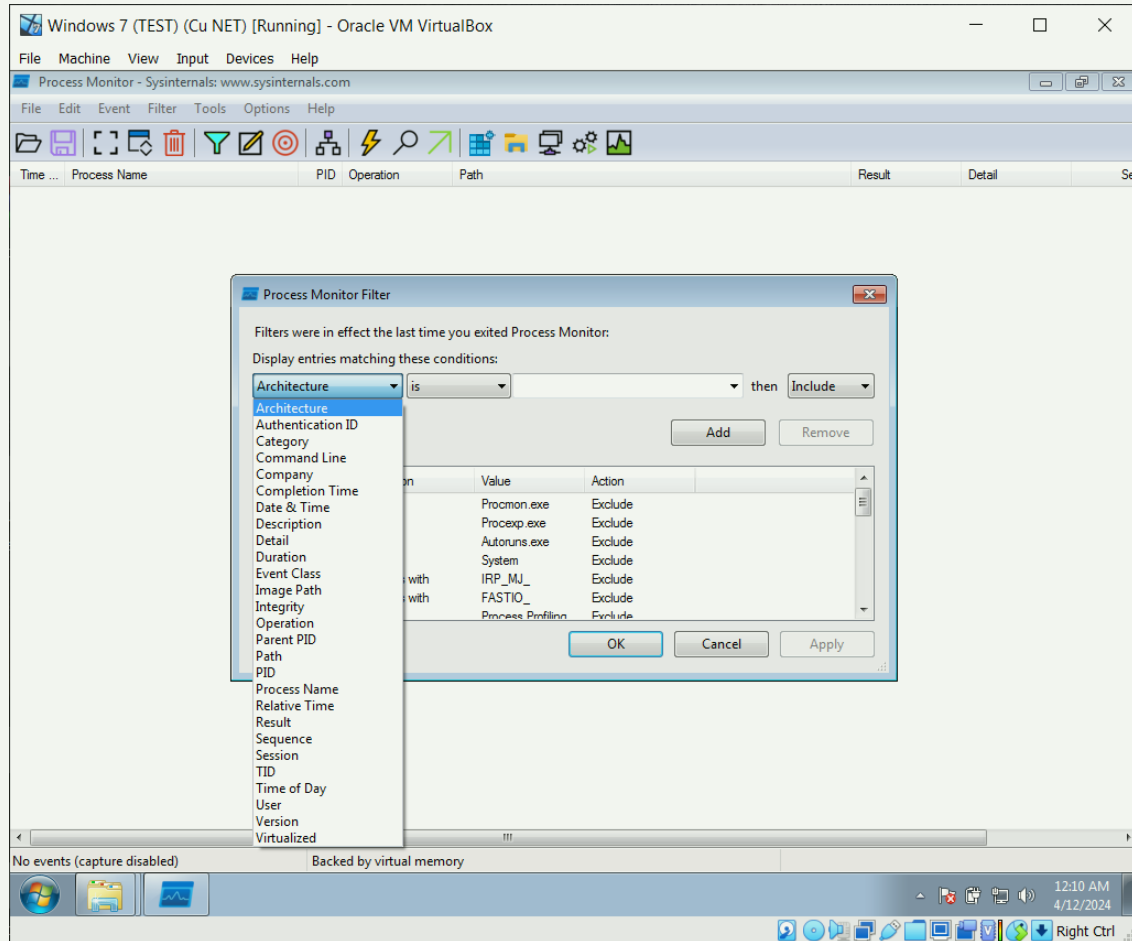
- C.11.1 FILTRE PE EVENIMENTELE SISTEMULUI DE OPERARE CU PROCESS MONITOR
- C.11.2 DETONAREA ȘI INSPECȚIA COMPORTAMENTALĂ A APLICAȚIILOR MALWARE
- C.11.3 CAPTAREA EVENIMENTELOR EFEMERE ȘI ELIMINAREA ZGOMOTULUI
- C.11.4 DETONAREA ȘI ANALIZA VIRUȘILOR PRIN UTILIZAREA UNUI HONEYPOT

C.10.1

# FILTRE PE EVENIMENTELE SISTEMULUI DE OPERARE CU PROCESS MONITOR

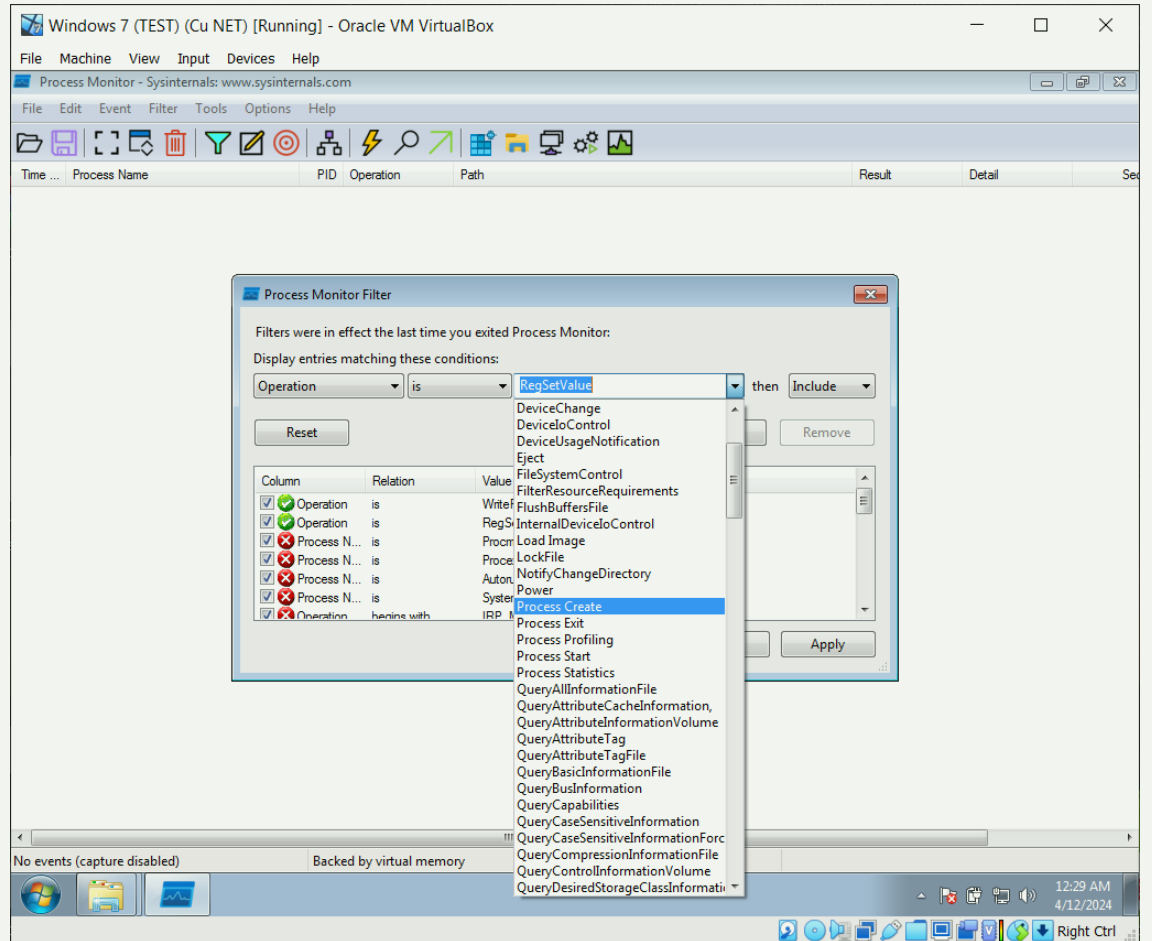
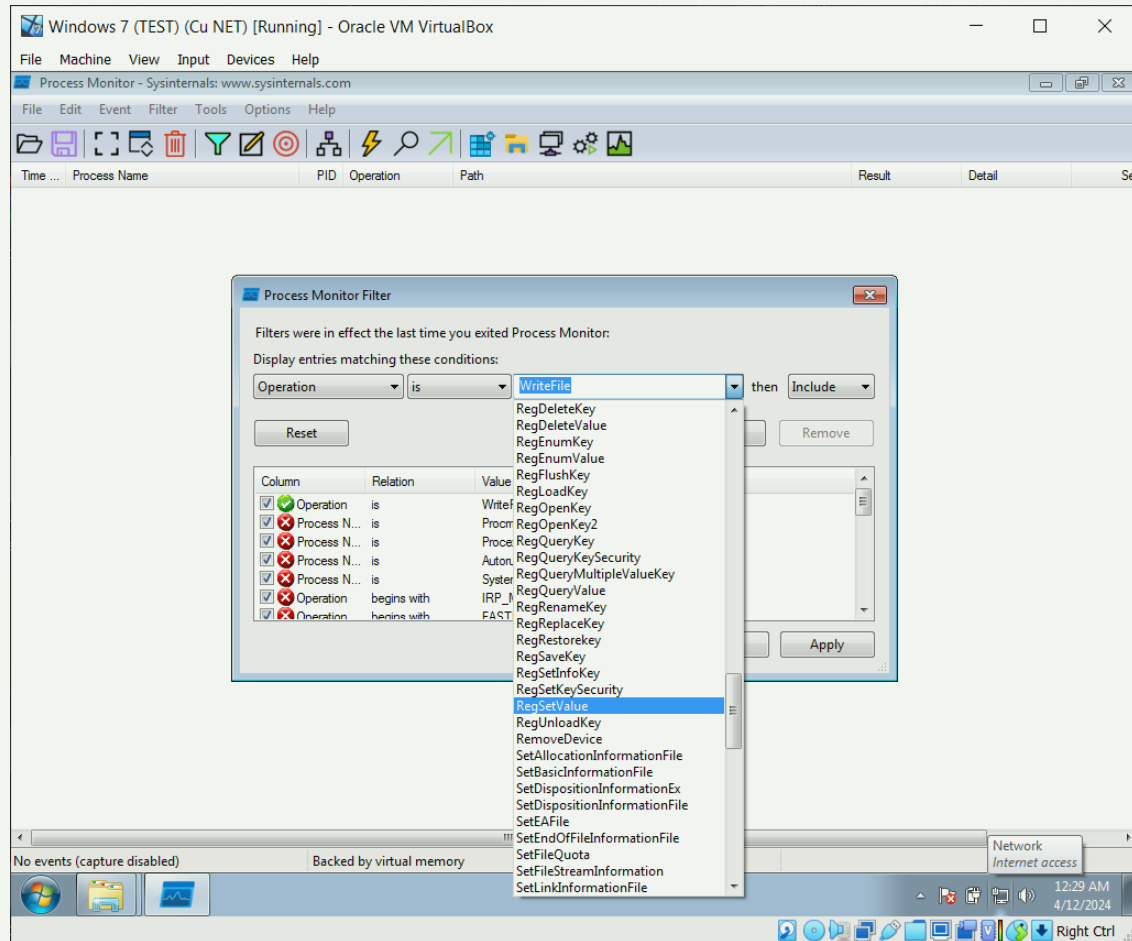


# PROCESS MONITOR FILTRE





# PROCESS MONITOR OPERATION





## Ingineria Inversă în Analiza Malware folosind Filtrele ProMon

### Filtrarea după Numele Procesului (Process Name Filtering)

- Analizează activitățile specifice ale proceselor suspecte
- Izolează comportamentul malware-ului

### Accesul la Registru (Registry Activity)

- Monitorizează modificările în registru
- Vizează locațiile frecvent accesate de malware

### Activitatea de Rețea (Network Activity)

- Filtrează și supraveghează conexiunile de rețea
- Detectează comunicațiile neobișnuite sau către adrese IP suspecte

### Modificările Fișierelor Sistem (File System Changes)

- Detectează crearea, modificarea sau ștergerea de fișiere
- Se concentrează pe locații critice care semnalează acțiuni de malware

### Operațiuni de Injecție de Cod (Code Injection Operations)

- Identifică încercările de injectare a codului în alte procese
- O metodă comună utilizată de malware pentru a se ascunde și a executa cod malițios

### Activități Suspicioase de Thread (Suspicious Thread Activity)

- Monitorizează crearea de noi thread-uri în procese
- Indică posibile activități malițioase desfășurate în secret

### Filtrarea după Căi și Tipuri de Fișiere Specifice (Path and File Type Filtering)

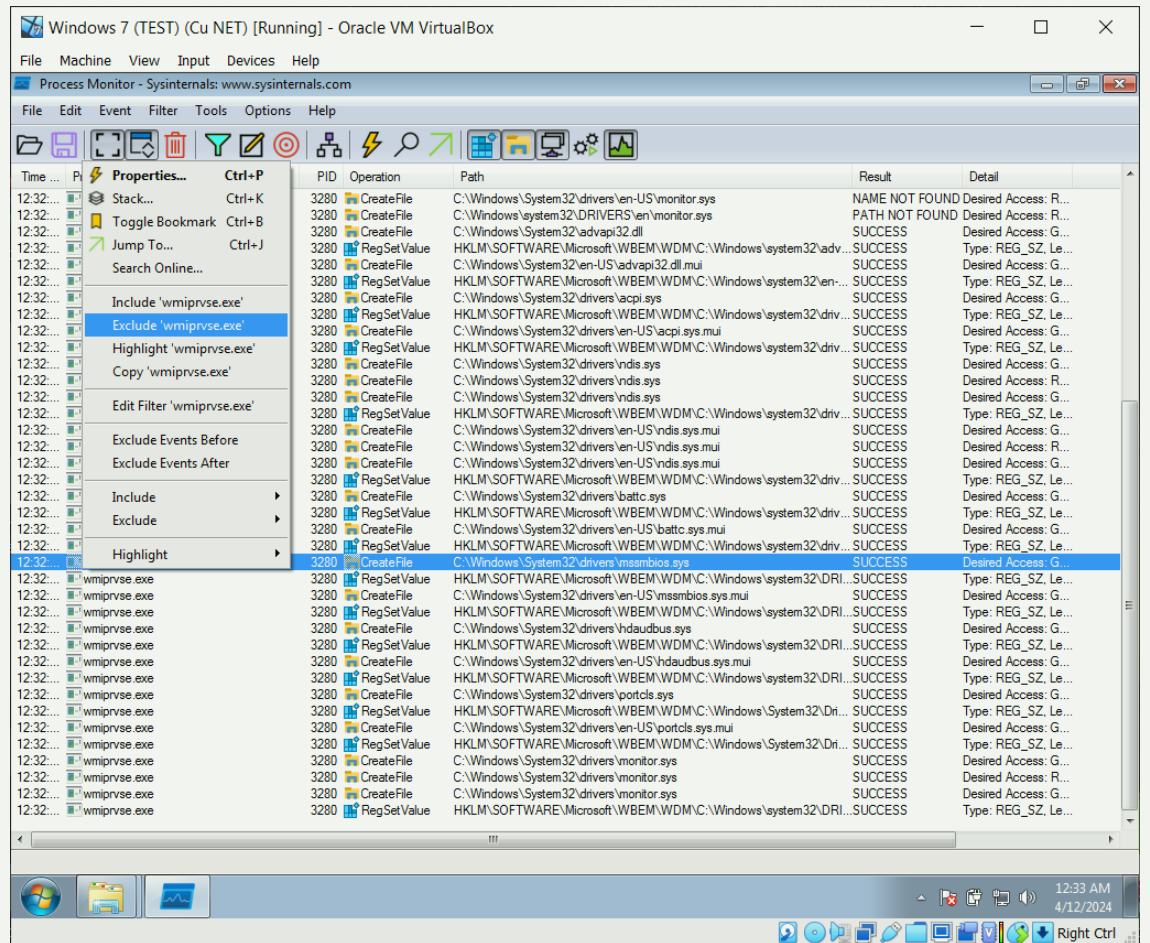
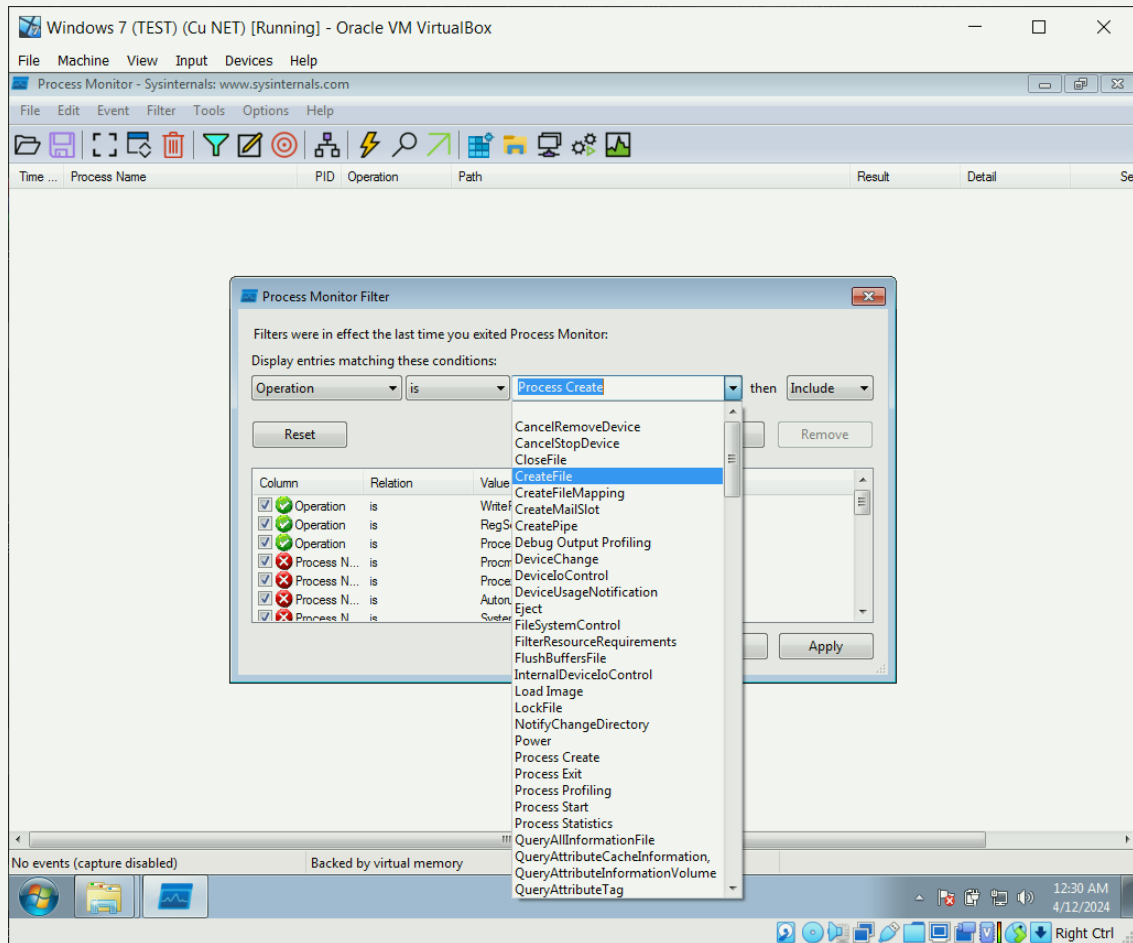
- Se concentrează pe accesul la fișiere executabile (.exe, .dll) sau scripturi (.js, .vbs) în locații neașteptate sau temporare
- Semnalează activitate malițioasă

### Modificările de Securitate (Security Changes)

- Supraveghează schimbările în permisiunile fișierelor sau directoarelor
- Sugerază încercări de evitare a măsurilor de securitate

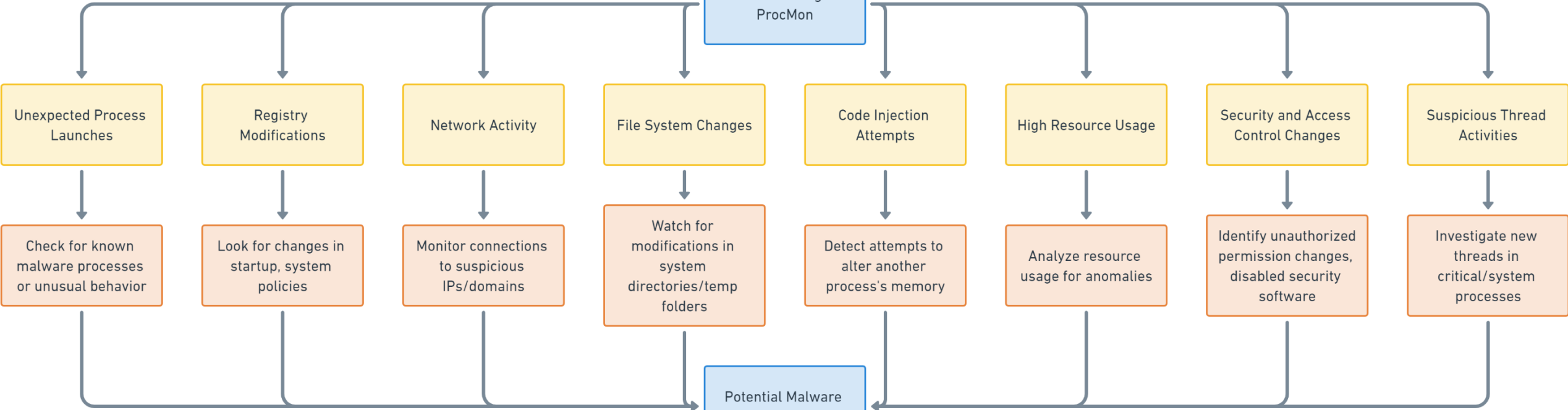
# PROCESS MONITOR

## OPERATIONS / INCLUDE / EXCLUDE





Start Monitoring with  
ProcMon





## Observații în Arborele de Procese cu ProcMon

```
graph LR; A[Observații în Arborele de Procese cu ProcMon] --- B[Relații Părinte-Copil]; A --- C[Anomalii ale Proceselor]; A --- D[Consumul de Resurse]; A --- E[Conexiuni de Rețea]; A --- F[Operațiuni cu Fișiere]; A --- G[Atribute Suspecte ale Proceselor]; A --- H[Durata de Viață a Proceselor]; B --- B1[Monitorizează procesele părinte neașteptate sau suspecte]; B --- B2[Urmărește procesele copil generate de aplicații comune]; C --- C1[Identifică procesele cu nume sau locații neobișnuite]; C --- C2[Verifică procesele ascunse sau de sistem care se comportă anormal]; D --- D1[Observă consumul mare de resurse de către procese necunoscute sau inerte]; E --- E1[Observă activitatea de rețea legată de procese specifice]; E --- E2[Urmărește conexiunile inițiate de procese suspecte]; F --- F1[Urmărește operațiunile excesive cu fișiere de către un singur proces]; F --- F2[Verifică tiparele de acces la fișiere pentru abateri de la activitatea normală]; G --- G1[Verifică procesele cu steaguri de execuție neobișnuite sau module încărcate]; H --- H1[Monitorizează procesele cu durate de viață foarte scurte sau neobișnuit de lungi];
```

### Relații Părinte-Copil

Monitorizează procesele părinte neașteptate sau suspecte

Urmărește procesele copil generate de aplicații comune

### Anomalii ale Proceselor

Identifică procesele cu nume sau locații neobișnuite

Verifică procesele ascunse sau de sistem care se comportă anormal

### Consumul de Resurse

Observă consumul mare de resurse de către procese necunoscute sau inerte

### Conexiuni de Rețea

Observă activitatea de rețea legată de procese specifice

Urmărește conexiunile inițiate de procese suspecte

### Operațiuni cu Fișiere

Urmărește operațiunile excesive cu fișiere de către un singur proces

Verifică tiparele de acces la fișiere pentru abateri de la activitatea normală

### Atribute Suspecte ale Proceselor

Verifică procesele cu steaguri de execuție neobișnuite sau module încărcate

### Durata de Viață a Proceselor

Monitorizează procesele cu durate de viață foarte scurte sau neobișnuit de lungi

## Opțiuni Importante ale Filtrului Monitorului de Procese

**Arhitectură:** Filtrează după arhitectura procesorului

**Operații:** Include sau exclude evenimente bazate pe tipul de procesor

**Linia de Comandă:** Filtrează procesele după argumentele liniei de comandă

**Operații:** Include sau exclude procese bazate pe argumentele specificate

**Companie:** Filtrează după compania care a semnat procesul

**Operații:** Include sau exclude procese bazate pe semnătura companiei

**Data & Timp:** Filtrează evenimentele după data și ora apariției

**Operații:** Include sau exclude evenimente bazate pe marcajul temporal

**Descriere:** Filtrează după câmpul descrierii în proprietățile procesului

**Operații:** Include sau exclude procese bazate pe descriere

**Durată:** Filtrează după durata de timp în care un eveniment are loc

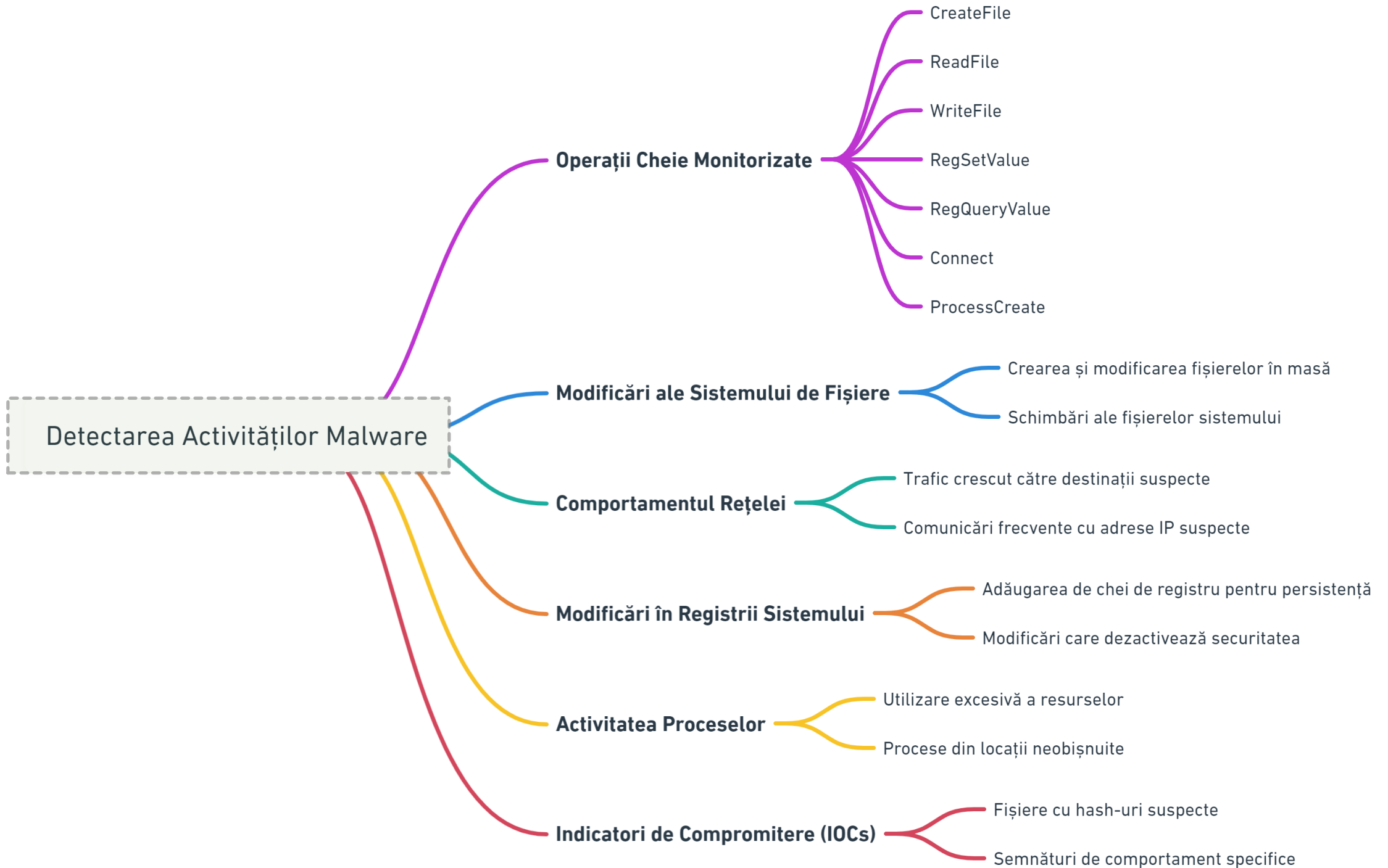
**Operații:** Include sau exclude evenimente bazate pe durată

**Cale:** Filtrează după calea fișierului asociat cu evenimentul

**Operații:** Include sau exclude evenimente bazate pe calea specificată

**Numele Procesului:** Filtrează după numele procesului

**Operații:** Include sau exclude procese bazate pe nume



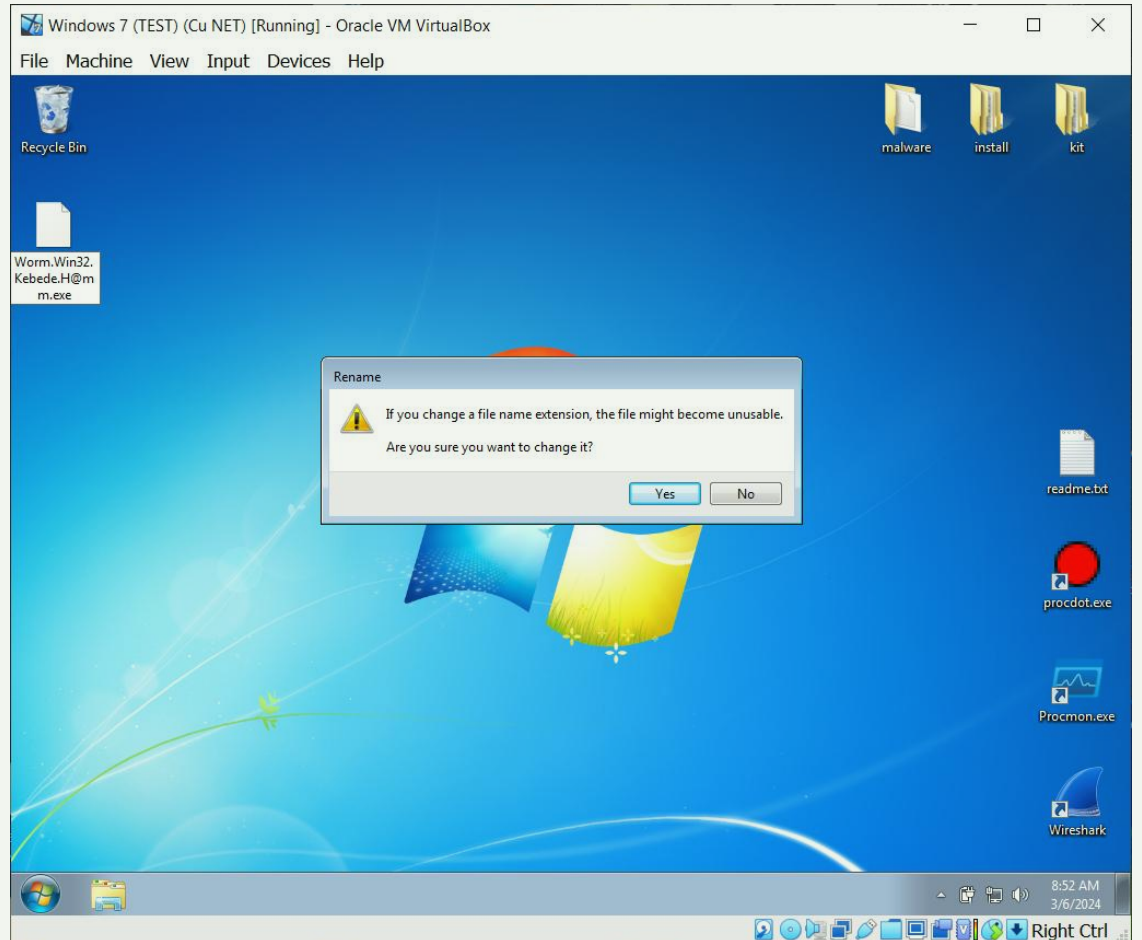
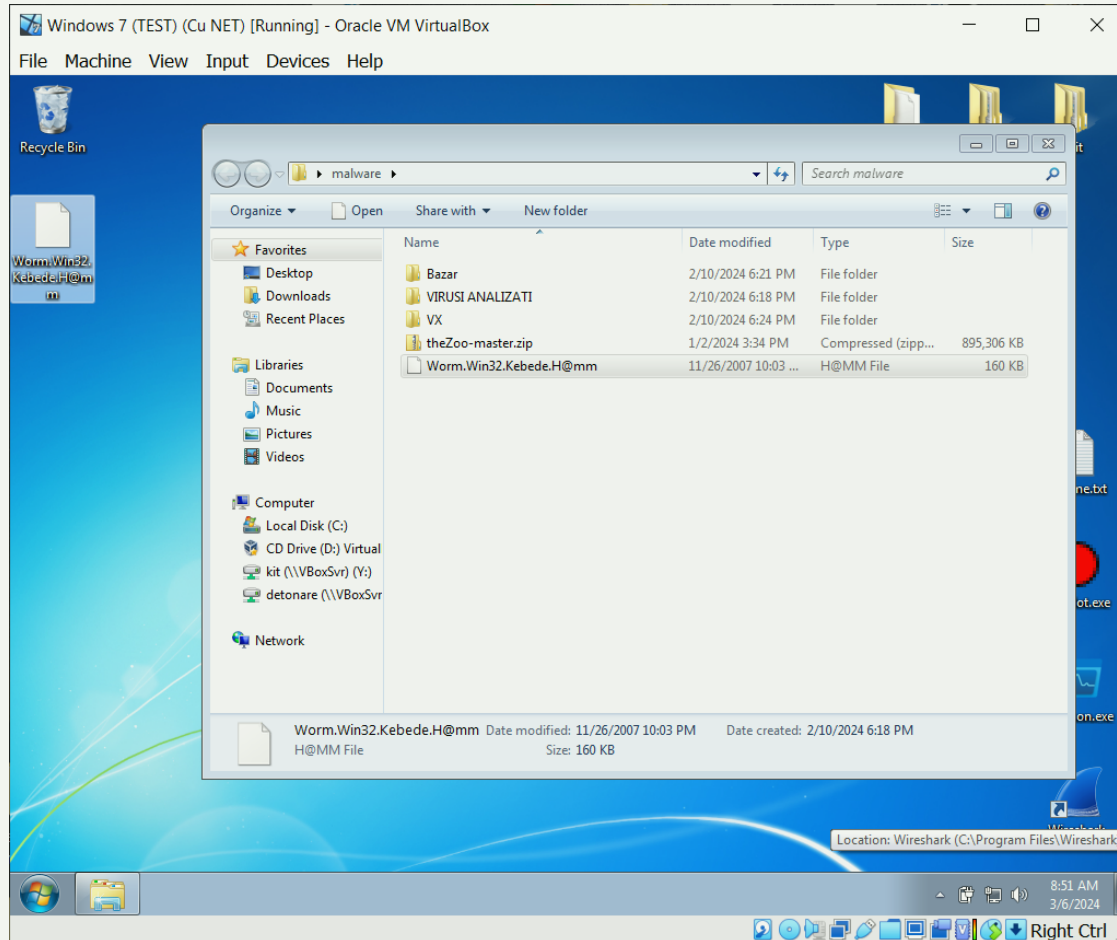
C.11.2

# DETONAREA ȘI INSPECȚIA COMPORTAMENTALĂ A APLICAȚIILOR MALWARE



# PROCESS MONITOR & WIRESHARK

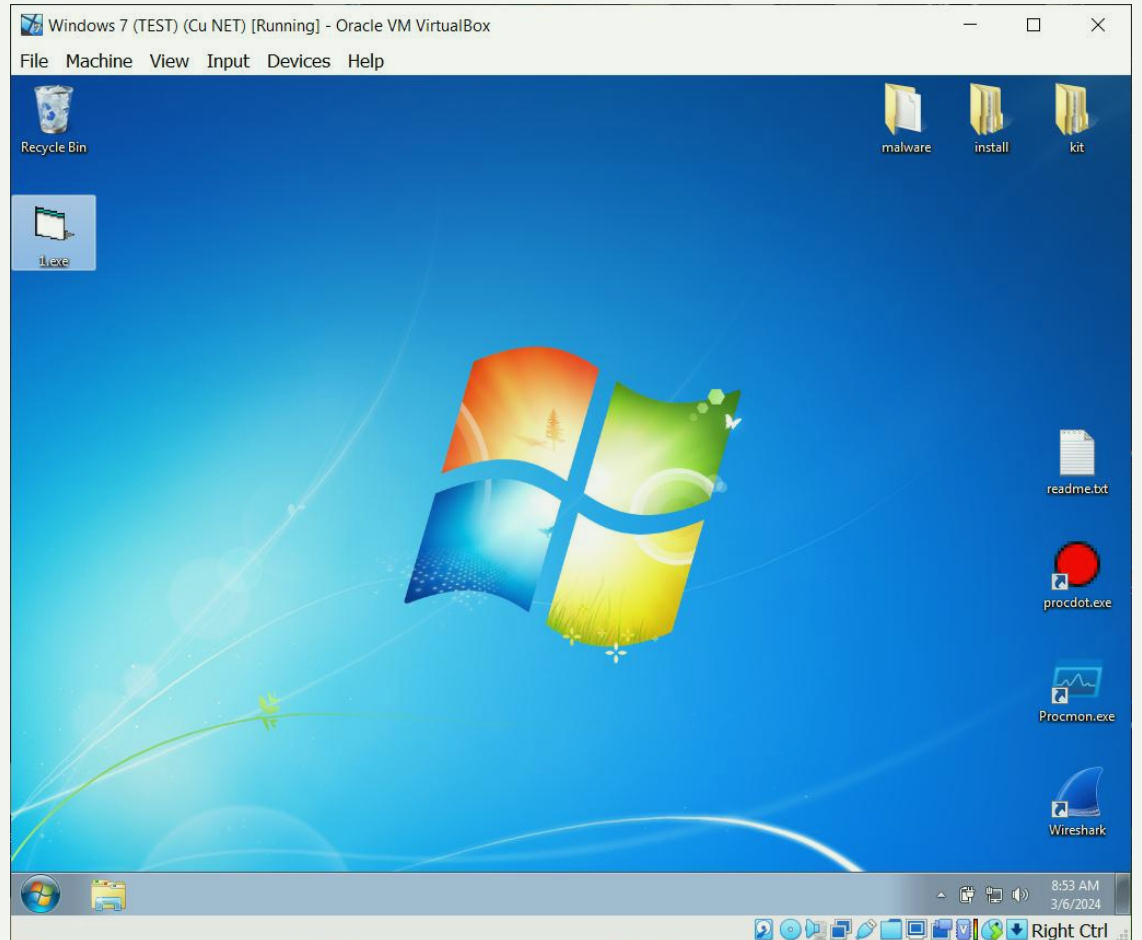
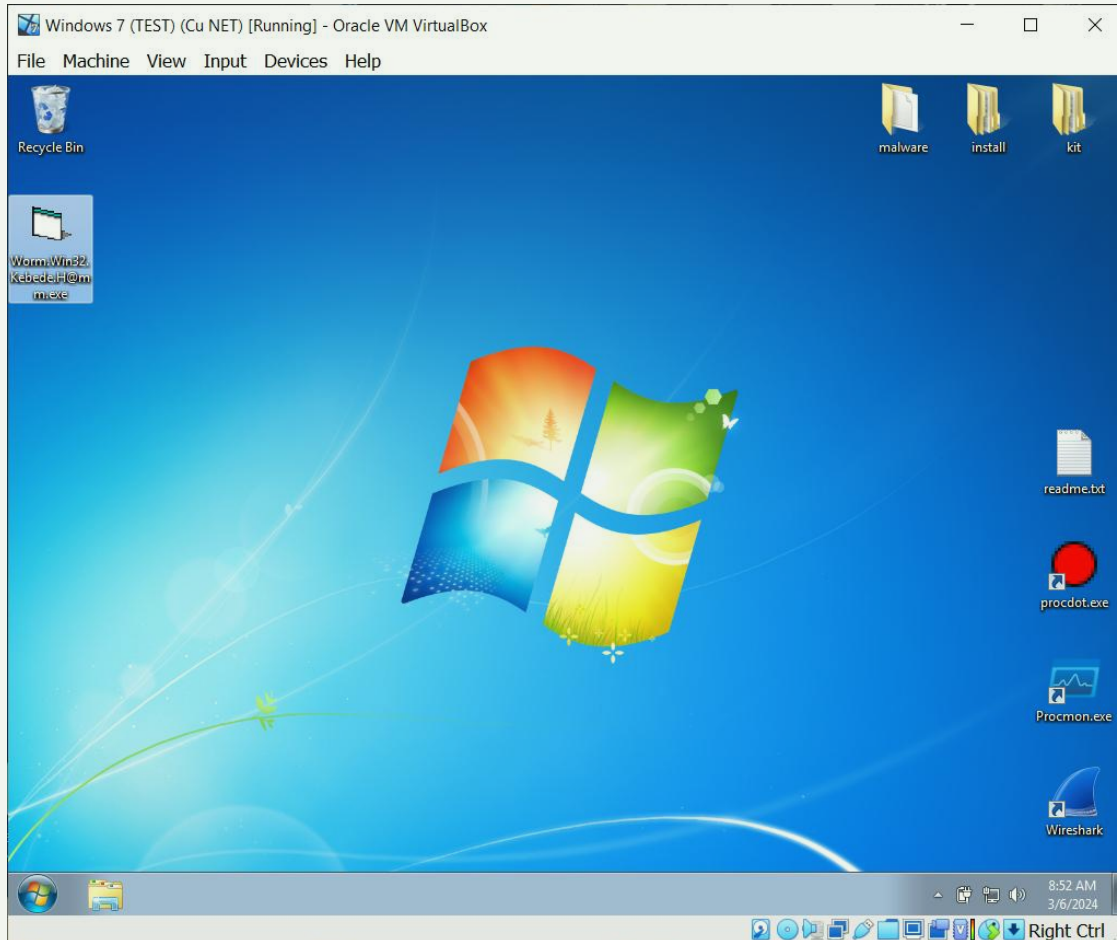
## DETONAREA UNUI TROIAN





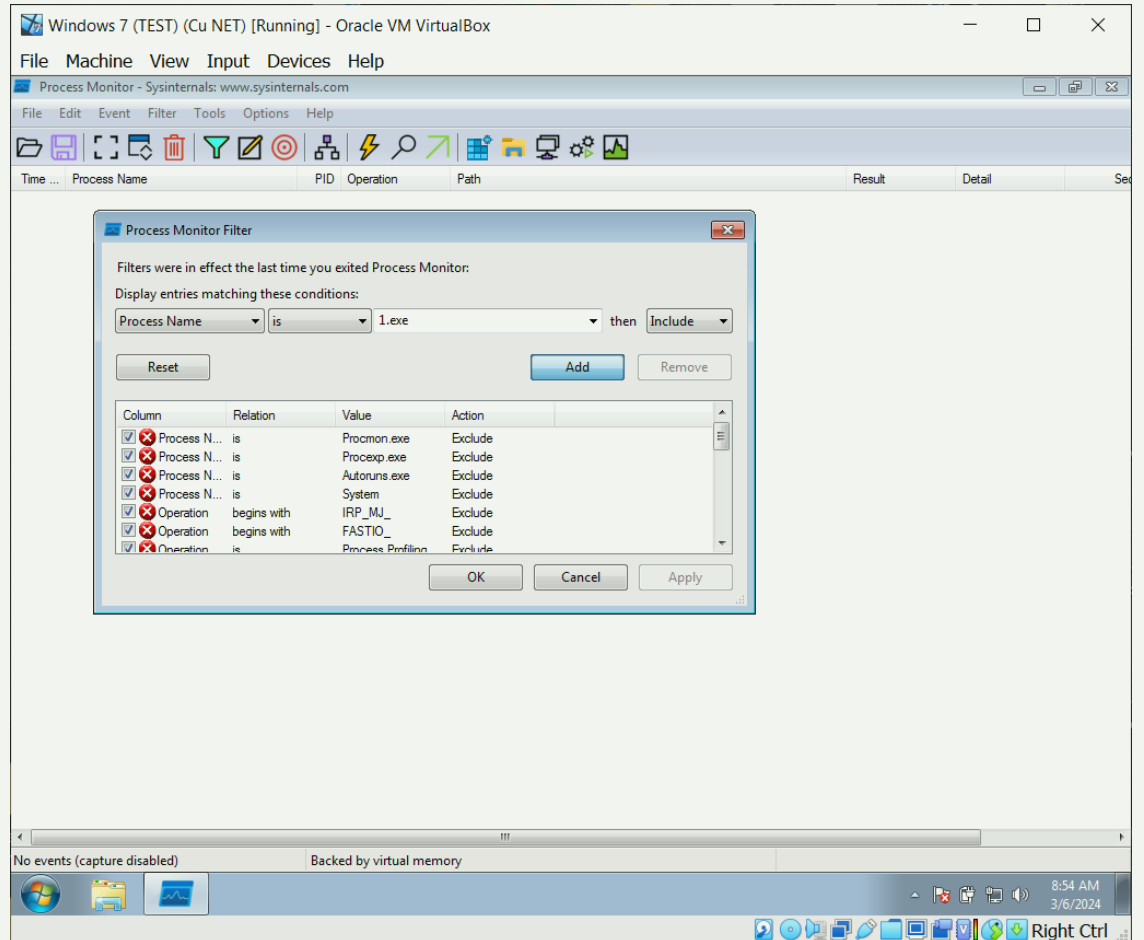
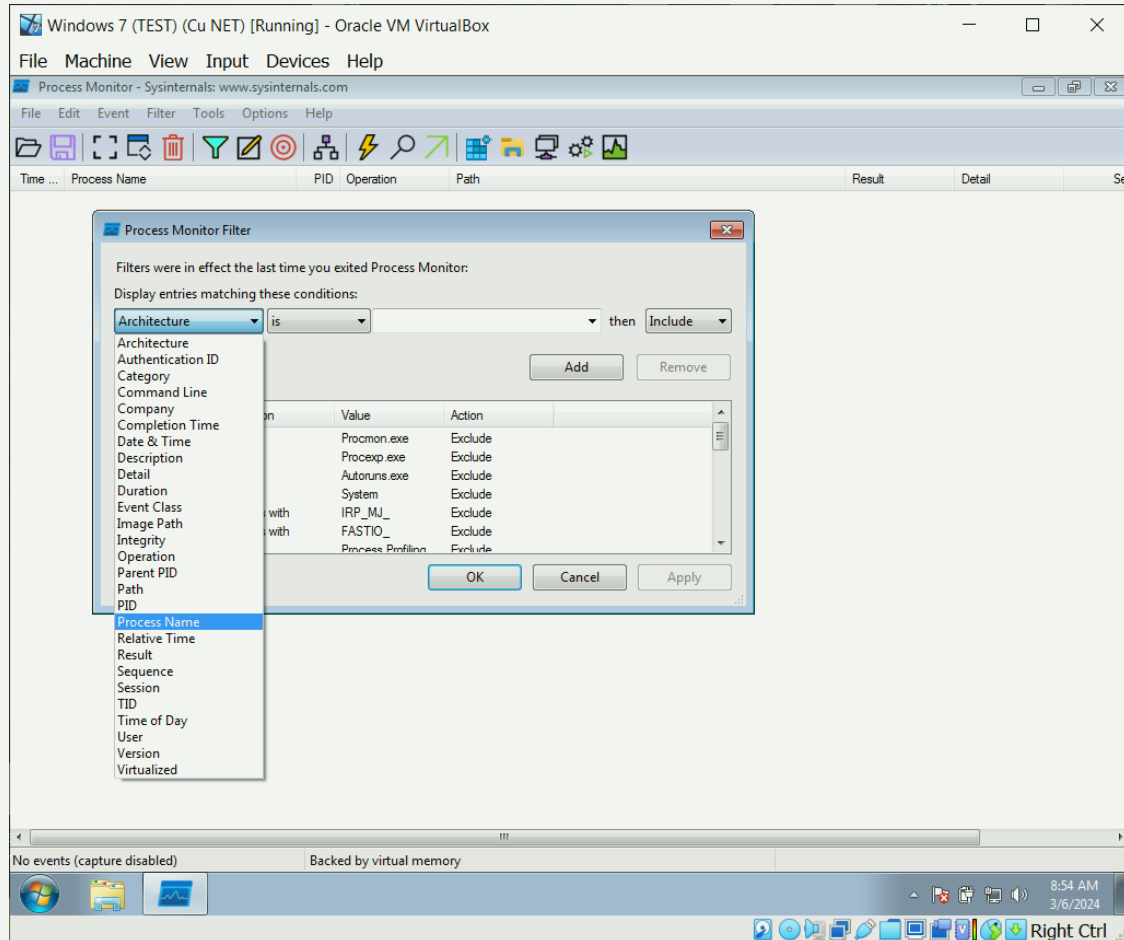
# PROCESS MONITOR & WIRESHARK

## DETONAREA UNUI TROIAN / PREGATIRE SCENA



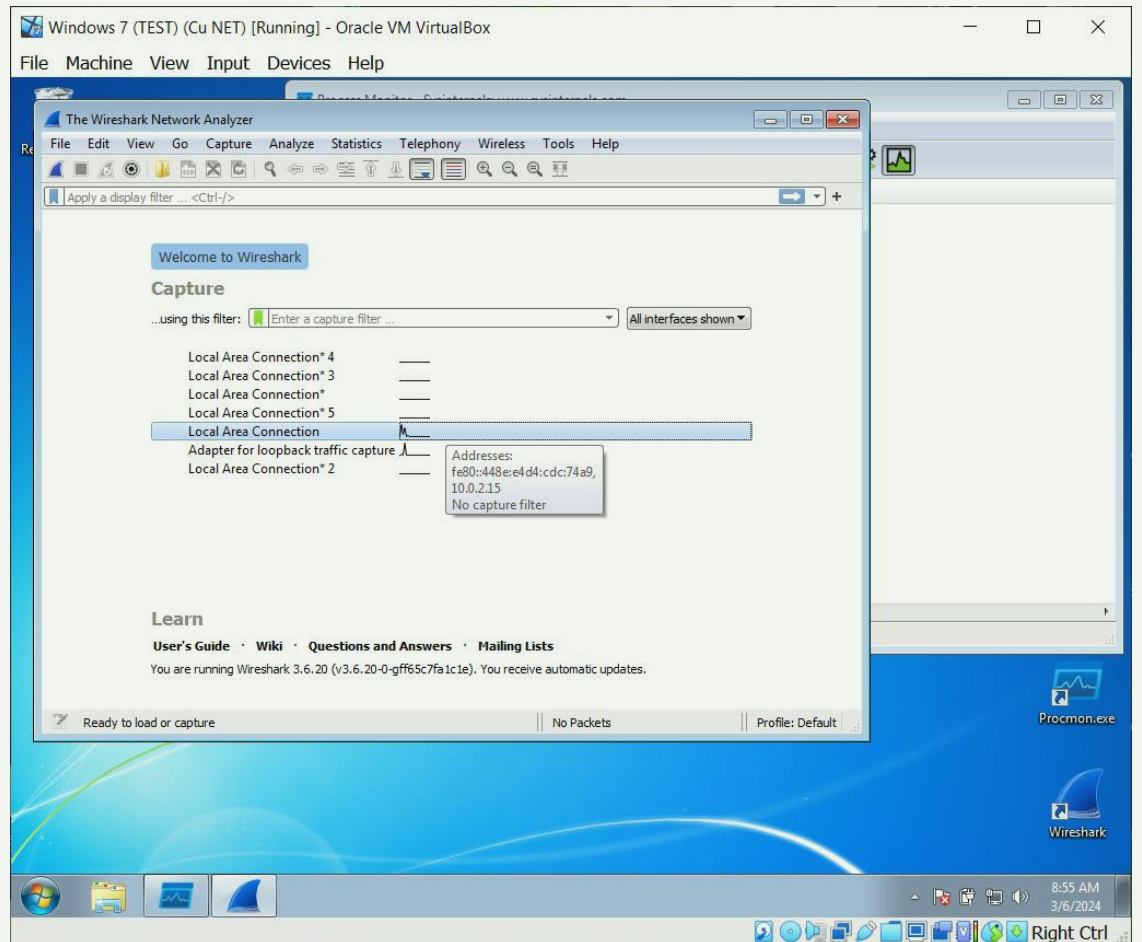
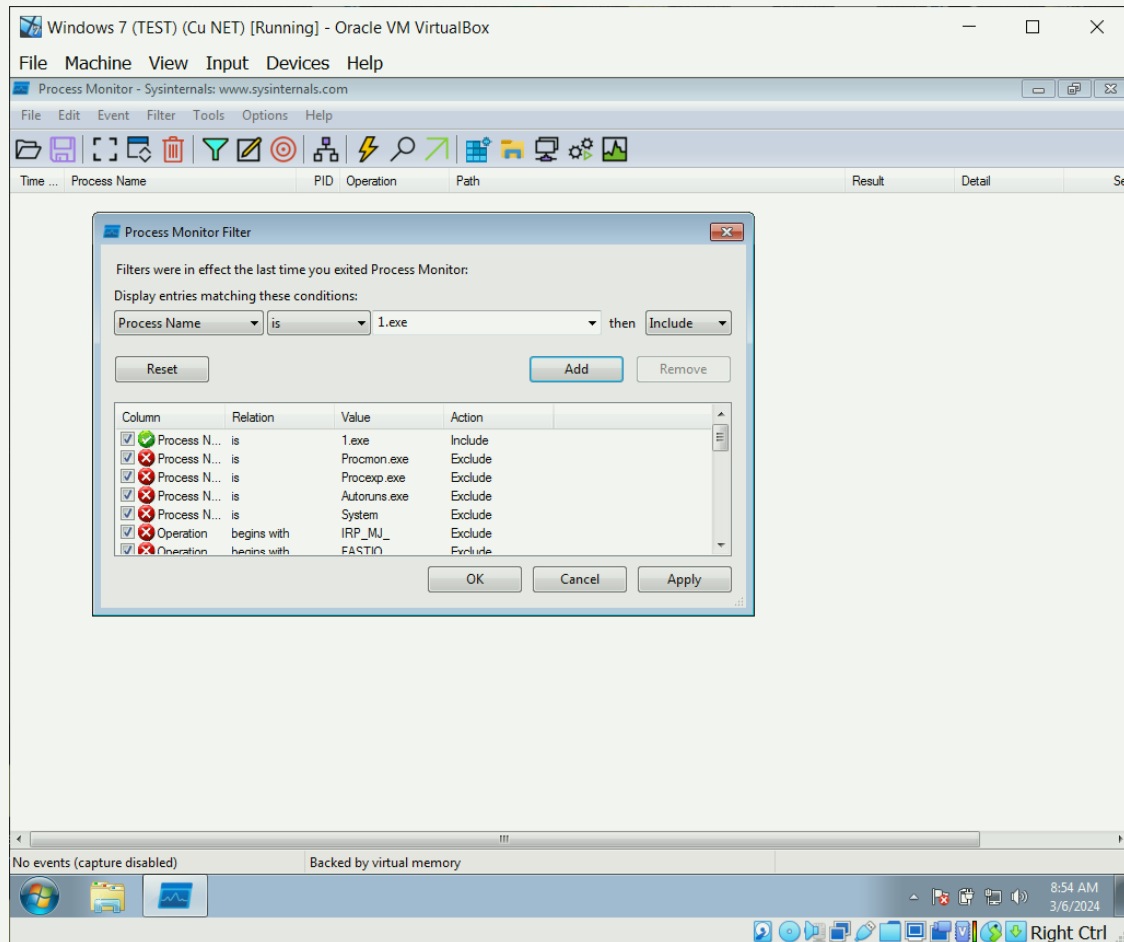
# PROCESS MONITOR & WIRESHARK

## DETONAREA UNUI TROIAN / FILTRU PE NUMELE PROCESULUI



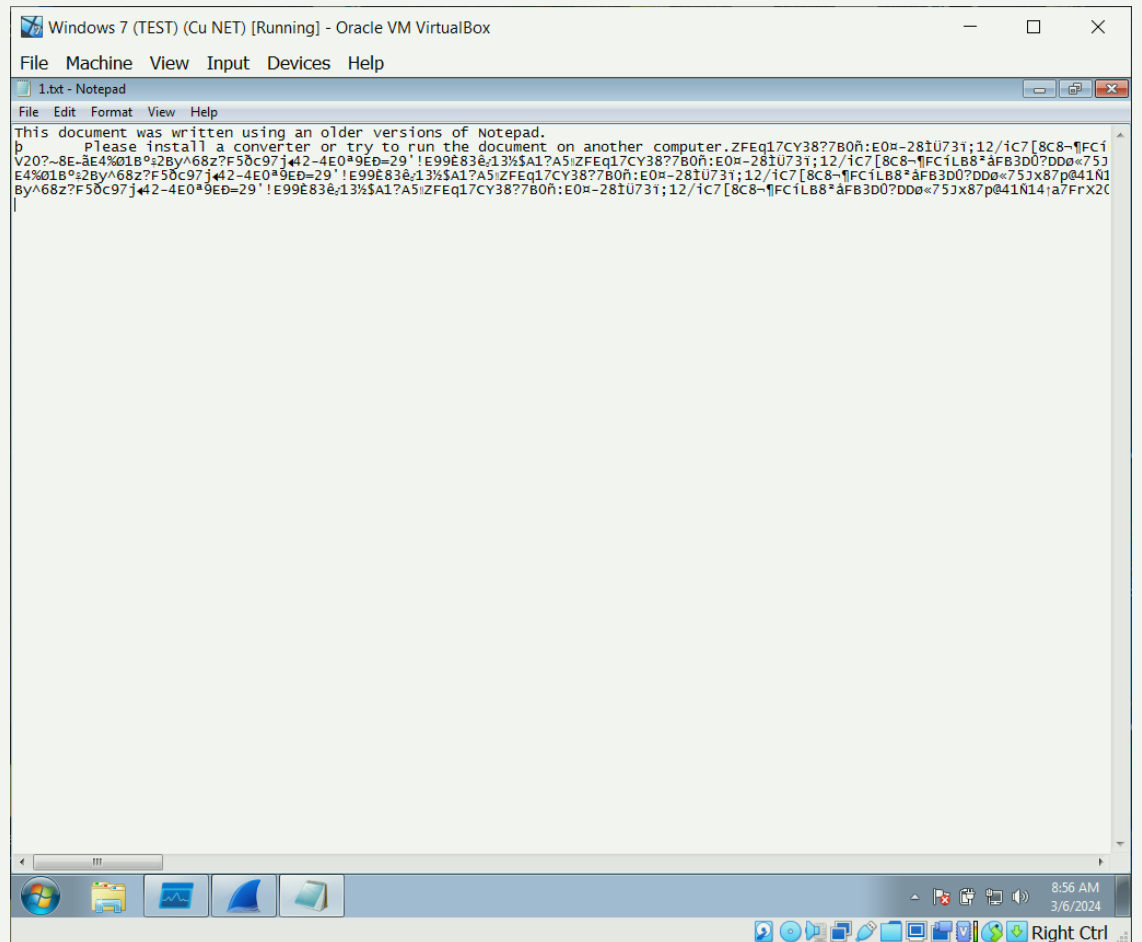
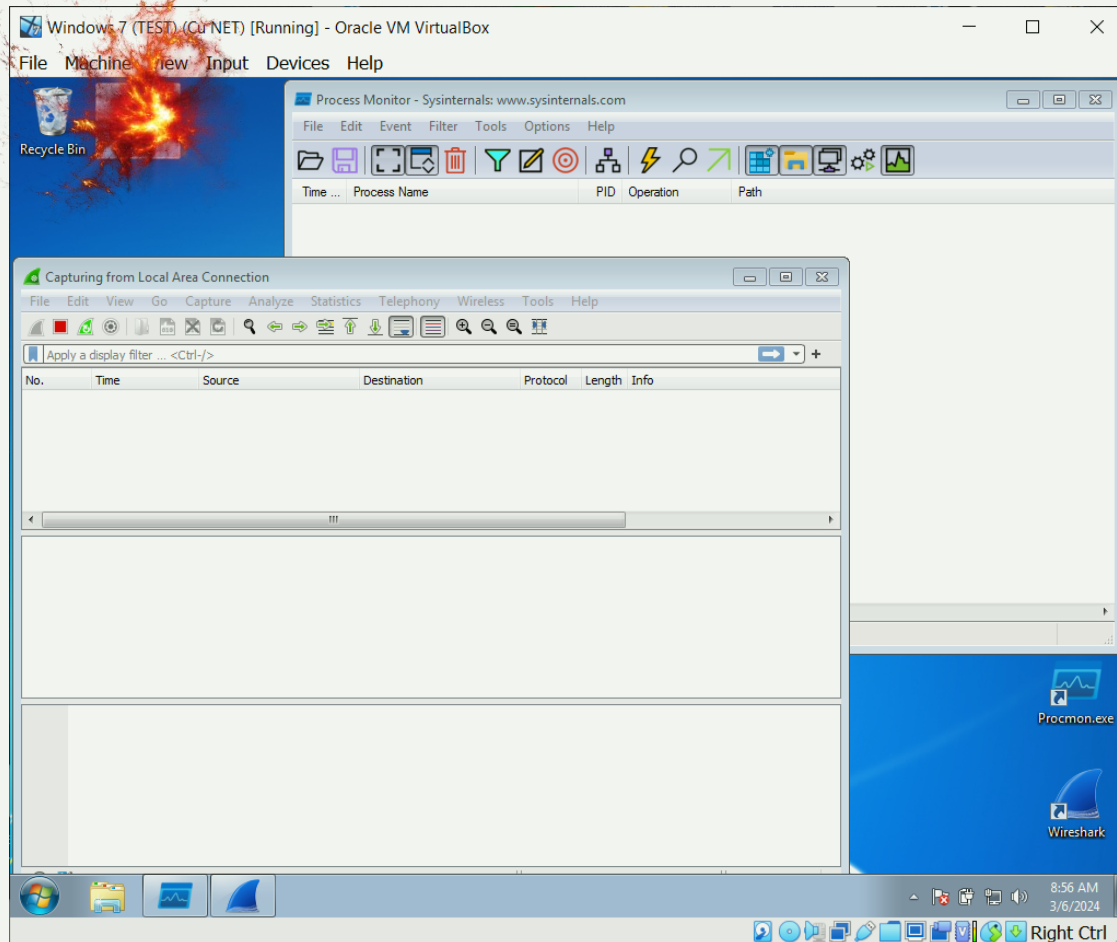
# PROCESS MONITOR & WIRESHARK

## DETONAREA UNUI TROIAN / ADAUGAREA FILTRULUI SI PORNIREA WIRESHARK



# PROCESS MONITOR & WIRESHARK

## SCENA PREGATITA - DETONAREA APLICATIEI MALWARE



# PROCESS MONITOR & WIRESHARK

## MONITORIZARE POST-DETONARE

The screenshot shows a Windows 7 virtual machine running on Oracle VM VirtualBox. The desktop has a Recycle Bin and a file named '1.exe'. Two applications are open: Process Monitor (Sysinternals) and Wireshark.

**Process Monitor - Sysinternals: www.sysinternals.com**

| Time ...  | Process Name | PID  | Operation          | Path                                              |
|-----------|--------------|------|--------------------|---------------------------------------------------|
| 8:57:0... | 1.exe        | 3344 | CloseFile          | C:\Users\Paul\AppData\Local\Microsoft\Web Results |
| 8:57:0... | 1.exe        | 3344 | CreateFile         | C:\Users                                          |
| 8:57:0... | 1.exe        | 3344 | QueryBasicInfor... | C:\Users                                          |

**Capturing from Local Area Connection**

| No. | Time     | Source        | Destination   | Protocol | Length | Info                      |
|-----|----------|---------------|---------------|----------|--------|---------------------------|
| 15  | 0.918308 | 10.0.2.15     | 104.208.16.93 | TCP      | 54     | 49195 → 443 [ACK] Seq=1 A |
| 16  | 0.918876 | 10.0.2.15     | 104.208.16.93 | SSLv3    | 112    | Client Hello              |
| 17  | 0.919142 | 104.208.16.93 | 10.0.2.15     | TCP      | 60     | 443 → 49195 [ACK] Seq=1 A |
| 18  | 1.136766 | 104.208.16.93 | 10.0.2.15     | TCP      | 60     | 443 → 49195 [RST, ACK] Se |
| 19  | 1.825169 | 10.0.2.15     | 193.231.252.1 | DNS      | 82     | Standard query 0xd737 A e |
| 20  | 2.021611 | 193.231.252.1 | 10.0.2.15     | DNS      | 155    | Standard query response 0 |

**Frame 1: 342 bytes on wire (2736 bits), 342 bytes captured (2736 bits) on interface \Device\NPF\_{B1B1794...}**

- Ethernet II, Src: PcsCompu\_f1:1a:20 (08:00:27:f1:1a:20), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
- Internet Protocol Version 4, Src: 10.0.2.15, Dst: 255.255.255.255
- User Datagram Protocol, Src Port: 68, Dst Port: 67
- Dynamic Host Configuration Protocol (Inform)

The taskbar shows the Start button, taskbar icons, and the system clock (8:57 AM, 3/6/2024). The taskbar also shows icons for Recycle Bin, 1.exe, Process Monitor (Procmon.exe), and Wireshark.

The screenshot shows a Windows 7 virtual machine running on Oracle VM VirtualBox. The desktop has a Recycle Bin and a file named '1.exe'. The Process Monitor (Sysinternals) application is open, displaying a list of events.

**Process Monitor - Sysinternals: www.sysinternals.com**

| Time ...  | Process Name | PID  | Operation          | Path                                                              | Result            | Detail                  |
|-----------|--------------|------|--------------------|-------------------------------------------------------------------|-------------------|-------------------------|
| 8:56:3... | 1.exe        | 3344 | CreateFile         | C:\Windows\Prefetch\1.EXE-55565196.pf                             | NAME NOT FOUND    | Desired Access: G...    |
| 8:56:3... | 1.exe        | 3344 | RegOpenKey         | HKLM\System\CurrentControlSet\Control\Session Manager             | REPARSE           | Desired Access: R...    |
| 8:56:3... | 1.exe        | 3344 | RegOpenKey         | HKLM\System\CurrentControlSet\Control\Session Manager             | SUCCESS           | Desired Access: R...    |
| 8:56:3... | 1.exe        | 3344 | RegQueryValue      | HKLM\System\CurrentControlSet\Control\Session Manager\CWDille...  | NAME NOT FOUND    | Length: 1,024           |
| 8:56:3... | 1.exe        | 3344 | RegCloseKey        | HKLM\System\CurrentControlSet\Control\Session Manager             | SUCCESS           |                         |
| 8:56:3... | 1.exe        | 3344 | CreateFile         | C:\Users\Paul\Desktop                                             | SUCCESS           | Desired Access: E...    |
| 8:56:3... | 1.exe        | 3344 | RegOpenKey         | HKLM\System\CurrentControlSet\Control\Terminal Server             | REPARSE           | Desired Access: R...    |
| 8:56:3... | 1.exe        | 3344 | RegQueryValue      | HKLM\System\CurrentControlSet\Control\Terminal Server\TSAppCo...  | NAME NOT FOUND    | Length: 548             |
| 8:56:3... | 1.exe        | 3344 | RegOpenKey         | HKLM\System\CurrentControlSet\Control\Terminal Server\TSUserEn... | SUCCESS           | Type: REG_DWO...        |
| 8:56:3... | 1.exe        | 3344 | RegCloseKey        | HKLM\System\CurrentControlSet\Control\Terminal Server             | SUCCESS           |                         |
| 8:56:3... | 1.exe        | 3344 | RegOpenKey         | HKLM\System\CurrentControlSet\Control\SafeBoot\Option             | REPARSE           | Desired Access: Q...    |
| 8:56:3... | 1.exe        | 3344 | RegOpenKey         | HKLM\System\CurrentControlSet\Control\SafeBoot\Option             | NAME NOT FOUND    | Desired Access: Q...    |
| 8:56:3... | 1.exe        | 3344 | RegOpenKey         | HKLM\System\CurrentControlSet\Control\Sip\GPD\DLL                 | REPARSE           | Desired Access: R...    |
| 8:56:3... | 1.exe        | 3344 | RegOpenKey         | HKLM\System\CurrentControlSet\Control\Sip\GPD\DLL                 | NAME NOT FOUND    | Desired Access: R...    |
| 8:56:3... | 1.exe        | 3344 | RegOpenKey         | HKLM\Software\Policies\Microsoft\Windows\Safer\CodeIdentifiers    | SUCCESS           | Desired Access: Q...    |
| 8:56:3... | 1.exe        | 3344 | RegQueryValue      | HKLM\SOFTWARE\Policies\Microsoft\Windows\safer\codeidentifiers... | NAME NOT FOUND    | Length: 80              |
| 8:56:3... | 1.exe        | 3344 | RegCloseKey        | HKLM\SOFTWARE\Policies\Microsoft\Windows\safer\codeidentifiers... | SUCCESS           |                         |
| 8:56:3... | 1.exe        | 3344 | RegOpenKey         | HKCU\Software\Policies\Microsoft\Windows\Safer\CodeIdentifiers    | NAME NOT FOUND    | Desired Access: Q...    |
| 8:56:3... | 1.exe        | 3344 | CreateFile         | C:\Users\Paul\Desktop\MSVBVM60.DLL                                | NAME NOT FOUND    | Desired Access: R...    |
| 8:56:3... | 1.exe        | 3344 | CreateFile         | C:\Windows\System32\msvbvm60.dll                                  | SUCCESS           | Desired Access: R...    |
| 8:56:3... | 1.exe        | 3344 | QueryBasicInfor... | C:\Windows\System32\msvbvm60.dll                                  | SUCCESS           | CreationTime: 6/10...   |
| 8:56:3... | 1.exe        | 3344 | CloseFile          | C:\Windows\System32\msvbvm60.dll                                  | SUCCESS           |                         |
| 8:56:3... | 1.exe        | 3344 | CreateFile         | C:\Windows\System32\msvbvm60.dll                                  | SUCCESS           | Desired Access: R...    |
| 8:56:3... | 1.exe        | 3344 | CreateFileMap...   | C:\Windows\System32\msvbvm60.dll                                  | FILE LOCKED WI... | SyncType: SyncTy...     |
| 8:56:3... | 1.exe        | 3344 | QueryStandardI...  | C:\Windows\System32\msvbvm60.dll                                  | SUCCESS           | AllocationSize: 1,3...  |
| 8:56:3... | 1.exe        | 3344 | ReadFile           | C:\Windows\System32\msvbvm60.dll                                  | SUCCESS           | Offset: 0, Length: 4... |
| 8:56:3... | 1.exe        | 3344 | CreateFileMap...   | C:\Windows\System32\msvbvm60.dll                                  | SUCCESS           | SyncType: SyncTy...     |
| 8:56:3... | 1.exe        | 3344 | CloseFile          | C:\Windows\System32\msvbvm60.dll                                  | SUCCESS           |                         |
| 8:56:3... | 1.exe        | 3344 | ReadFile           | C:\Windows\System32\msvbvm60.dll                                  | SUCCESS           | Offset: 1,118,208, ...  |
| 8:56:3... | 1.exe        | 3344 | ReadFile           | C:\Windows\System32\msvbvm60.dll                                  | SUCCESS           | Offset: 1,003,520, ...  |
| 8:56:3... | 1.exe        | 3344 | ReadFile           | C:\Windows\System32\msvbvm60.dll                                  | SUCCESS           | Offset: 4,096, Leng...  |
| 8:56:3... | 1.exe        | 3344 | CreateFile         | C:\Windows\System32\sechost.dll                                   | SUCCESS           | Desired Access: R...    |
| 8:56:3... | 1.exe        | 3344 | QueryBasicInfor... | C:\Windows\System32\sechost.dll                                   | SUCCESS           | CreationTime: 8/15...   |
| 8:56:3... | 1.exe        | 3344 | CloseFile          | C:\Windows\System32\sechost.dll                                   | SUCCESS           |                         |
| 8:56:3... | 1.exe        | 3344 | CreateFile         | C:\Windows\System32\sechost.dll                                   | SUCCESS           | Desired Access: R...    |
| 8:56:3... | 1.exe        | 3344 | CreateFileMap...   | C:\Windows\System32\sechost.dll                                   | FILE LOCKED WI... | SyncType: SyncTy...     |
| 8:56:3... | 1.exe        | 3344 | CreateFileMap...   | C:\Windows\System32\sechost.dll                                   | SUCCESS           | SyncType: SyncTy...     |
| 8:56:3... | 1.exe        | 3344 | CloseFile          | C:\Windows\System32\sechost.dll                                   | SUCCESS           |                         |

The taskbar shows the Start button, taskbar icons, and the system clock (8:59 AM, 3/6/2024). The taskbar also shows icons for Recycle Bin, 1.exe, and Process Monitor.



# PROCESS MONITOR & WIRESHARK

## CE URMARIM IN WIRESHARK?

Windows 7 (TEST) (Cu NET) [Running] - Oracle VM VirtualBox

File Machine View Input Devices Help

Capturing from Local Area Connection

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter ... <Ctrl-F>

| No. | Time     | Source        | Destination     | Protocol | Length | Info                                                              |
|-----|----------|---------------|-----------------|----------|--------|-------------------------------------------------------------------|
| 1   | 0.000000 | 10.0.2.15     | 255.255.255.255 | DHCP     | 342    | DHCP Inform - Transaction ID 0x677d0a3d                           |
| 2   | 0.000323 | 10.0.2.2      | 10.0.2.15       | DHCP     | 590    | DHCP ACK - Transaction ID 0x677d0a3d                              |
| 3   | 0.008138 | 10.0.2.15     | 193.231.252.1   | DNS      | 80     | Standard query 0x039a A watson.microsoft.com                      |
| 4   | 0.071533 | 193.231.252.1 | 10.0.2.15       | DNS      | 201    | Standard query response 0x039a A watson.microsoft.com CNAME le... |
| 5   | 0.073295 | 10.0.2.15     | 104.208.16.93   | TCP      | 66     | 49194 → 443 [SYN] Seq=0 Win=8192 Len=0 MSS=1460 WS=256 SACK_P...  |
| 6   | 0.382347 | 104.208.16.93 | 10.0.2.15       | TCP      | 60     | 443 → 49194 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460       |
| 7   | 0.382464 | 10.0.2.15     | 104.208.16.93   | TCP      | 54     | 49194 → 443 [ACK] Seq=1 Ack=1 Win=64240 Len=0                     |
| 8   | 0.384236 | 10.0.2.15     | 104.208.16.93   | TLSv1    | 186    | Client Hello                                                      |
| 9   | 0.384598 | 104.208.16.93 | 10.0.2.15       | TCP      | 60     | 443 → 49194 [ACK] Seq=1 Ack=133 Win=65535 Len=0                   |
| 10  | 0.598310 | 104.208.16.93 | 10.0.2.15       | TCP      | 60     | 443 → 49194 [RST, ACK] Seq=1 Ack=133 Win=65535 Len=0              |

Frame 16: 112 bytes on wire (896 bits), 112 bytes captured (896 bits) on interface \Device\NPF\_{B181794A-0F49-497E-BBE9-E4E6B0484814}, id 0

Ethernet II, Src: PcsCompu\_f1:1a:20 (08:00:27:f1:1a:20), Dst: RealtekU\_12:35:02 (52:54:00:12:35:02)

Internet Protocol Version 4, Src: 10.0.2.15, Dst: 104.208.16.93

Transmission Control Protocol, Src Port: 49195, Dst Port: 443, Seq: 1, Ack: 1, Len: 58

Transport Layer Security

0000 52 54 00 12 35 02 08 00 27 f1 1a 20 08 00 45 00 RT...5... '...' E-

0010 00 62 01 27 40 08 08 06 00 00 0a 00 02 0f 68 d0 .b'@... ..h-

0020 10 5d c0 2b 01 bb a9 b7 dc a4 01 0c 8e 02 50 18 .]'+... ..P-

0030 fa f0 85 90 00 00 16 03 00 00 35 01 00 00 31 03 .e'..5...-1-

0040 00 65 e8 a0 53 76 5c 4c b4 37 90 5c 26 f6 95 ba .e'..Sv\..7-&...

0050 30 35 8a ef 81 e5 ed 30 19 9b d4 84 9e 93 15 ea 05... ..0 .....

0060 85 00 00 0a 00 0a 00 00 05 00 04 00 ff 01 00 .....

Local Area Connection: <live capture in progress> | Packets: 30 · Displayed: 30 (100.0%) | Profile: Default

9:00 AM 3/6/2024

Windows 7 (TEST) (Cu NET) [Running] - Oracle VM VirtualBox

File Machine View Input Devices Help

\*Local Area Connection

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter ... <Ctrl-F>

| No. | Time       | Source                 | Destination       | Protocol | Length | Info                                                           |
|-----|------------|------------------------|-------------------|----------|--------|----------------------------------------------------------------|
| 1   | 0.000000   | 10.0.2.15              | 255.255.255.255   | DHCP     | 342    | DHCP Inform - Transaction ID 0x677d0a3d                        |
| 2   | 0.000323   | 10.0.2.2               | 10.0.2.15         | DHCP     | 590    | DHCP ACK - Transaction ID 0x677d0a3d                           |
| 3   | 0.008138   | 10.0.2.15              | 193.231.252.1     | DNS      | 80     | Standard query 0x039a A watson.microsoft.com                   |
| 4   | 0.071533   | 193.231.252.1          | 10.0.2.15         | DNS      | 201    | Standard query response 0x039a A watson.microsoft.com CNAME    |
| 5   | 0.073295   | 10.0.2.15              | 104.208.16.93     | TCP      | 66     | 49194 → 443 [SYN] Seq=0 Win=8192 Len=0 MSS=1460 WS=256 SACK... |
| 6   | 0.382347   | 104.208.16.93          | 10.0.2.15         | TCP      | 60     | 443 → 49194 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460    |
| 7   | 0.382464   | 10.0.2.15              | 104.208.16.93     | TCP      | 54     | 49194 → 443 [ACK] Seq=1 Ack=1 Win=64240 Len=0                  |
| 8   | 0.384236   | 10.0.2.15              | 104.208.16.93     | TLSv1    | 186    | Client Hello                                                   |
| 9   | 0.384598   | 104.208.16.93          | 10.0.2.15         | TCP      | 60     | 443 → 49194 [ACK] Seq=1 Ack=133 Win=65535 Len=0                |
| 10  | 0.598310   | 104.208.16.93          | 10.0.2.15         | TCP      | 60     | 443 → 49194 [RST, ACK] Seq=1 Ack=133 Win=65535 Len=0           |
| 11  | 0.598751   | 10.0.2.15              | 104.208.16.93     | TCP      | 66     | 49195 → 443 [SYN] Seq=0 Win=8192 Len=0 MSS=1460 WS=256 SACK... |
| 12  | 0.626416   | 10.0.2.15              | 193.231.252.1     | DNS      | 86     | Standard query 0xb109 PTR 93.16.208.104.in-addr.arpa           |
| 13  | 0.783037   | 193.231.252.1          | 10.0.2.15         | DNS      | 163    | Standard query response 0xb109 No such name PTR 93.16.208.10   |
| 14  | 0.918218   | 104.208.16.93          | 10.0.2.15         | TCP      | 60     | 443 → 49195 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460    |
| 15  | 0.918308   | 10.0.2.15              | 104.208.16.93     | TCP      | 54     | 49195 → 443 [ACK] Seq=1 Ack=1 Win=64240 Len=0                  |
| 16  | 0.918876   | 10.0.2.15              | 104.208.16.93     | SSLv3    | 112    | Client Hello                                                   |
| 17  | 0.919142   | 104.208.16.93          | 10.0.2.15         | TCP      | 60     | 443 → 49195 [ACK] Seq=1 Ack=59 Win=65535 Len=0                 |
| 18  | 1.136766   | 104.208.16.93          | 10.0.2.15         | TCP      | 60     | 443 → 49195 [RST, ACK] Seq=1 Ack=59 Win=65535 Len=0            |
| 19  | 1.825169   | 10.0.2.15              | 193.231.252.1     | DNS      | 82     | Standard query 0xd737 A email.people.yahoo.com                 |
| 20  | 2.021611   | 193.231.252.1          | 10.0.2.15         | DNS      | 155    | Standard query response 0xd737 No such name A email.people.y   |
| 21  | 40.648957  | 10.0.2.15              | 172.67.75.39      | TCP      | 54     | 49190 → 443 [FIN, ACK] Seq=1 Ack=1 Win=64240 Len=0             |
| 22  | 40.649342  | 172.67.75.39           | 10.0.2.15         | TCP      | 60     | 443 → 49190 [ACK] Seq=1 Ack=2 Win=65535 Len=0                  |
| 23  | 40.698265  | 172.67.75.39           | 10.0.2.15         | TCP      | 60     | 443 → 49190 [FIN, ACK] Seq=1 Ack=2 Win=65535 Len=0             |
| 24  | 40.698311  | 10.0.2.15              | 172.67.75.39      | TCP      | 54     | 49190 → 443 [ACK] Seq=2 Ack=2 Win=64240 Len=0                  |
| 25  | 71.110426  | 10.0.2.15              | 92.122.17.28      | TCP      | 54     | 49192 → 80 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0                  |
| 26  | 71.110509  | 10.0.2.15              | 92.122.17.28      | TCP      | 54     | 49193 → 80 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0                  |
| 27  | 101.329145 | 192.229.221.95         | 10.0.2.15         | TCP      | 60     | 80 → 49191 [FIN, ACK] Seq=1 Ack=1 Win=65535 Len=0              |
| 28  | 101.329240 | PcsCompu_f1:1a:20      | Broadcast         | ARP      | 42     | Who has 10.0.2.2? Tell 10.0.2.15                               |
| 29  | 101.329397 | RealtekU_12:35:02      | PcsCompu_f1:1a:20 | ARP      | 60     | 10.0.2.2 is at 52:54:00:12:35:02                               |
| 30  | 101.329418 | 10.0.2.15              | 192.229.221.95    | TCP      | 54     | 49191 → 80 [ACK] Seq=1 Ack=2 Win=64240 Len=0                   |
| 31  | 315.481022 | fe80::448e:e4d4:cdc... | ff02::1:2         | DHCPv6   | 149    | Solicit XID: 0xc9bbea CID: 000100012d5a4a43080027f11a20        |
| 32  | 316.405868 | fe80::448e:e4d4:cdc... | ff02::1:2         | DHCPv6   | 149    | Solicit XID: 0xc9bbea CID: 000100012d5a4a43080027f11a20        |

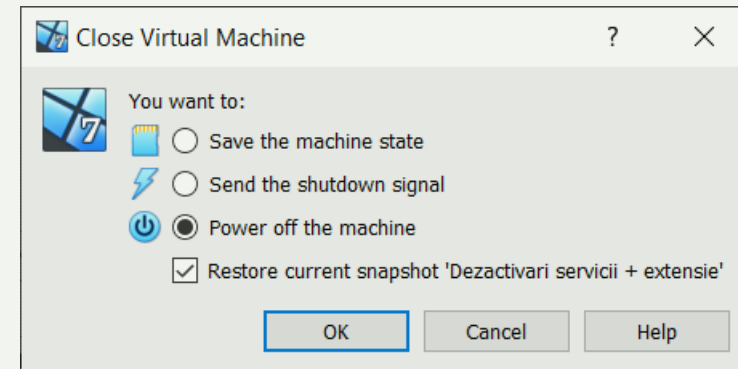
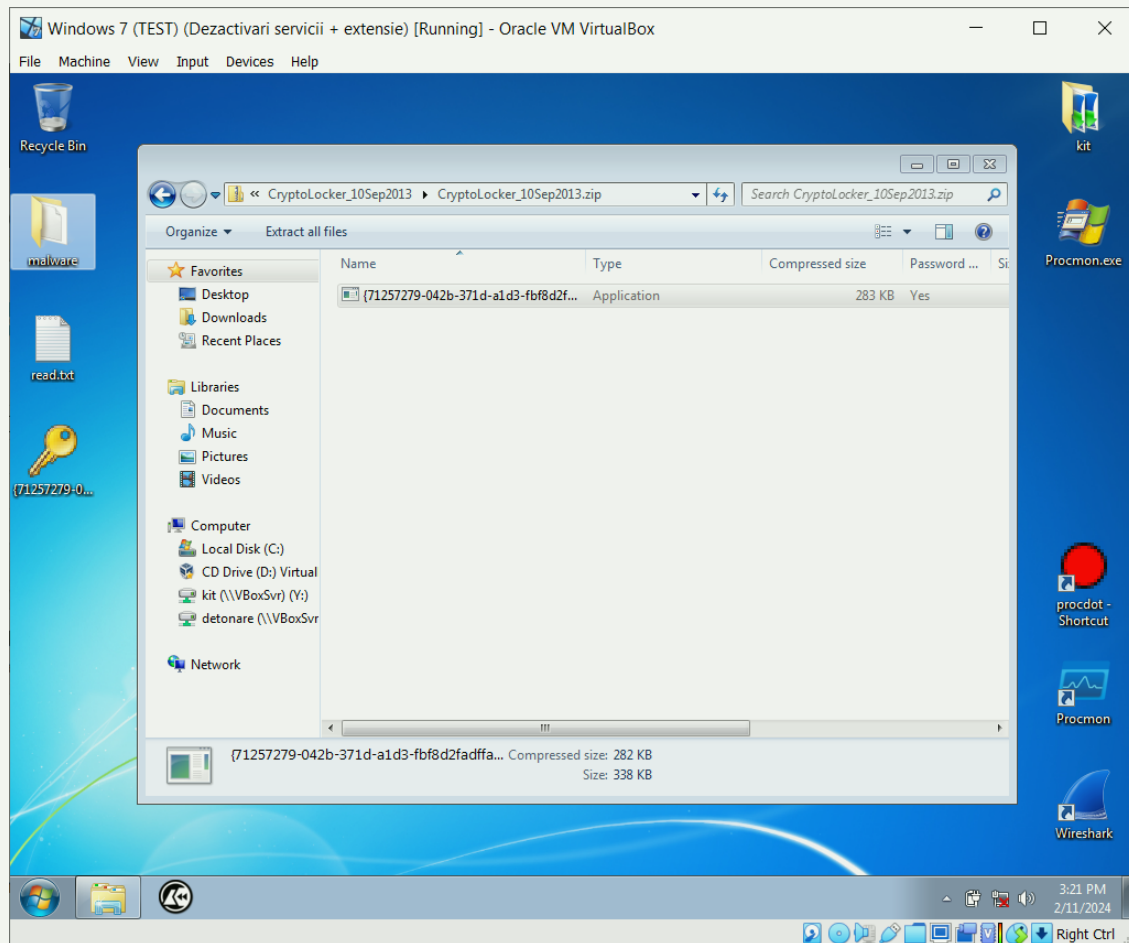
Frame 27: 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface \Device\NPF\_{B181794A-0F49-497E-BBE9-E4E6B0484814}, id 0

wireshark\_Local Area Connection1.BICK2.pcapng | Packets: 34 · Displayed: 34 (100.0%) · Dropped: 0 (0.0%) | Profile: Default

9:04 AM 3/6/2024

# POST-ANALIZA

## RESETAREA MASINII VIRTUALE



Dupa un experiment Masina Virtuala trebuie Resetata prin selectarea : Restore current shapshot ...

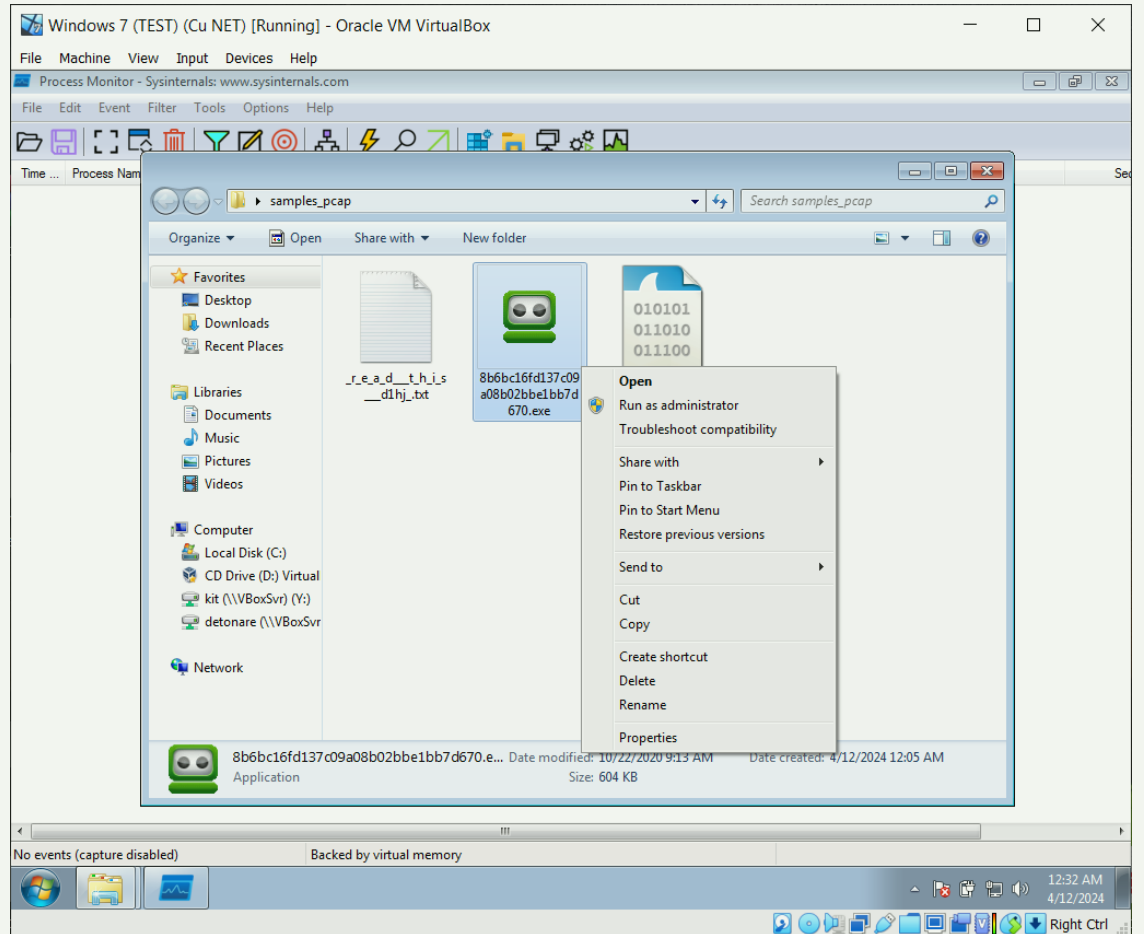
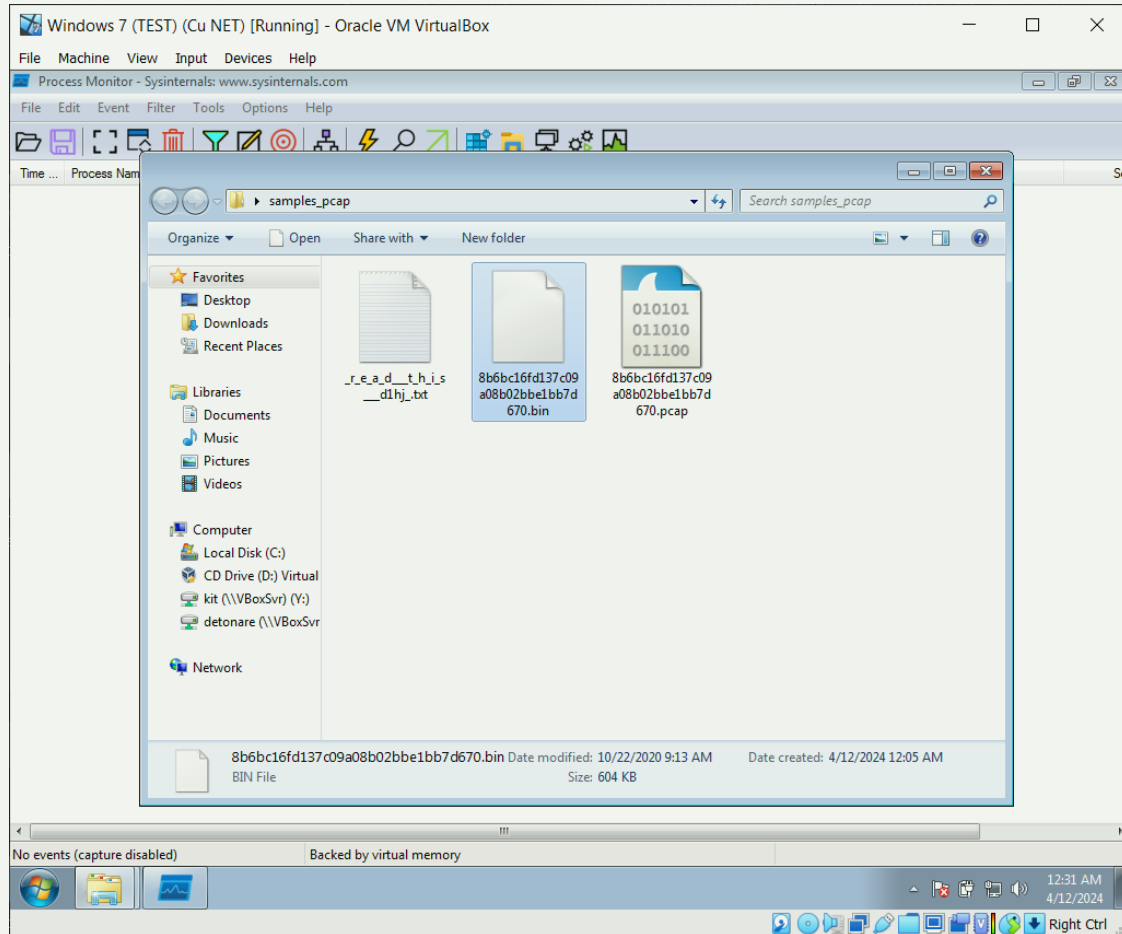
C.11.3

# CAPTAREA EVENIMENTELOR EFEMERE ȘI ELIMINAREA ' ZGOMOTULUI



# PROCESS MONITOR & WIRESHARK

## DETONAREA UNUI FISIER SUSPECT





# PROCESS MONITOR & WIRESHARK

## OPERATIUNI DE SCRIERE & FILTRELE WIRESHARK !

Windows 7 (TEST) (Cu NET) [Running] - Oracle VM VirtualBox

File Machine View Input Devices Help

\*Local Area Connection

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

www

| No.  | Time      | Source        | Destination   | Protocol | Length | Info                                                                |
|------|-----------|---------------|---------------|----------|--------|---------------------------------------------------------------------|
| 4409 | 76.544638 | 10.0.2.15     | 104.20.20.251 | HTTP     | 421    | GET /v1/btc/main/addr/17gd1msp5FnMcEMF1MitTNSsYs7w7AQyCt?_u=1712... |
| 4413 | 76.808220 | 104.20.20.251 | 10.0.2.15     | HTTP/1.1 | 403    | Forbidden , JavaScript Object Notation (application/...             |
| 4460 | 79.733013 | 10.0.2.15     | 104.18.21.226 | HTTP     | 299    | GET /rootr6/MFEwTzBNMEswSTAJBgUrDgMCGgUABBRi%2B7TJbHyn9Em39u031e... |
| 4463 | 79.752741 | 104.18.21.226 | 10.0.2.15     | OCSP     | 746    | Response                                                            |
| 4470 | 79.815131 | 10.0.2.15     | 104.18.21.226 | HTTP     | 307    | GET /gsgccr6alphasslca2023/ME0wSzB3MEcwRTAJBgUrDgMCGgUABBTYuQbxg... |
| 4473 | 80.010554 | 104.18.21.226 | 10.0.2.15     | OCSP     | 495    | Response                                                            |

Frame 4460: 299 bytes on wire (2392 bits), 299 bytes captured (2392 bits) on interface \Device\NPF\_{B1B1794A-0F49-497E-BBE9-E4E6D0484814}, i...

Ethernet II, Src: PcsCompu\_f1:1a:20 (08:00:27:f1:1a:20), Dst: RealtekU\_12:35:02 (52:54:00:12:35:02)

Internet Protocol Version 4, Src: 10.0.2.15, Dst: 104.18.21.226

Transmission Control Protocol, Src Port: 40310, Dst Port: 80, Seq: 1, Len: 245

0000 52 54 00 12 35 02 08 00 27 f1 1a 20 08 00 45 00 RT--5...E  
0010 01 1d 0e 7f 40 00 00 06 00 00 0a 00 02 0f 68 12 ...@...h  
0020 15 e2 c0 3a 00 50 61 d9 bc cf 01 be 4a 02 50 18 ...Pa...J-P  
0030 fa f0 8b 12 00 00 47 45 54 20 2f 72 6f 6f 74 72 ...GE T /rootr  
0040 36 2f 4d 46 45 77 54 7a 42 4e 4d 45 73 77 53 54 6/MFEwTz BNMEswST  
0050 41 4a 42 67 55 72 44 67 4d 43 47 67 55 41 42 42 AJBgUrDg MCGgUA8B  
0060 52 69 25 32 42 37 54 4a 62 48 59 6e 39 45 6d 4a Ri%2B7TJ bHyn9EmJ  
0070 39 57 30 33 6c 65 63 42 37 50 25 32 42 47 37 51 9u031ecB 7P%2BGTQ  
0080 51 55 72 6d 77 46 6f 35 4d 54 34 71 4c 6e 34 74 QUmwFo5 MT4qLn4t  
0090 63 63 31 73 66 77 66 38 6e 55 36 41 43 45 48 ccisfwf8 hnUACEH  
0100 38 66 4c 4a 41 75 67 39 44 6a 74 76 73 37 37 6b 8fLJAug9 Djtvs77k  
0110 65 4c 58 6f 41 25 33 44 20 48 54 54 50 2f 31 2e eLXoAX3D HTTP/1.  
0120 31 0d 0a 43 6f 6e 6e 65 63 74 69 6f 6e 3a 20 40 1...Conne ction: K

"www" is neither a field nor a protocol name. | Packets: 4517 · Displayed: 6 (0.1%) | Profile: Default

12:37 AM  
4/12/2024

Windows 7 (TEST) (Cu NET) [Running] - Oracle VM VirtualBox

File Machine View Input Devices Help

Process Monitor - Sysinternals: www.sysinternals.com

File Edit Event Filter Tools Options Help

| Time ... | Process Name          | PID  | Operation  | Path                                                    | Result            | Detail                 | Sequence |
|----------|-----------------------|------|------------|---------------------------------------------------------|-------------------|------------------------|----------|
| 12:38... | \\BoxSvr\detonare\... | 4088 | WriteFile  | \\BoxSvr\detonare\...\email-worm\email-worm.win32\Em... | MEDIA WRITE PR... | Offset: 20,480, Len... | 60581    |
| 12:38... | \\BoxSvr\detonare\... | 4088 | WriteFile  | \\BoxSvr\detonare\...\email-worm\email-worm.win32\Em... | MEDIA WRITE PR... | Offset: 20,480, Len... | 60582    |
| 12:38... | \\BoxSvr\detonare\... | 4088 | CreateFile | \\BoxSvr\detonare\...\email-worm\email-worm.win32\Em... | SUCCESS           | Desired Access: R...   | 60583    |
| 12:38... | \\BoxSvr\detonare\... | 4088 | CreateFile | \\BoxSvr\detonare\...\email-worm\Email-Worm.Win32       | SUCCESS           | Desired Access: W...   | 60584    |
| 12:38... | \\BoxSvr\detonare\... | 4088 | CreateFile | \\BoxSvr\detonare\...\email-worm\email-worm.win32\Em... | SUCCESS           | Desired Access: G...   | 60585    |
| 12:38... | \\BoxSvr\detonare\... | 4088 | CreateFile | \\BoxSvr\detonare\...\email-worm\email-worm.win32\4K... | NAME NOT FOUND    | Desired Access: R...   | 60586    |
| 12:38... | \\BoxSvr\detonare\... | 4088 | WriteFile  | \\BoxSvr\detonare\...\email-worm\email-worm.win32\Em... | MEDIA WRITE PR... | Offset: 1,852, Leng... | 60587    |
| 12:38... | \\BoxSvr\detonare\... | 4088 | WriteFile  | \\BoxSvr\detonare\...\email-worm\email-worm.win32\Em... | MEDIA WRITE PR... | Offset: 1,792, Leng... | 60588    |
| 12:38... | \\BoxSvr\detonare\... | 4088 | WriteFile  | \\BoxSvr\detonare\...\email-worm\email-worm.win32\Em... | MEDIA WRITE PR... | Offset: 81,920, Len... | 60589    |
| 12:38... | \\BoxSvr\detonare\... | 4088 | WriteFile  | \\BoxSvr\detonare\...\email-worm\email-worm.win32\Em... | MEDIA WRITE PR... | Offset: 81,920, Len... | 60590    |
| 12:38... | \\BoxSvr\detonare\... | 4088 | CreateFile | \\BoxSvr\detonare\...\email-worm\email-worm.win32\Em... | SUCCESS           | Desired Access: R...   | 60591    |
| 12:38... | \\BoxSvr\detonare\... | 4088 | CreateFile | \\BoxSvr\detonare\...\email-worm\Email-Worm.Win32       | SUCCESS           | Desired Access: W...   | 60592    |
| 12:38... | \\BoxSvr\detonare\... | 4088 | CreateFile | \\BoxSvr\detonare\...\email-worm\email-worm.win32\Em... | SUCCESS           | Desired Access: G...   | 60593    |
| 12:38... | \\BoxSvr\detonare\... | 4088 | CreateFile | \\BoxSvr\detonare\...\email-worm\email-worm.win32\Em... | SUCCESS           | Desired Access: G...   | 60594    |
| 12:38... | \\BoxSvr\detonare\... | 4088 | WriteFile  | \\BoxSvr\detonare\...\email-worm\email-worm.win32\Em... | NAME NOT FOUND    | Desired Access: R...   | 60595    |
| 12:38... | \\BoxSvr\detonare\... | 4088 | WriteFile  | \\BoxSvr\detonare\...\email-worm\email-worm.win32\Em... | MEDIA WRITE PR... | Offset: 1,852, Leng... | 60596    |
| 12:38... | \\BoxSvr\detonare\... | 4088 | WriteFile  | \\BoxSvr\detonare\...\email-worm\email-worm.win32\Em... | MEDIA WRITE PR... | Offset: 1,792, Leng... | 60597    |
| 12:38... | \\BoxSvr\detonare\... | 4088 | WriteFile  | \\BoxSvr\detonare\...\email-worm\email-worm.win32\Em... | MEDIA WRITE PR... | Offset: 39,494, Len... | 60598    |
| 12:38... | \\BoxSvr\detonare\... | 4088 | WriteFile  | \\BoxSvr\detonare\...\email-worm\email-worm.win32\Em... | MEDIA WRITE PR... | Offset: 39,494, Len... | 60599    |
| 12:38... | \\BoxSvr\detonare\... | 4088 | WriteFile  | \\BoxSvr\detonare\...\email-worm\email-worm.win32\Em... | MEDIA WRITE PR... | Offset: 39,494, Len... | 60600    |
| 12:38... | \\BoxSvr\detonare\... | 4088 | CreateFile | \\BoxSvr\detonare\...\email-worm\email-worm.win32\Em... | SUCCESS           | Desired Access: R...   | 60601    |
| 12:38... | \\BoxSvr\detonare\... | 4088 | CreateFile | \\BoxSvr\detonare\...\email-worm\Email-Worm.Win32       | SUCCESS           | Desired Access: W...   | 60602    |
| 12:38... | \\BoxSvr\detonare\... | 4088 | CreateFile | \\BoxSvr\detonare\...\email-worm\email-worm.win32\Em... | SUCCESS           | Desired Access: G...   | 60603    |
| 12:38... | \\BoxSvr\detonare\... | 4088 | CreateFile | \\BoxSvr\detonare\...\email-worm\email-worm.win32\Em... | NAME NOT FOUND    | Desired Access: R...   | 60604    |
| 12:38... | \\BoxSvr\detonare\... | 4088 | WriteFile  | \\BoxSvr\detonare\...\email-worm\email-worm.win32\Em... | MEDIA WRITE PR... | Offset: 1,852, Leng... | 60605    |
| 12:38... | \\BoxSvr\detonare\... | 4088 | WriteFile  | \\BoxSvr\detonare\...\email-worm\email-worm.win32\Em... | MEDIA WRITE PR... | Offset: 1,792, Leng... | 60606    |
| 12:38... | \\BoxSvr\detonare\... | 4088 | WriteFile  | \\BoxSvr\detonare\...\email-worm\email-worm.win32\Em... | MEDIA WRITE PR... | Offset: 20,992, Len... | 60607    |
| 12:38... | \\BoxSvr\detonare\... | 4088 | WriteFile  | \\BoxSvr\detonare\...\email-worm\email-worm.win32\Em... | MEDIA WRITE PR... | Offset: 20,992, Len... | 60608    |
| 12:38... | \\BoxSvr\detonare\... | 4088 | CreateFile | \\BoxSvr\detonare\...\email-worm\email-worm.win32\Em... | SUCCESS           | Desired Access: R...   | 60609    |
| 12:38... | \\BoxSvr\detonare\... | 4088 | CreateFile | \\BoxSvr\detonare\...\email-worm\Email-Worm.Win32       | SUCCESS           | Desired Access: W...   | 60610    |
| 12:38... | \\BoxSvr\detonare\... | 4088 | CreateFile | \\BoxSvr\detonare\...\email-worm\email-worm.win32\Em... | SUCCESS           | Desired Access: G...   | 60611    |
| 12:38... | \\BoxSvr\detonare\... | 4088 | CreateFile | \\BoxSvr\detonare\...\email-worm\email-worm.win32\Em... | NAME NOT FOUND    | Desired Access: R...   | 60612    |
| 12:38... | \\BoxSvr\detonare\... | 4088 | WriteFile  | \\BoxSvr\detonare\...\email-worm\email-worm.win32\Em... | MEDIA WRITE PR... | Offset: 1,852, Leng... | 60613    |
| 12:38... | \\BoxSvr\detonare\... | 4088 | WriteFile  | \\BoxSvr\detonare\...\email-worm\email-worm.win32\Em... | MEDIA WRITE PR... | Offset: 1,792, Leng... | 60614    |
| 12:38... | \\BoxSvr\detonare\... | 4088 | WriteFile  | \\BoxSvr\detonare\...\email-worm\email-worm.win32\Em... | MEDIA WRITE PR... | Offset: 12,288, Len... | 60615    |
| 12:38... | \\BoxSvr\detonare\... | 4088 | WriteFile  | \\BoxSvr\detonare\...\email-worm\email-worm.win32\Em... | MEDIA WRITE PR... | Offset: 12,288, Len... | 60616    |
| 12:38... | \\BoxSvr\detonare\... | 4088 | WriteFile  | \\BoxSvr\detonare\...\email-worm\email-worm.win32\Em... | MEDIA WRITE PR... | Offset: 12,288, Len... | 60617    |
| 12:38... | \\BoxSvr\detonare\... | 4088 | WriteFile  | \\BoxSvr\detonare\...\email-worm\email-worm.win32\Em... | MEDIA WRITE PR... | Offset: 12,288, Len... | 60618    |

Showing 60,619 of 503,455 events (12%) | Backed by virtual memory

12:38 AM  
4/12/2024



# PROCESS MONITOR

## EVALUAREA EVENIMENTELOR EFEMERE POST-DETONARE

Windows 7 (TEST) (Cu NET) [Running] - Oracle VM VirtualBox

File Machine View Input Devices Help

Process Monitor - Sysinternals: www.sysinternals.com

File Edit Event Filter Tools Options Help

Process Tree

Only show processes still running at end of current trace

Timelines cover displayed events only

| Process            | Description           | Image Path           | Life Time | Company               | Owner            | Comm     |
|--------------------|-----------------------|----------------------|-----------|-----------------------|------------------|----------|
| dumpcap.exe (3476) | Dumpcap               | C:\Program Files\... |           | The Wireshark de...   | Paul-PC\Paul     | "C:\P... |
| dumpcap.exe (2868) | Dumpcap               | C:\Program Files\... |           | The Wireshark de...   | Paul-PC\Paul     | "C:\P... |
| dumpcap.exe (3048) | Dumpcap               | C:\Program Files\... |           | The Wireshark de...   | Paul-PC\Paul     | "C:\P... |
| dumpcap.exe (2872) | Dumpcap               | C:\Program Files\... |           | The Wireshark de...   | Paul-PC\Paul     | "C:\P... |
| dumpcap.exe (3876) | Dumpcap               | C:\Program Files\... |           | The Wireshark de...   | Paul-PC\Paul     | "C:\P... |
| mshta.exe (4228)   | Microsoft (R) HTM...  | C:\Windows\Syst...   |           | Microsoft Corporat... | Paul-PC\Paul     | "C:\W... |
| NOTEPAD.EXE (4240) | Notepad               | C:\Windows\sys...    |           | Microsoft Corporat... | Paul-PC\Paul     | "C:\W... |
| Idle (0)           | Idle                  |                      |           |                       |                  |          |
| System (4)         | System                |                      |           | NT AUTHORITY\...      |                  |          |
| smss.exe (272)     | Windows Session ...   | C:\Windows\Syst...   |           | Microsoft Corporat... | NT AUTHORITY\... |          |
| csrss.exe (352)    | Client Server Runt... | C:\Windows\sys...    |           | Microsoft Corporat... | NT AUTHORITY\... |          |
| wininit.exe (404)  | Windows Start-Up...   | C:\Windows\sys...    |           | Microsoft Corporat... | NT AUTHORITY\... |          |
| services.exe (504) | Services and Cont...  | C:\Windows\sys...    |           | Microsoft Corporat... | NT AUTHORITY\... |          |
| svchost.exe (752)  | Host Process for ...  | C:\Windows\sys...    |           | Microsoft Corporat... | NT AUTHORITY\... |          |
| svchost.exe (804)  | Host Process for ...  | C:\Windows\sys...    |           | Microsoft Corporat... | NT AUTHORITY\... |          |
| AUDIODG.EXE (3988) | Windows Audio D...    | C:\Windows\sys...    |           | Microsoft Corporat... | NT AUTHORITY\... |          |
| svchost.exe (924)  | Host Process for ...  | C:\Windows\sys...    |           | Microsoft Corporat... | NT AUTHORITY\... |          |

Description: Elaborate Bytes AG

Company: Elaborate Bytes AG

Path: C:\Users\Paul\Desktop\samples\_pcap\8b6bc16fd137c09a08b02bbe1bb7d670.exe

Command: "C:\Users\Paul\Desktop\samples\_pcap\8b6bc16fd137c09a08b02bbe1bb7d670.exe"

User: Paul-PC\Paul

PID: 4088 Started: 4/12/2024 12:35:28 AM

Go To Event Include Process Include Subtree Close

Showing 110,977

Windows 7 (TEST) (Cu NET) [Running] - Oracle VM VirtualBox

File Machine View Input Devices Help

Process Monitor - Sysinternals: www.sysinternals.com

File Edit Event Filter Tools Options Help

| Time ... | Process Name | PID  | Operation  | Path                                      | Result         | Detail               | Sequence |
|----------|--------------|------|------------|-------------------------------------------|----------------|----------------------|----------|
| 12:36... | mshta.exe    | 4228 | CreateFile | C:\Windows\Prefetch\MSHTA.EXE-A970B441.pf | NAME NOT FOUND | Desired Access: G... | 26041    |
| 12:36... | mshta.exe    | 4228 | CreateFile | C:\Users\Paul\Desktop                     | SUCCESS        | Desired Access: E... | 26042    |
| 12:36... | mshta.exe    | 4228 | CreateFile | C:\Windows\System32\sechost.dll           | SUCCESS        | Desired Access: R... | 26043    |
| 12:36... | mshta.exe    | 4228 | CreateFile | C:\Windows\System32\sechost.dll           | SUCCESS        | Desired Access: R... | 26044    |
| 12:36... | mshta.exe    | 4228 | CreateFile | C:\Windows\System32\version.dll           | SUCCESS        | Desired Access: R... | 26045    |
| 12:36... | mshta.exe    | 4228 | CreateFile | C:\Windows\System32\version.dll           | SUCCESS        | Desired Access: R... | 26046    |
| 12:36... | mshta.exe    | 4228 | CreateFile | C:\Windows\System32\imm32.dll             | SUCCESS        | Desired Access: R... | 26047    |
| 12:36... | mshta.exe    | 4228 | CreateFile | C:\Windows\System32\imm32.dll             | SUCCESS        | Desired Access: R... | 26048    |
| 12:36... | mshta.exe    | 4228 | CreateFile | C:\Windows\System32\imm32.dll             | SUCCESS        | Desired Access: R... | 26049    |
| 12:36... | mshta.exe    | 4228 | CreateFile | C:\Windows\System32\imm32.dll             | SUCCESS        | Desired Access: R... | 26050    |
| 12:36... | mshta.exe    | 4228 | CreateFile | C:\Windows\System32\imm32.dll             | SUCCESS        | Desired Access: R... | 26051    |
| 12:36... | mshta.exe    | 4228 | CreateFile | C:\Windows\System32\imm32.dll             | SUCCESS        | Desired Access: R... | 26052    |
| 12:36... | mshta.exe    | 4228 | CreateFile | C:\Windows\System32\imm32.dll             | SUCCESS        | Desired Access: R... | 26053    |

Process Tree

Only show processes still running at end of current trace

Timelines cover displayed events only

| Process              | Description          | Image Path           | Life Time | Company               | Owner        | Comm     |
|----------------------|----------------------|----------------------|-----------|-----------------------|--------------|----------|
| Explorer.EXE (340)   | Windows Explorer     | C:\Windows\Expl...   |           | Microsoft Corporat... | Paul-PC\Paul | C:\Wi... |
| VBTray.exe (1644)    | VirtualBox Guest ... | C:\Windows\Syst...   |           | Oracle and/or its ... | Paul-PC\Paul | "C:\W... |
| Process Monitor      | Process Monitor      | C:\Users\Paul\De...  |           | Sysinternals: ww...   | Paul-PC\Paul | "C:\U... |
| Wireshark.exe (3128) | Wireshark            | C:\Program Files\... |           | The Wireshark de...   | Paul-PC\Paul | "C:\P... |
| dumpcap.exe (3316)   | Dumpcap              | C:\Program Files\... |           | The Wireshark de...   | Paul-PC\Paul | "C:\P... |
| dumpcap.exe (3396)   | Dumpcap              | C:\Program Files\... |           | The Wireshark de...   | Paul-PC\Paul | "C:\P... |
| dumpcap.exe (2812)   | Dumpcap              | C:\Program Files\... |           | The Wireshark de...   | Paul-PC\Paul | "C:\P... |
| dumpcap.exe (2744)   | Dumpcap              | C:\Program Files\... |           | The Wireshark de...   | Paul-PC\Paul | "C:\P... |
| dumpcap.exe (3368)   | Dumpcap              | C:\Program Files\... |           | The Wireshark de...   | Paul-PC\Paul | "C:\P... |
| dumpcap.exe (3476)   | Dumpcap              | C:\Program Files\... |           | The Wireshark de...   | Paul-PC\Paul | "C:\P... |
| dumpcap.exe (2868)   | Dumpcap              | C:\Program Files\... |           | The Wireshark de...   | Paul-PC\Paul | "C:\P... |
| dumpcap.exe (3048)   | Dumpcap              | C:\Program Files\... |           | The Wireshark de...   | Paul-PC\Paul | "C:\P... |
| dumpcap.exe (2872)   | Dumpcap              | C:\Program Files\... |           | The Wireshark de...   | Paul-PC\Paul | "C:\P... |
| dumpcap.exe (3876)   | Dumpcap              | C:\Program Files\... |           | The Wireshark de...   | Paul-PC\Paul | "C:\P... |
| mshta.exe (4228)     | Microsoft (R) HTM... | C:\Users\Paul\De...  |           | Microsoft Corporat... | Paul-PC\Paul | "C:\U... |
| NOTEPAD.EXE (4240)   | Notepad              | C:\Windows\sys...    |           | Microsoft Corporat... | Paul-PC\Paul | "C:\W... |
| Idle (0)             | Idle                 |                      |           |                       |              |          |

Go To Event

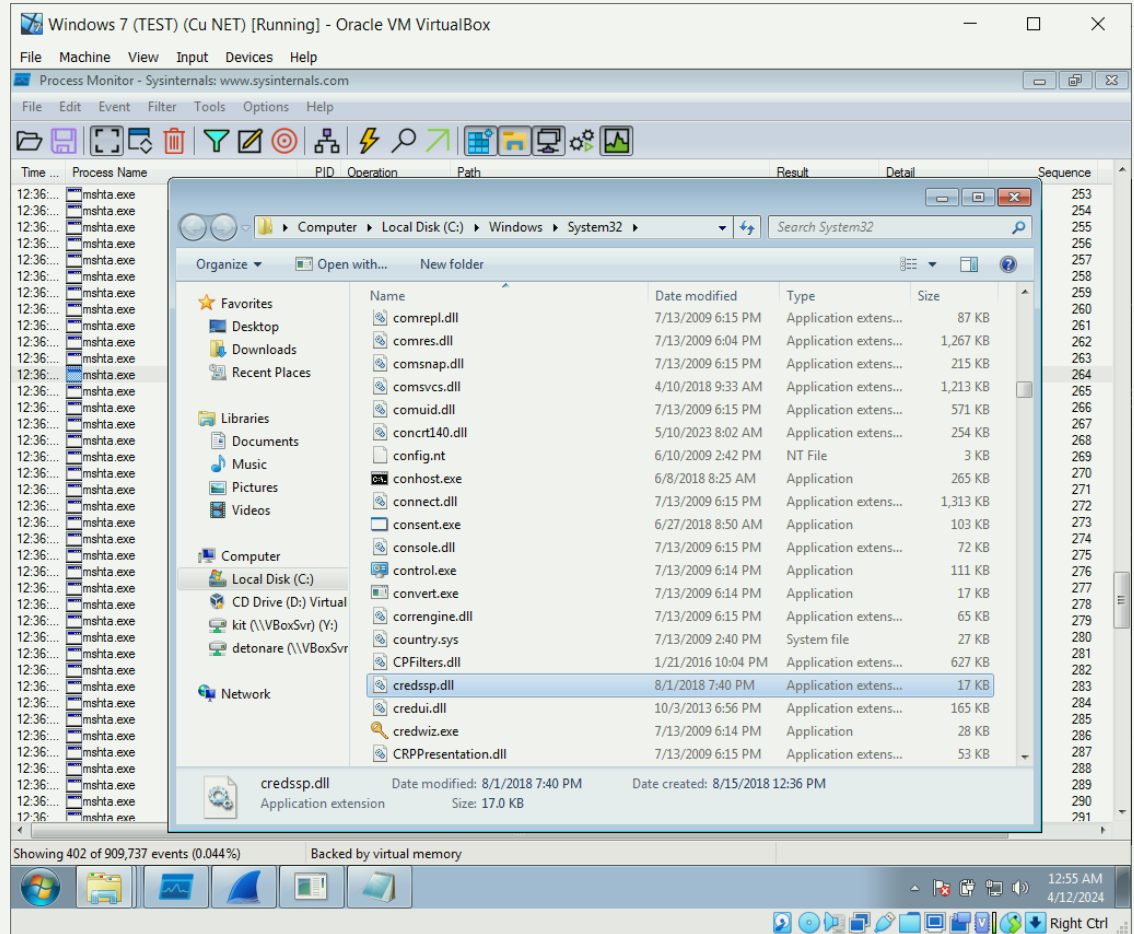
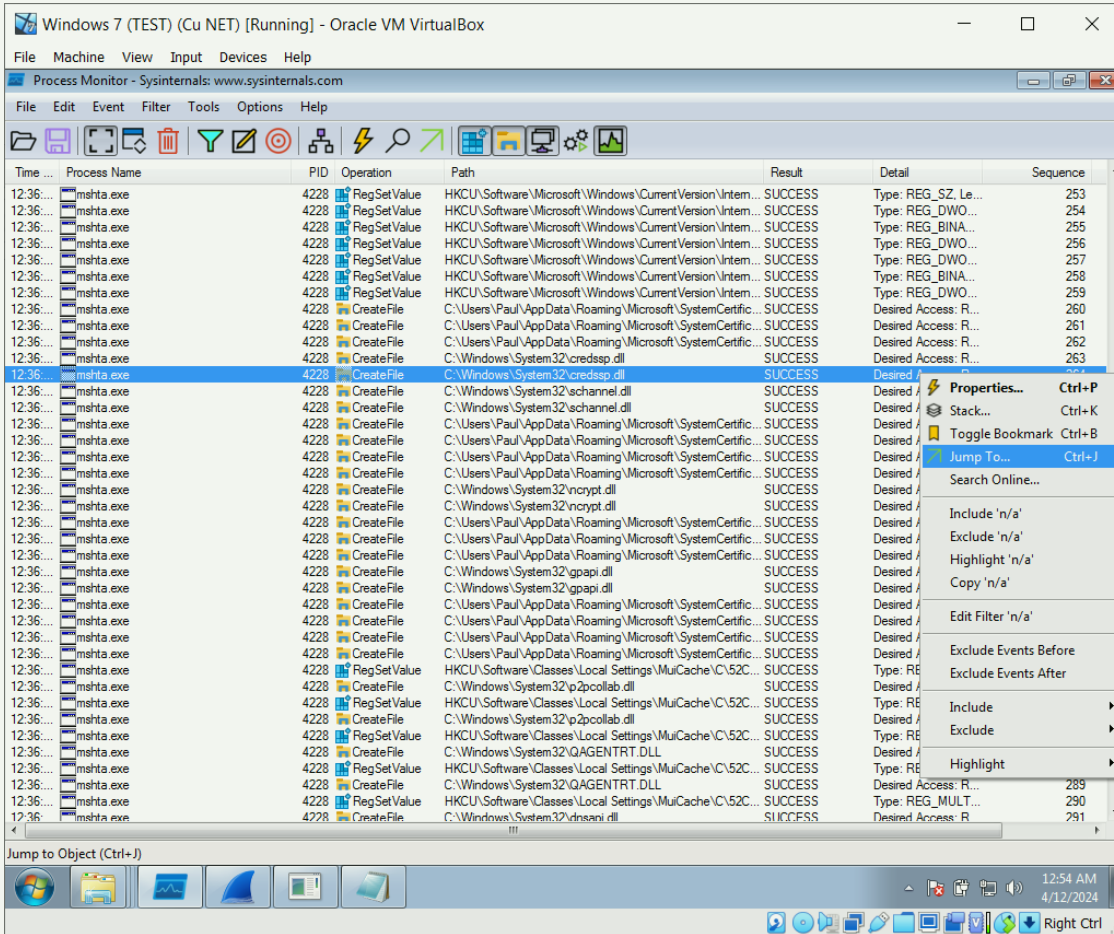
Add process to Include filter

Add process and children to Include filter

12:52 AM 4/12/2024

# PROCESS MONITOR & WIRESHARK

## ACCESAREA FISIERELOR SCRISE DE MALWARE

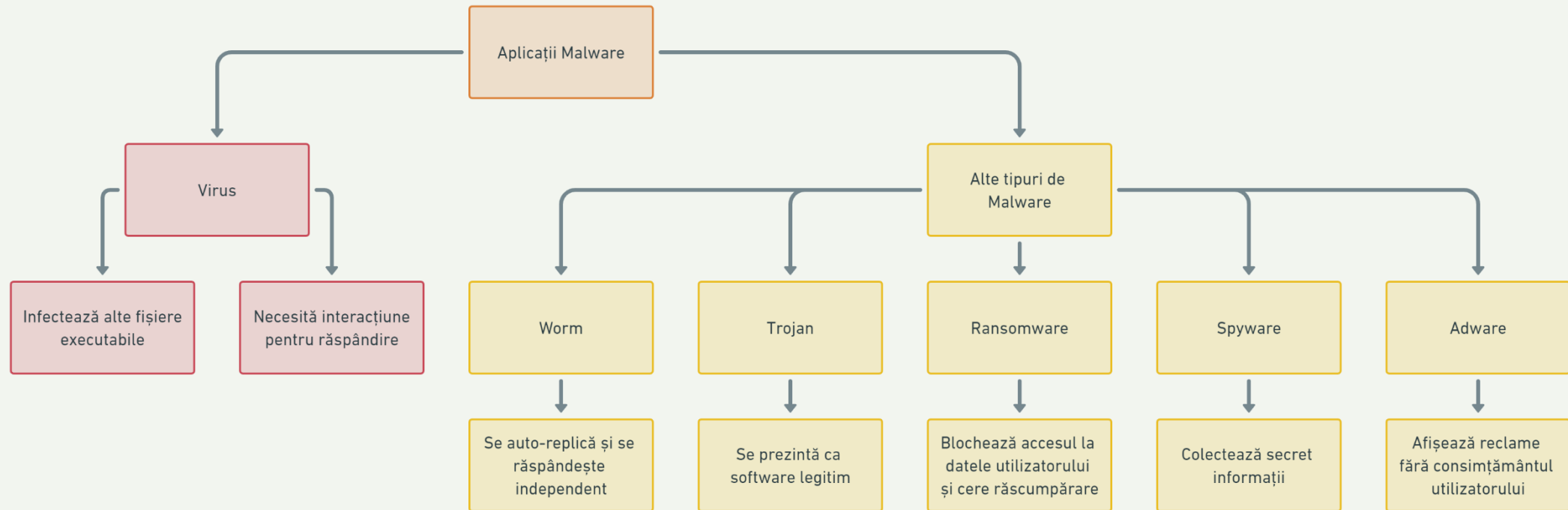


C.11.4

# DETONAREA ȘI ANALIZA VIRUȘILOR PRIN UTILIZAREA HONEYPOT



# Două puncte principale:



## Monitorizarea Virusului în Honeypot

```
graph LR; Root[Monitorizarea Virusului în Honeypot] --- A[Identificarea Fișierelor Infectate]; Root --- B[Analiza Comportamentului de Replicare]; Root --- C[Monitorizarea Modificărilor de Sistem]; Root --- D[Evaluarea Comunicărilor de Rețea]; Root --- E[Extragerea Semnăturilor]; Root --- F[Analiza Impactului]; A --- A1[Verifică tipurile de fișiere afectate]; A --- A2[Identifică modificările de conținut ale fișierelor]; B --- B1[Determină metodele de replicare]; B --- B2[Monitorizează răspândirea în alte directoare]; C --- C1[Verifică modificările în registri]; C --- C2[Detectează schimbări în setările de sistem]; D --- D1[Monitorizează traficul de ieșire]; D --- D2[Identifică conexiuni la servere de comandă și control]; E --- E1[Obține hash-uri ale fișierelor infectate]; E --- E2[Identifică semnături comportamentale]; F --- F1[Evaluează potențialele pierderi de date]; F --- F2[Estimează afectarea performanței sistemului];
```

### Identificarea Fișierelor Infectate

- Verifică tipurile de fișiere afectate
- Identifică modificările de conținut ale fișierelor

### Analiza Comportamentului de Replicare

- Determină metodele de replicare
- Monitorizează răspândirea în alte directoare

### Monitorizarea Modificărilor de Sistem

- Verifică modificările în registri
- Detectează schimbări în setările de sistem

### Evaluarea Comunicărilor de Rețea

- Monitorizează traficul de ieșire
- Identifică conexiuni la servere de comandă și control

### Extragerea Semnăturilor

- Obține hash-uri ale fișierelor infectate
- Identifică semnături comportamentale

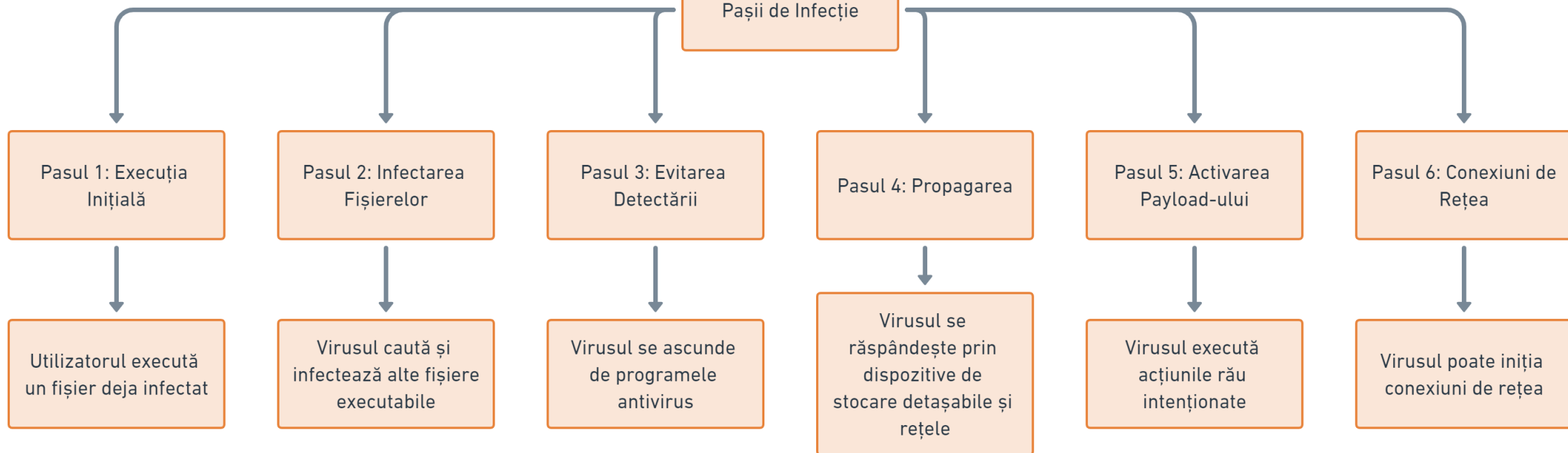
### Analiza Impactului

- Evaluează potențialele pierderi de date
- Estimează afectarea performanței sistemului

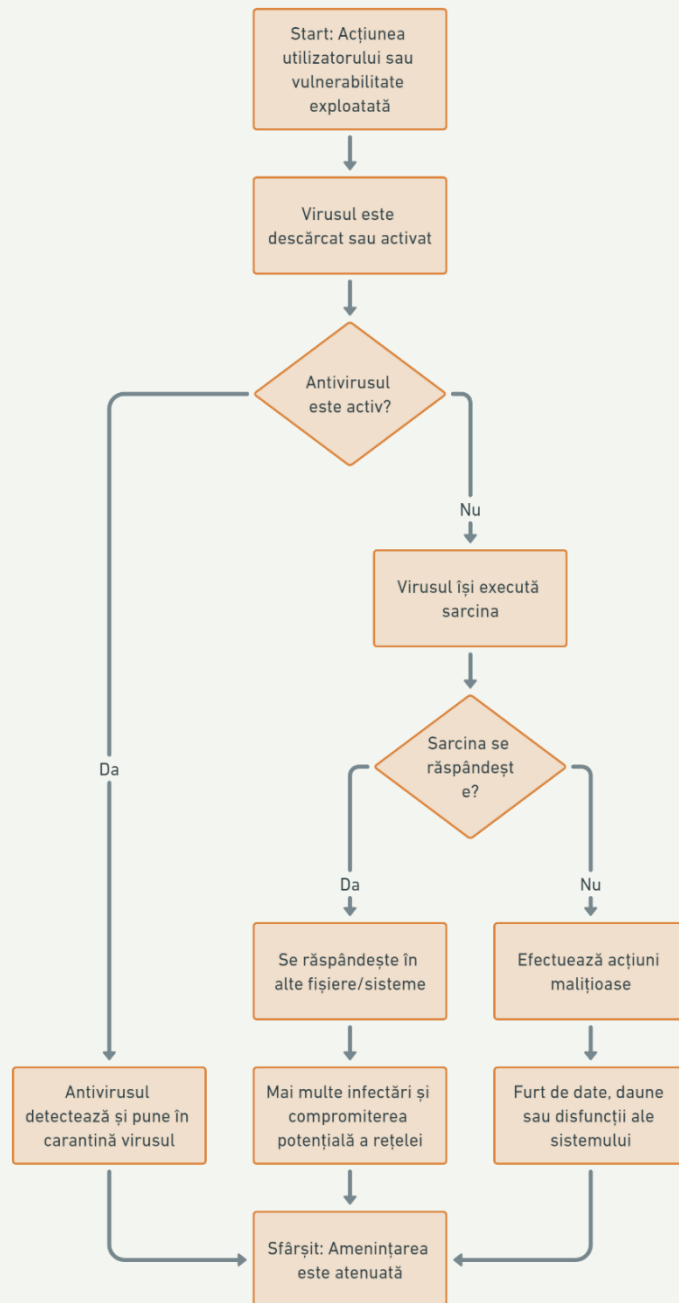
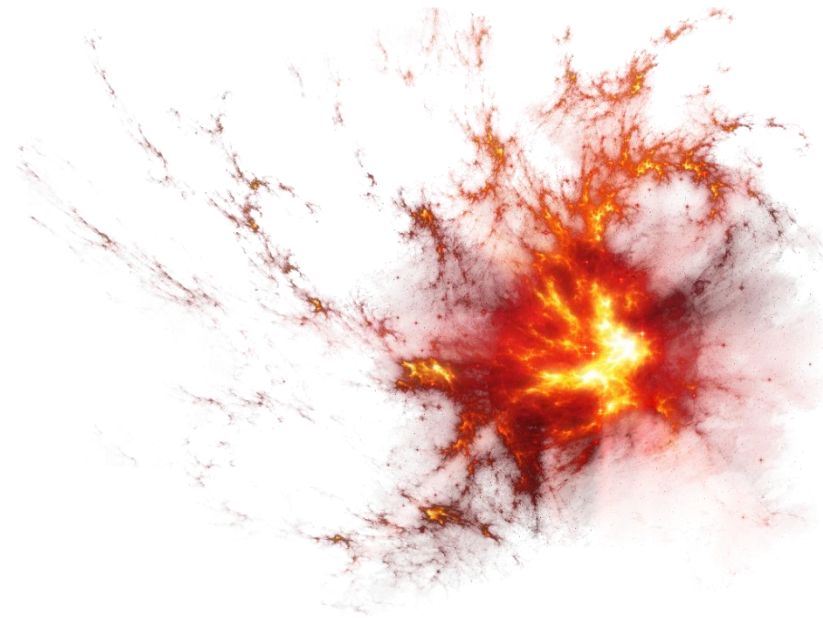




## Pașii de Infecție



# La ce ne așteptăm post-infecție?

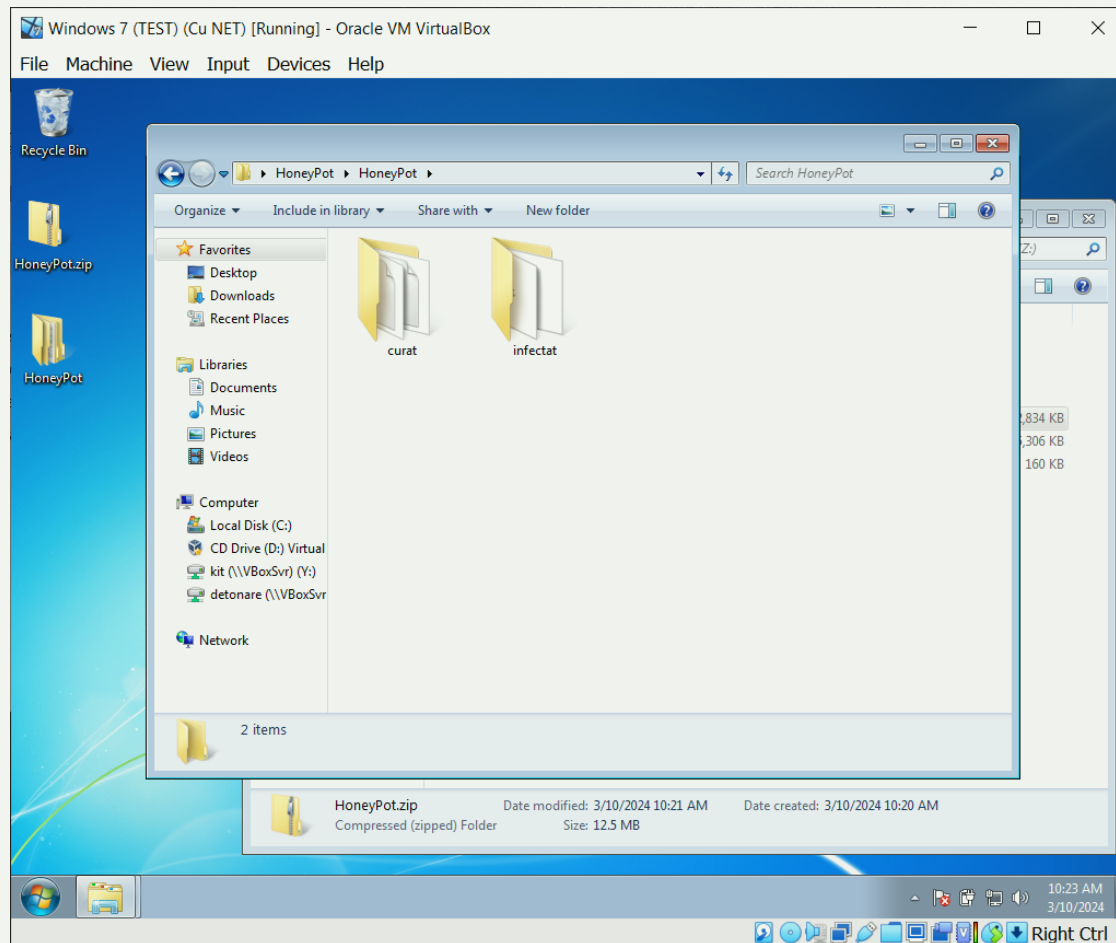
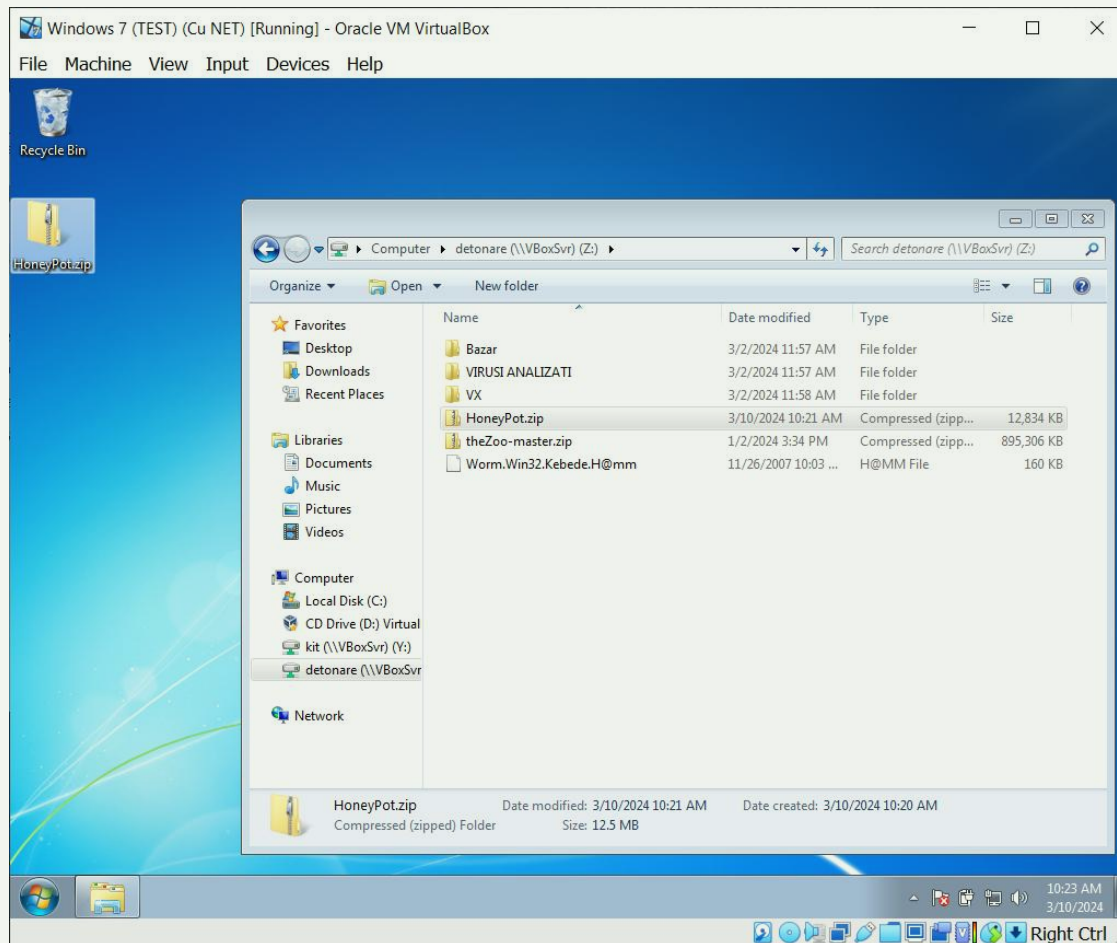


ȚINTE TENTANTE

# HONEYPOT (BORCANUL CU MIERE)

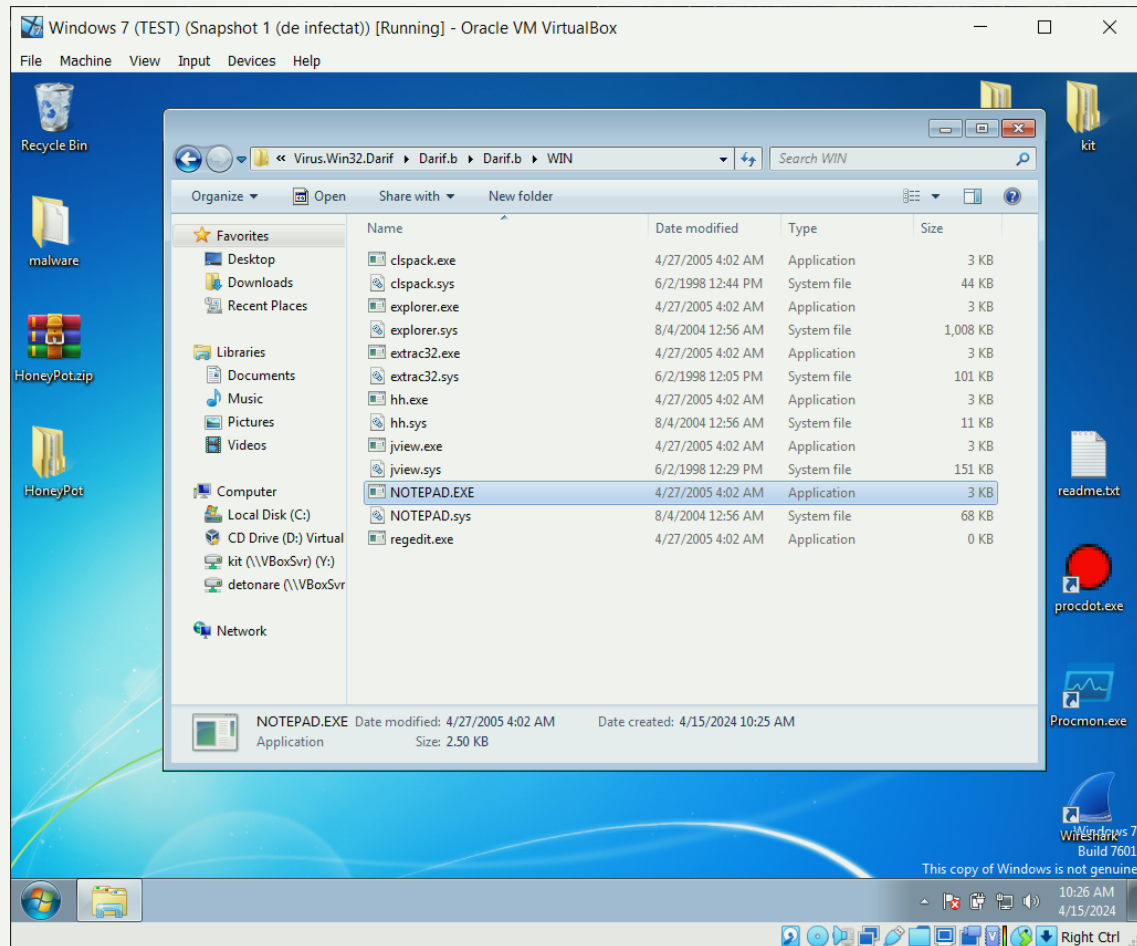
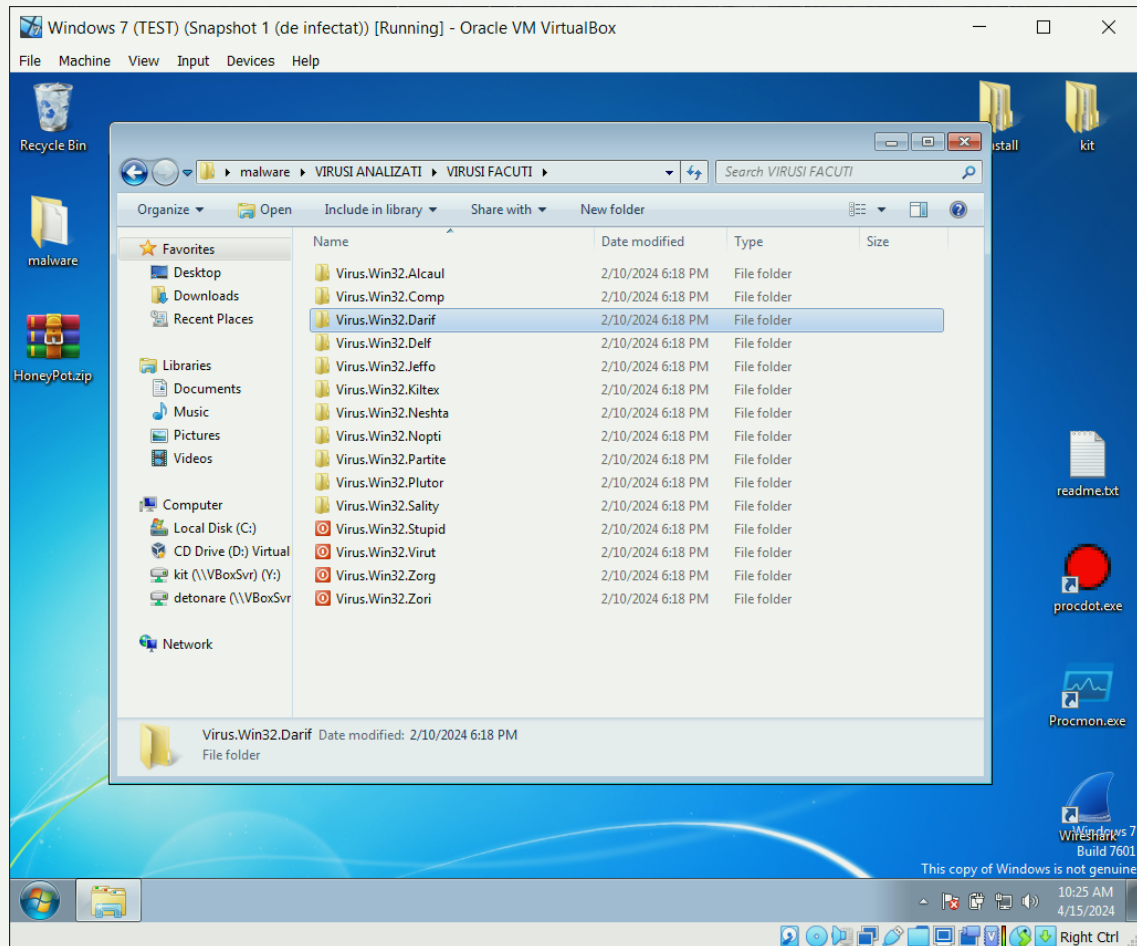


# INSTALARE HONEYPOT DE PE MOODLE CE ESTE UN HONEYPOT PENTRU INFECTIE?



# VERIFICAREA TIPULUI DE INFECTIE

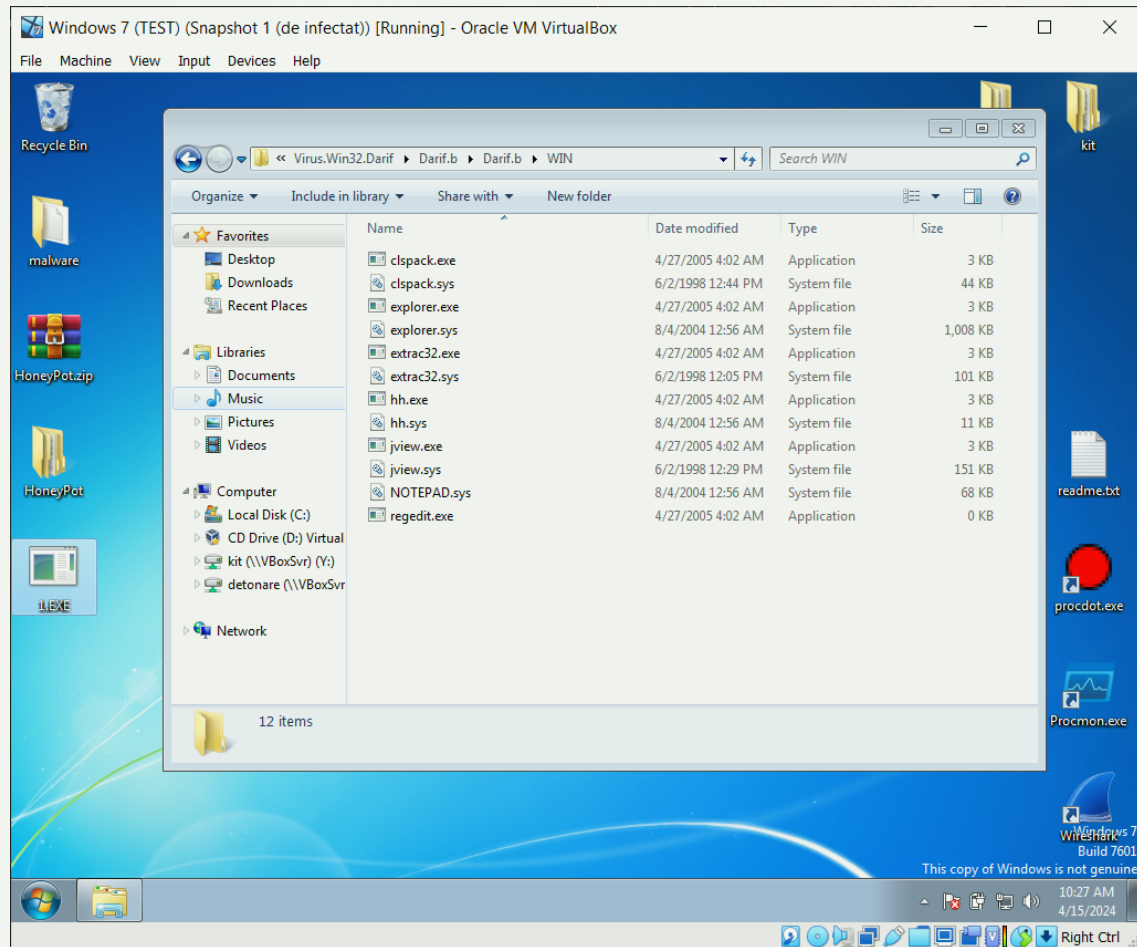
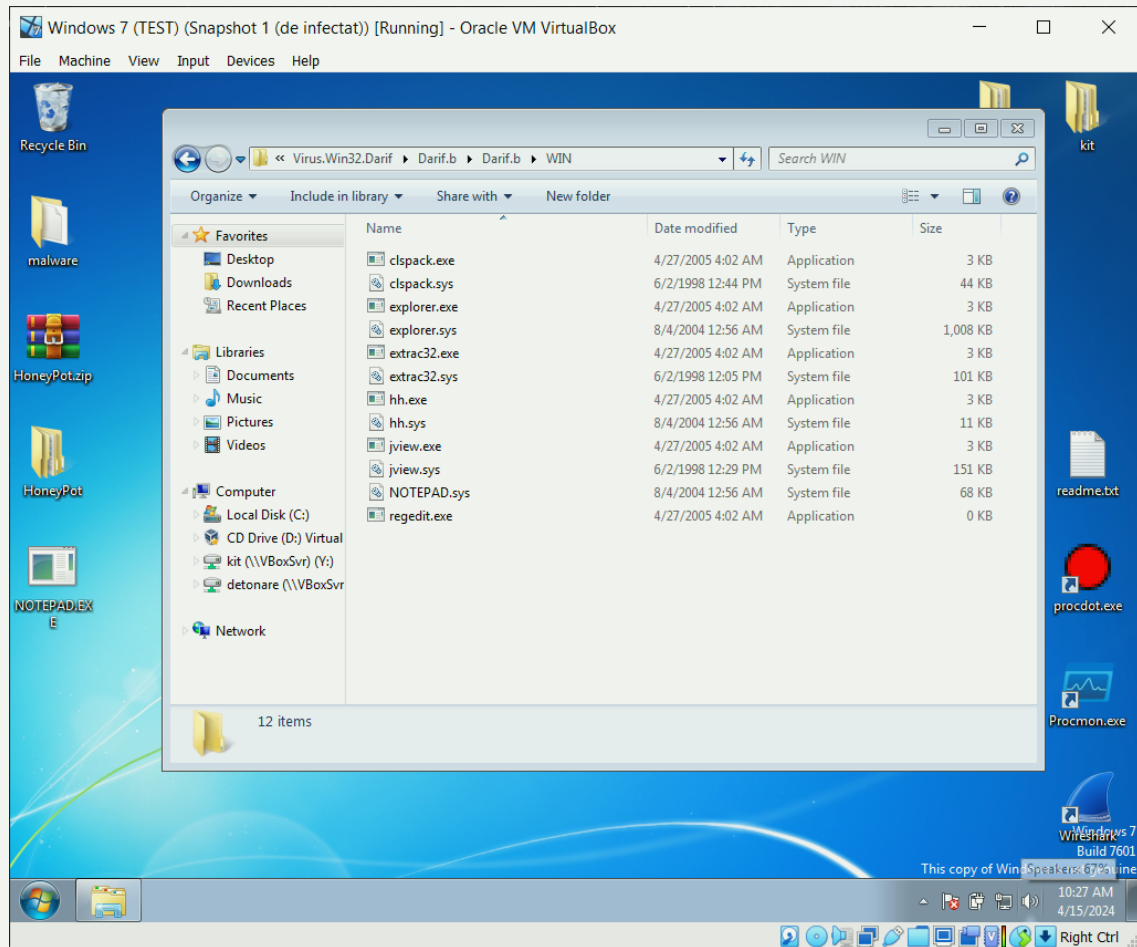
## ALEGEREA UNEI MOSTRE DE VIRUS (FISIER INFECTAT)



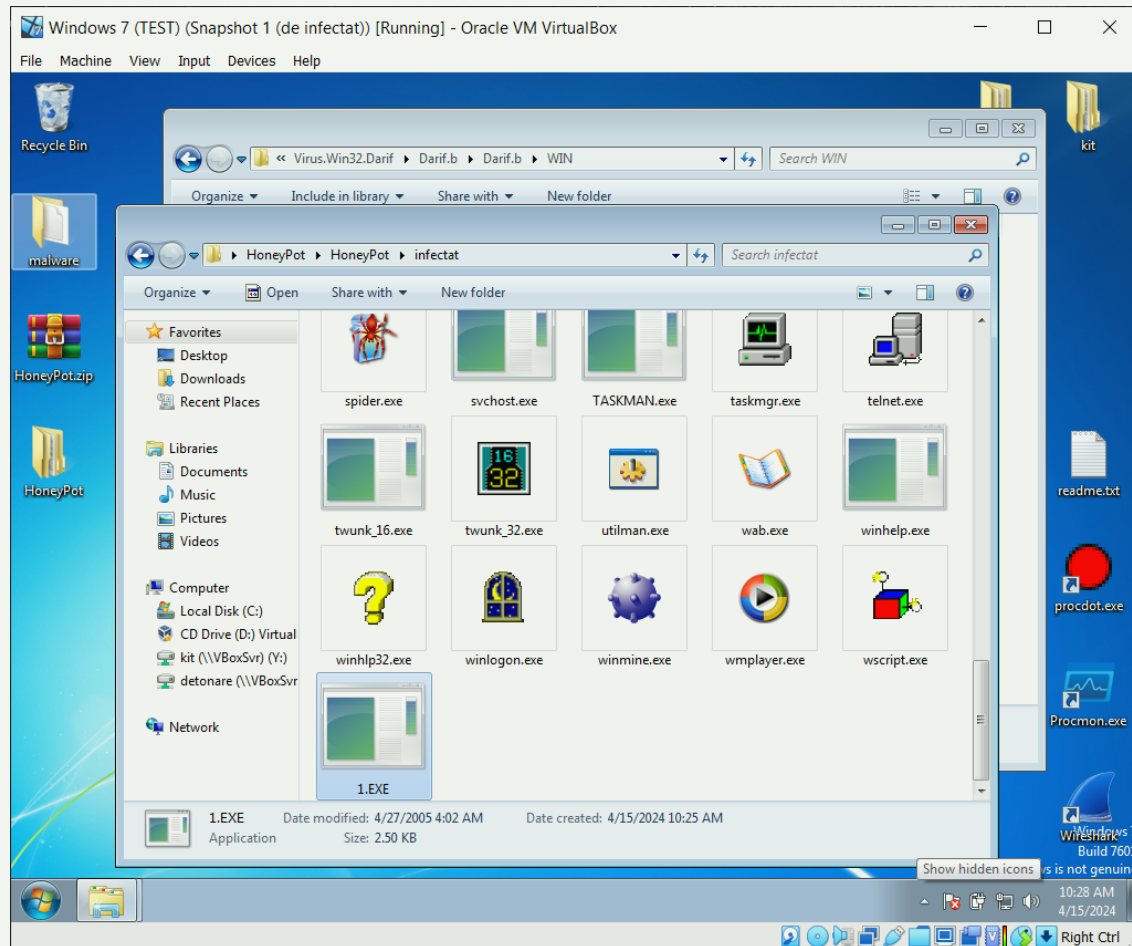
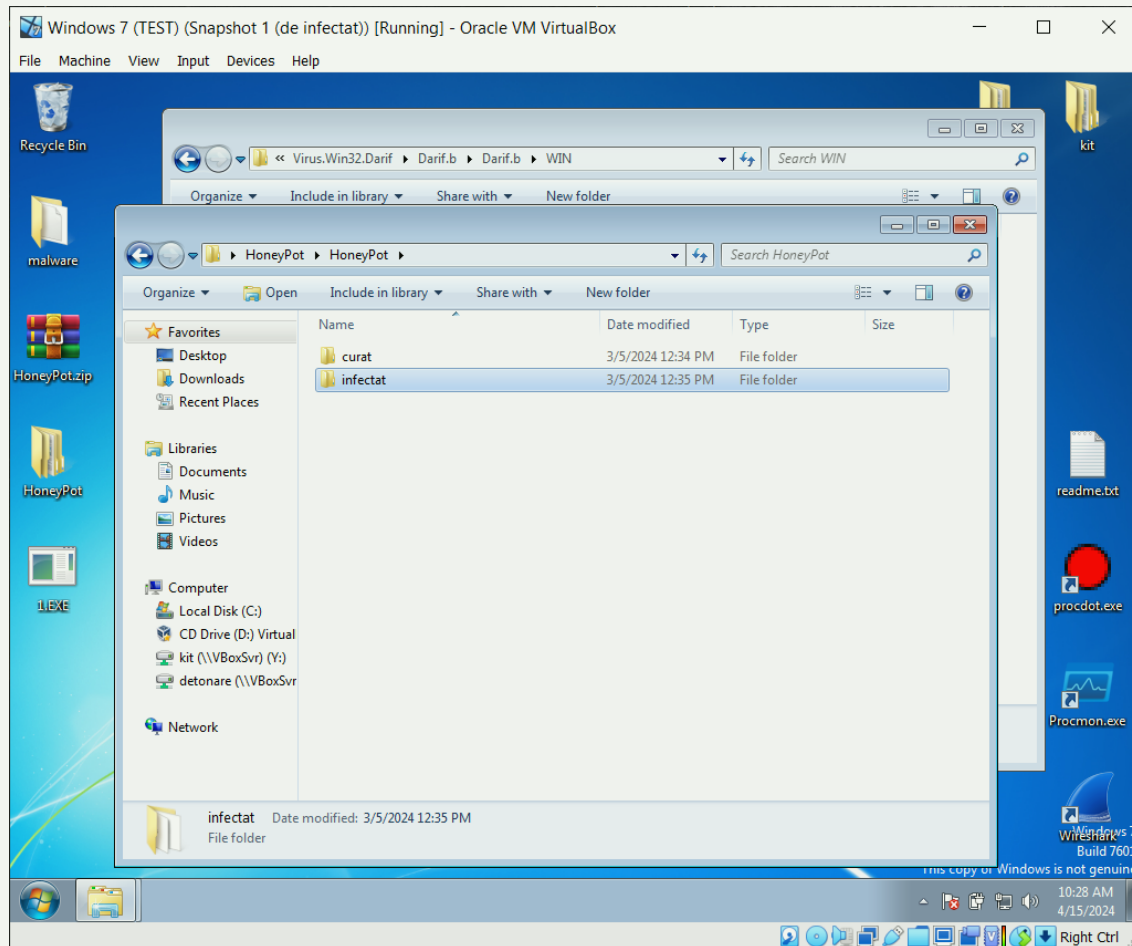


# MODIFICAREA NUMELUI FISIEREI INFECTAT

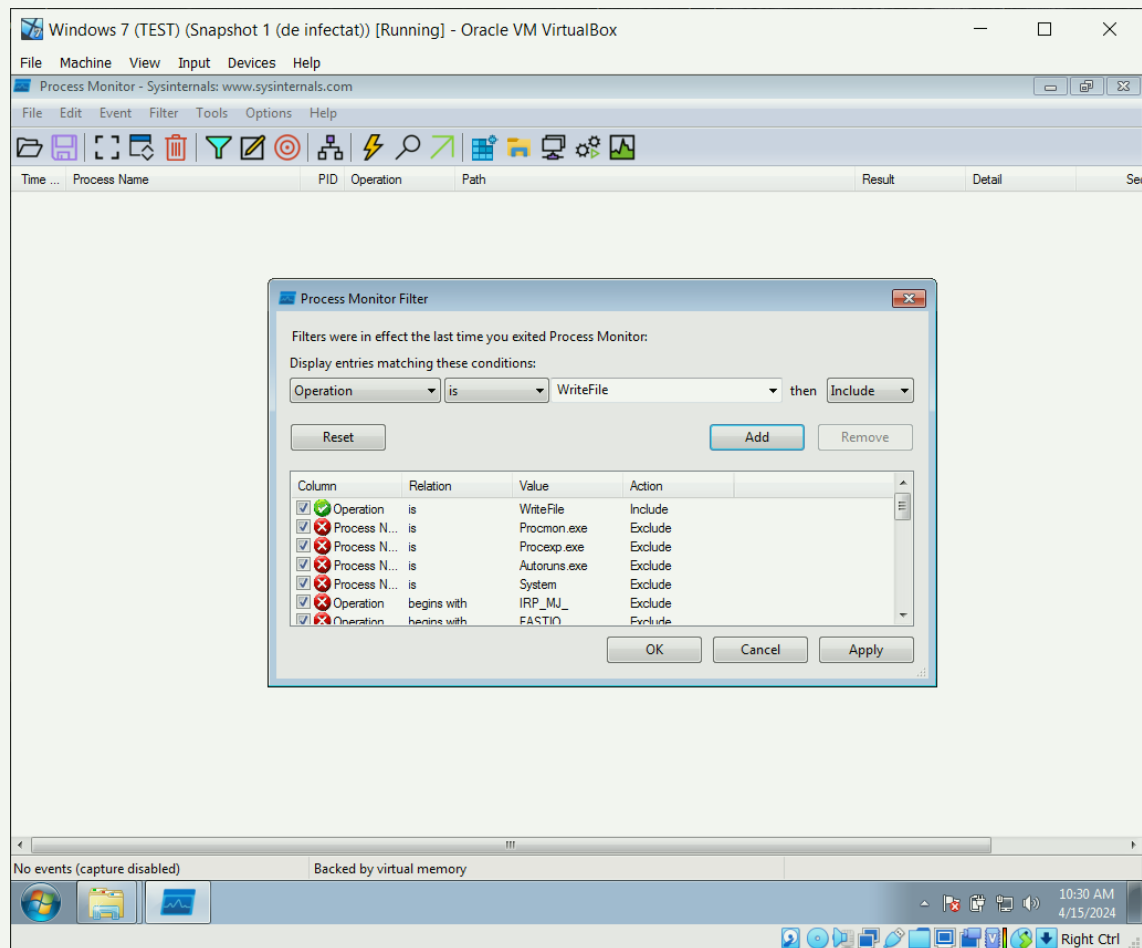
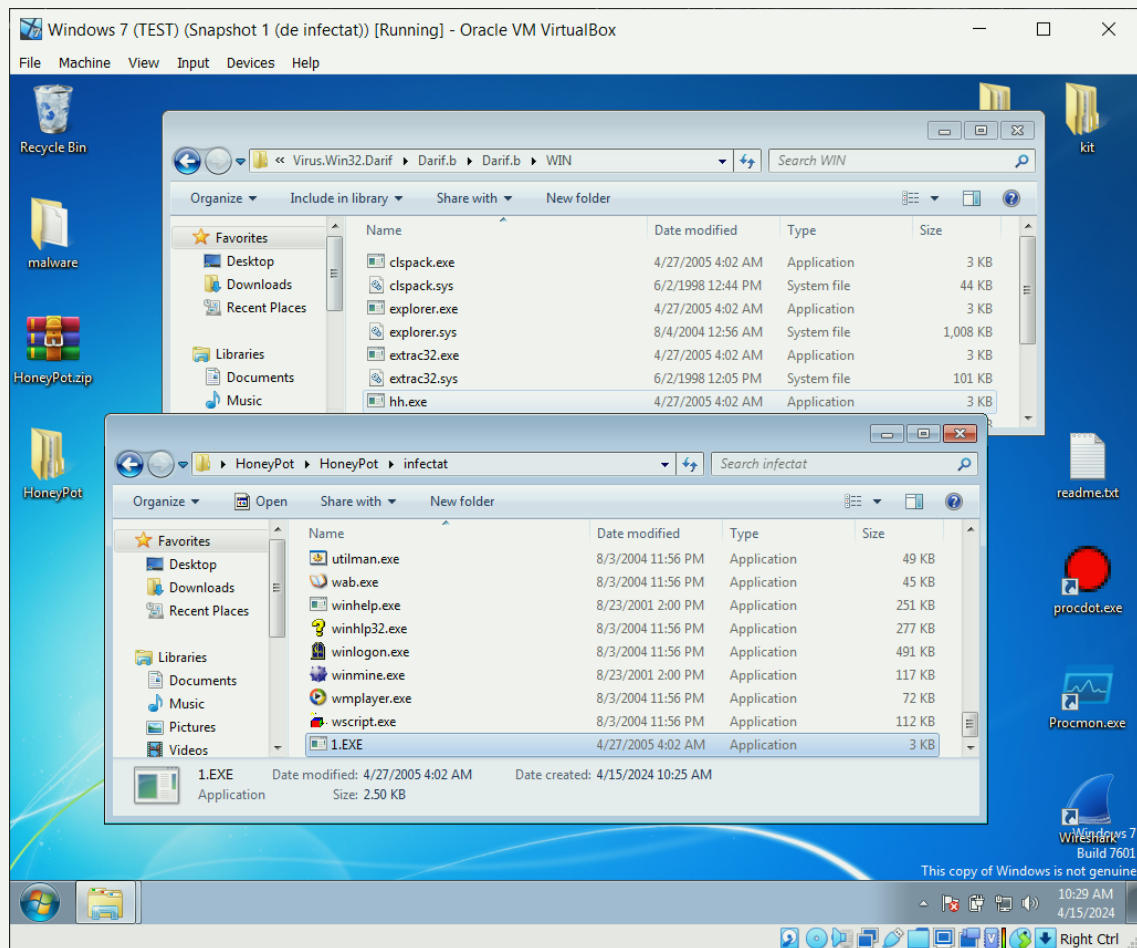
## COPIEREA TEMPORARA PE DESKTOP



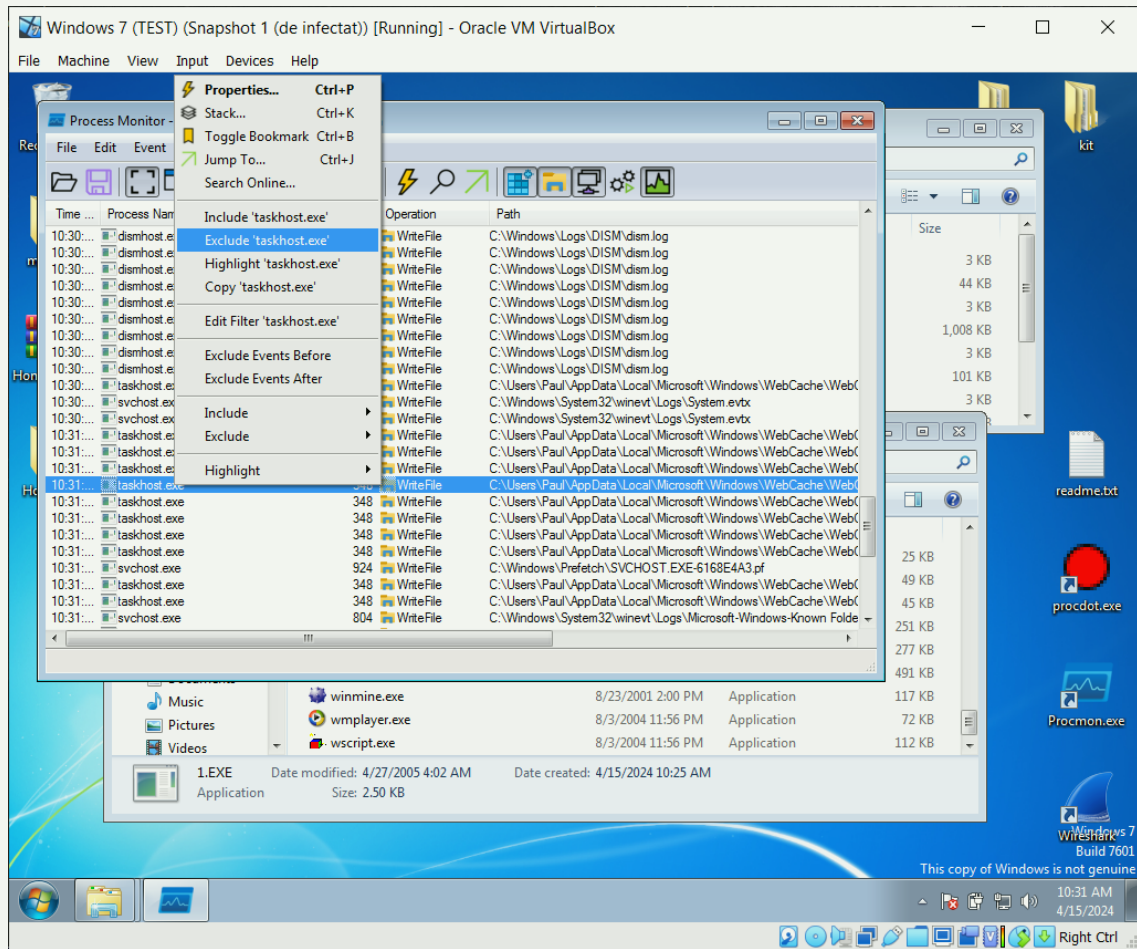
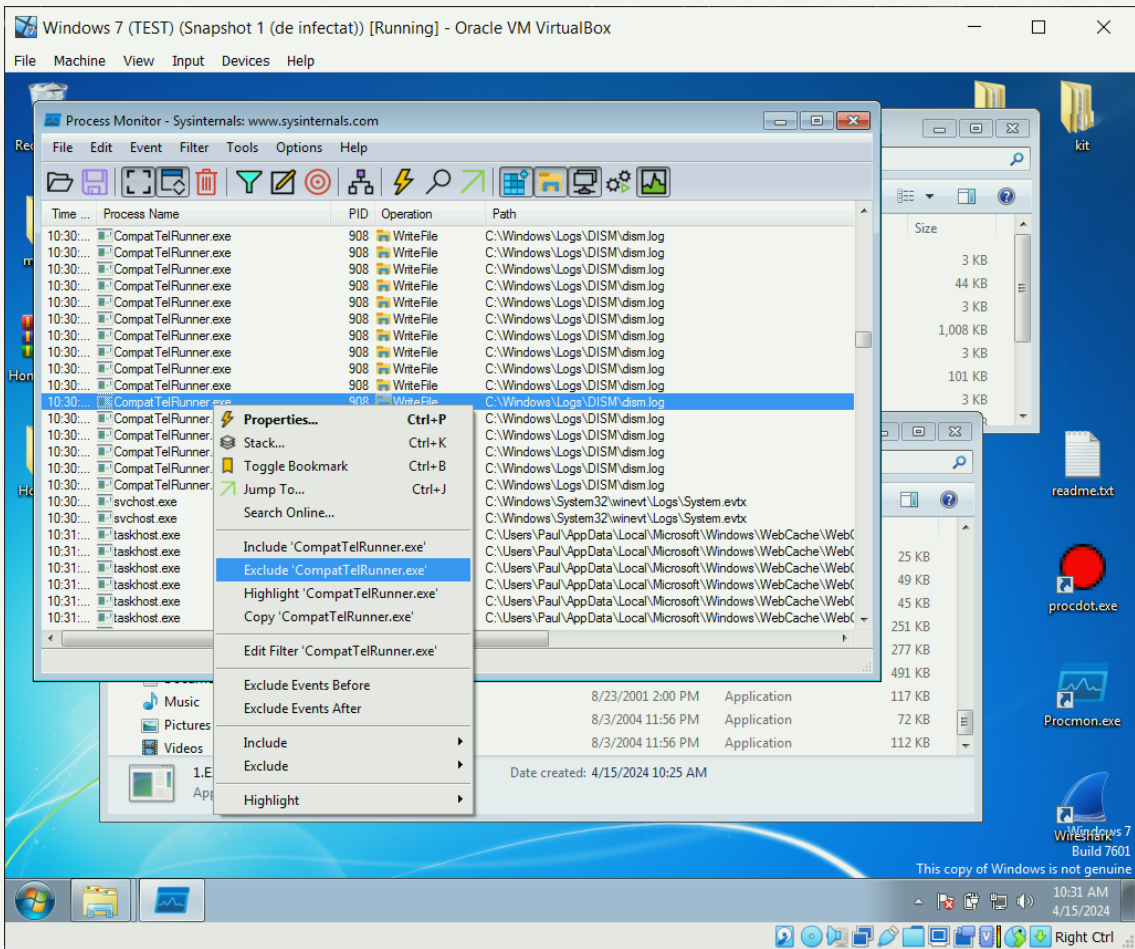
# MUTAREA VIRUSULUI DE PE DESKTOP IN DIRECTORUL DESKTOP AL HONEYPOT



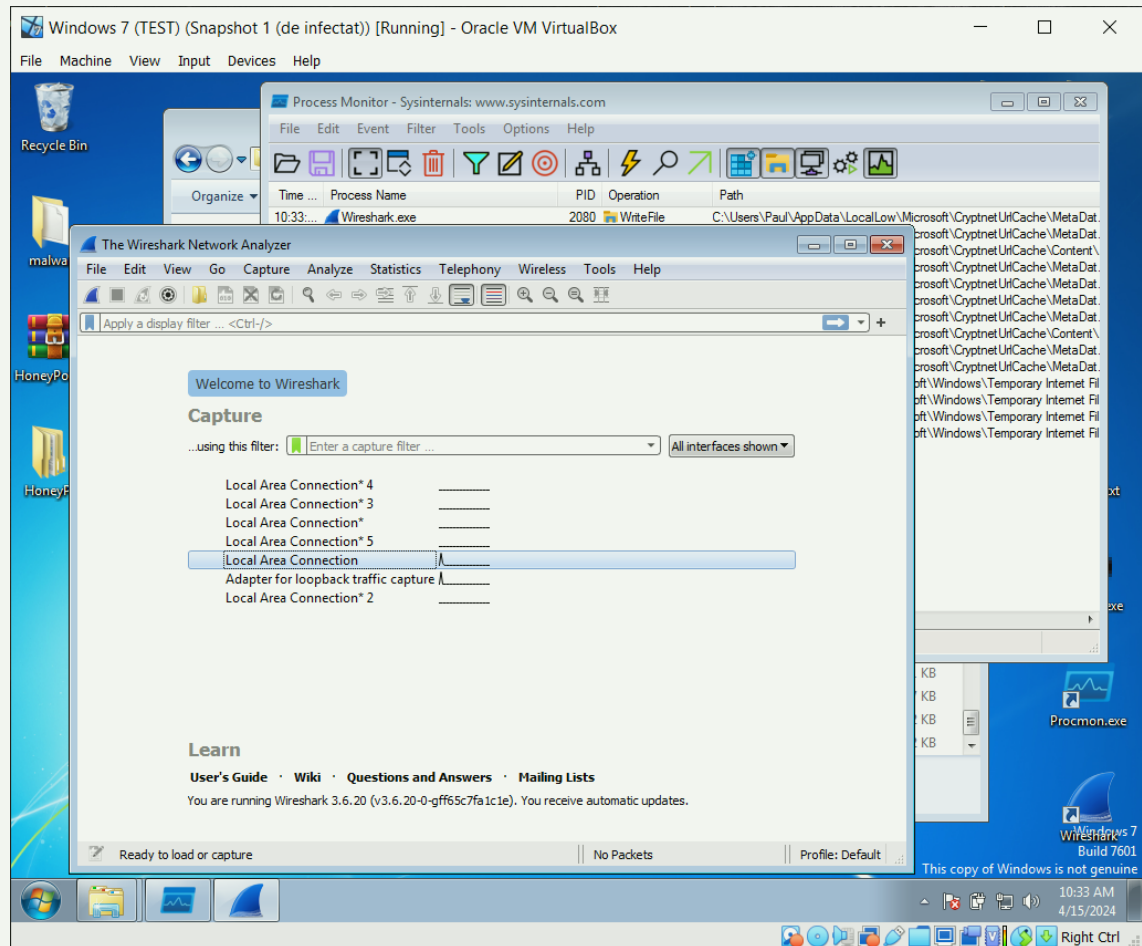
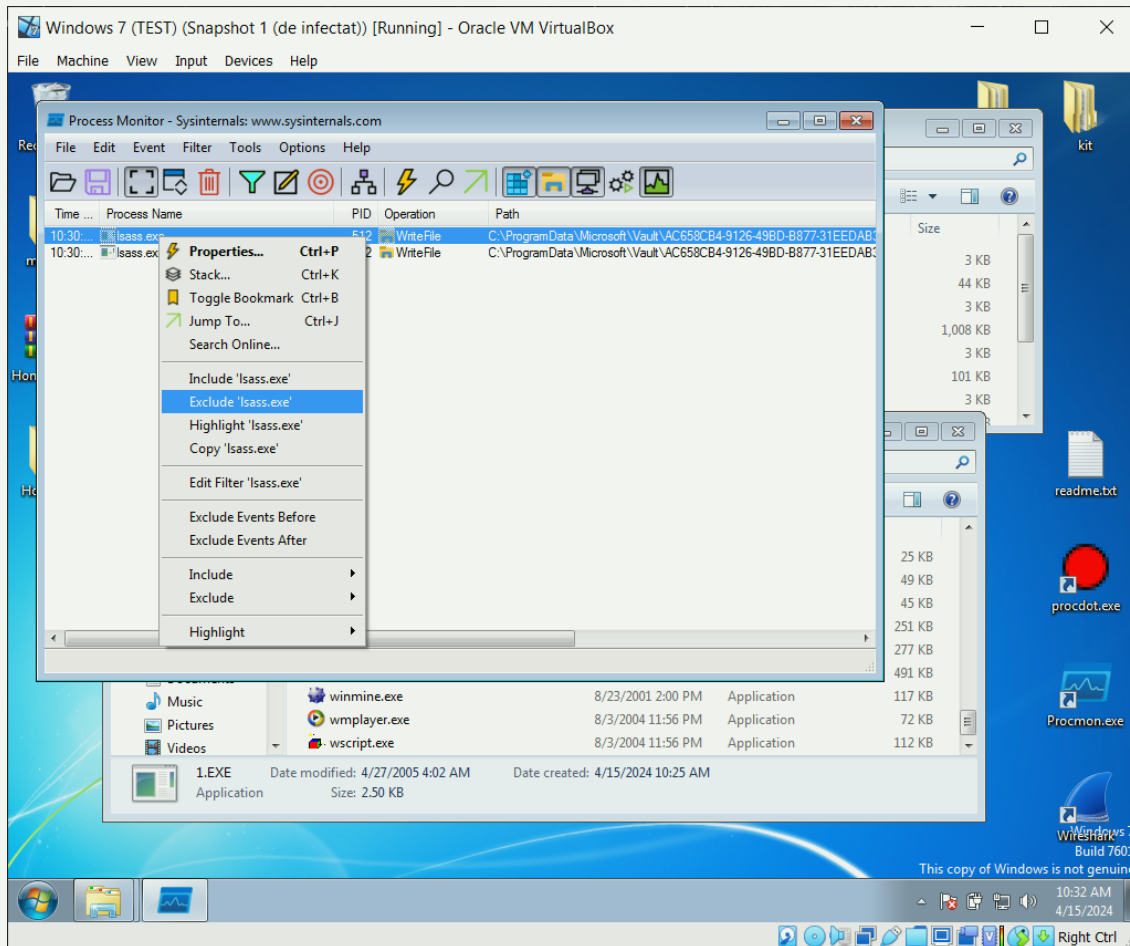
# PREGATIREA FILTRELOR PENTRU PROCESS MONITOR



# ELIMINAREA MANUALA A ZGOMOTUII DIN PROCESS MONITOR

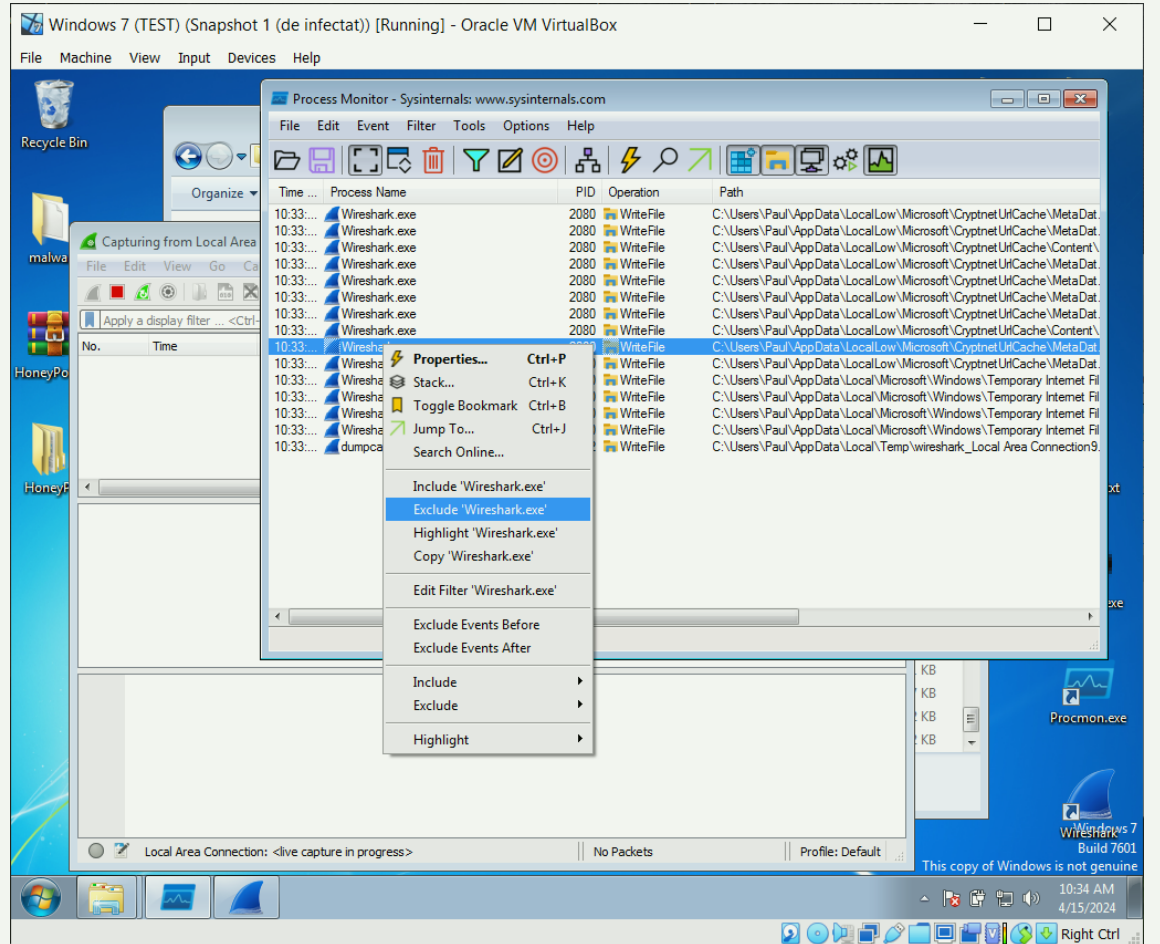
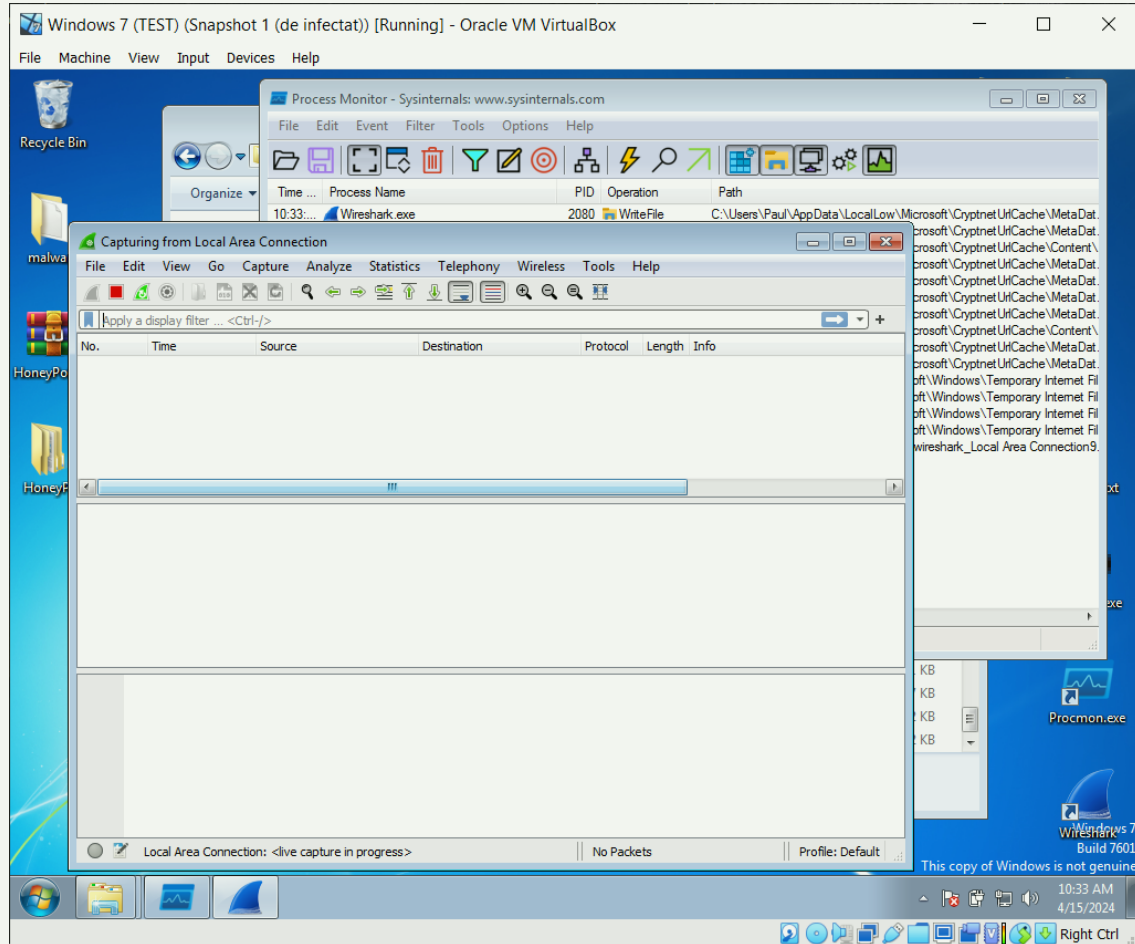


# PORNIREA WIRESHARK IN PARALEL CU PROCESS MONITOR

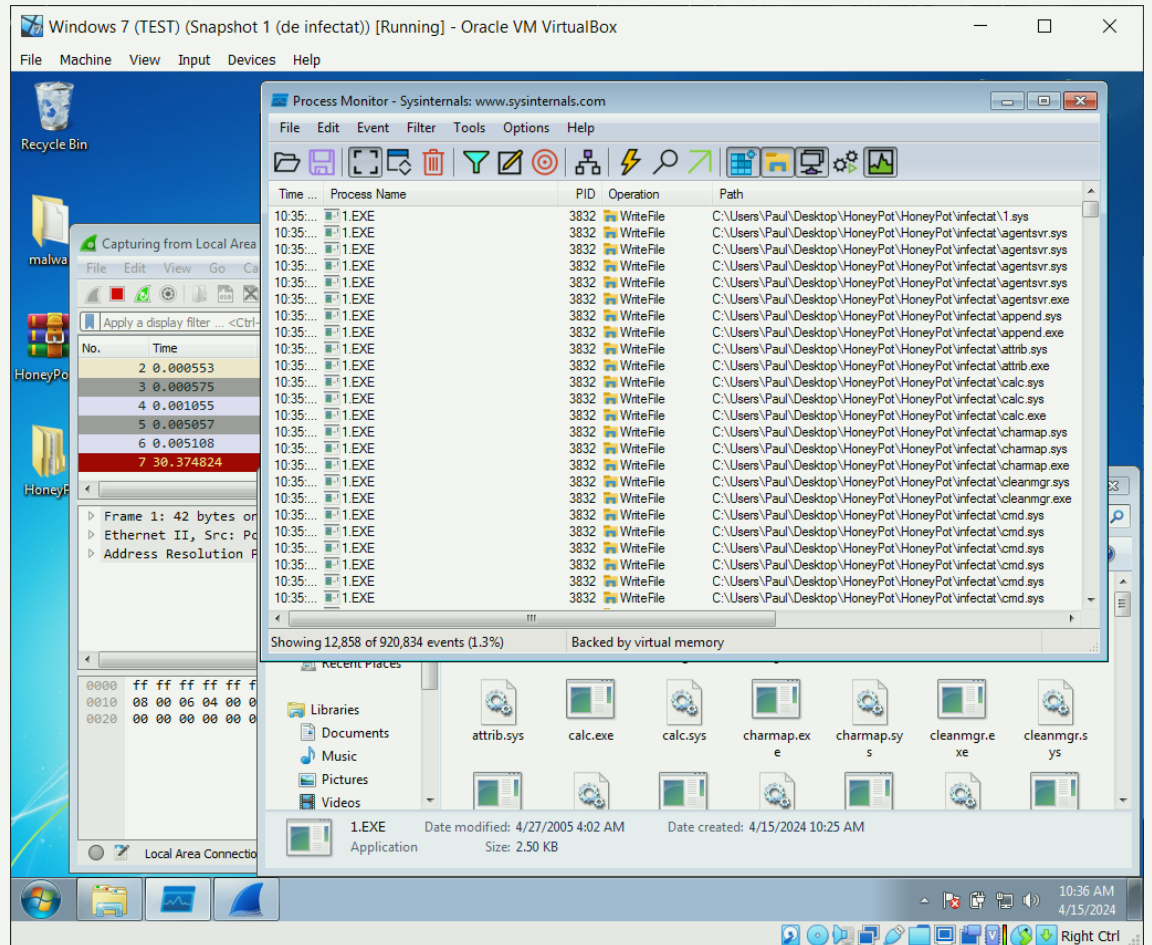
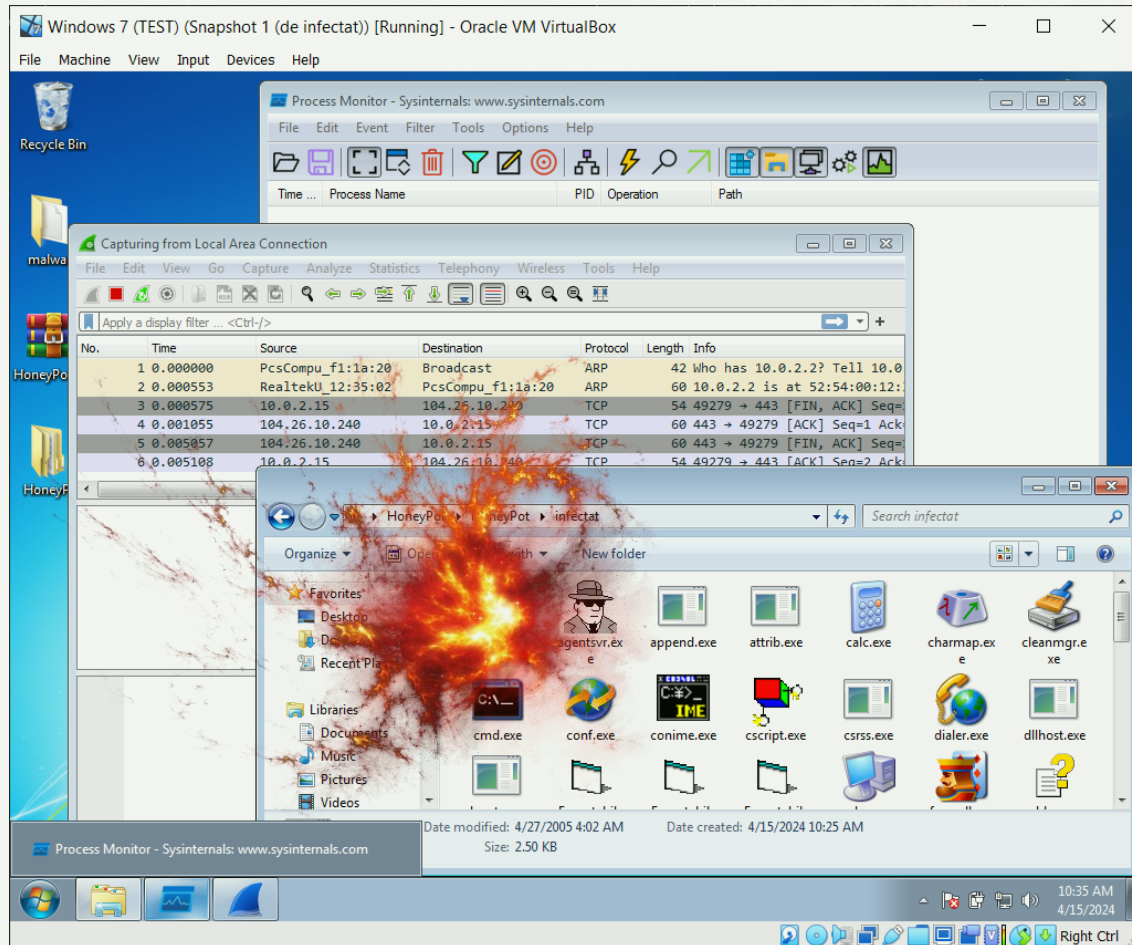




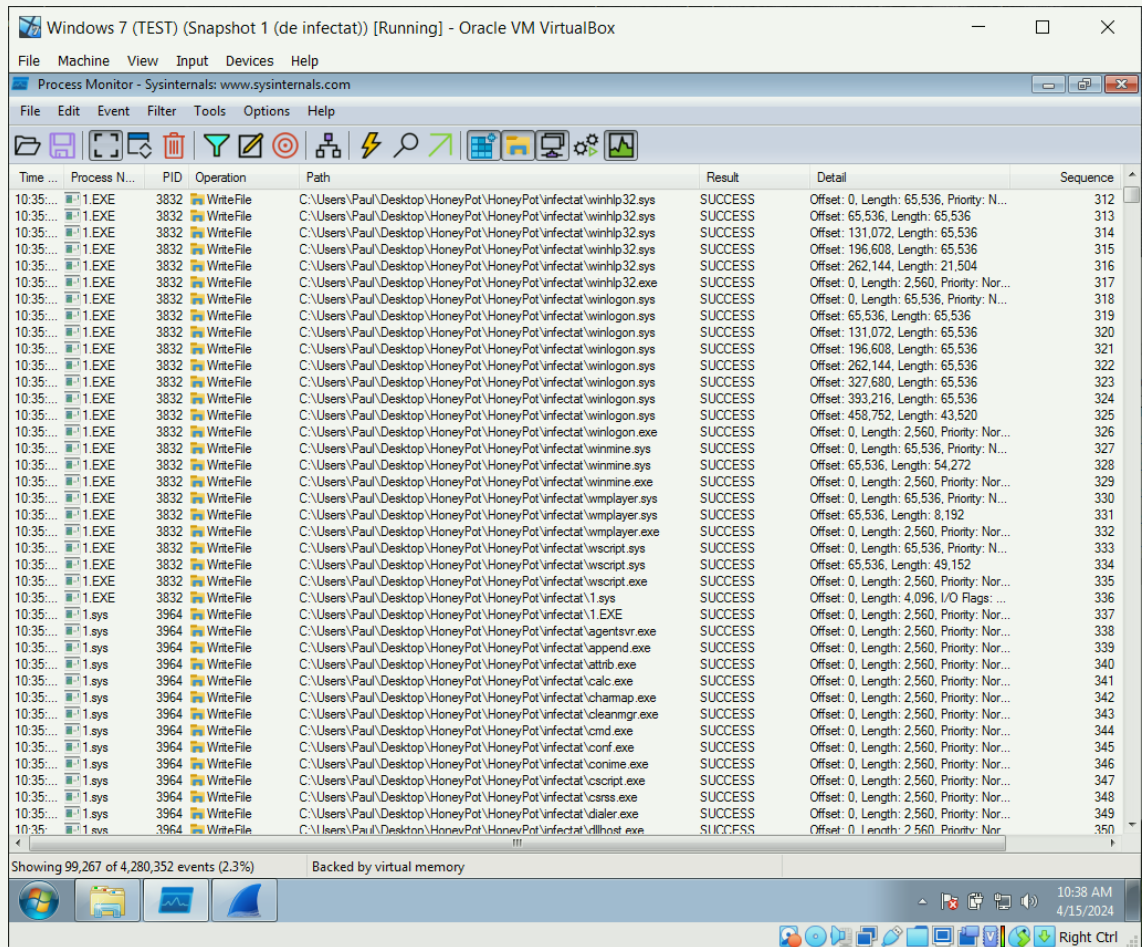
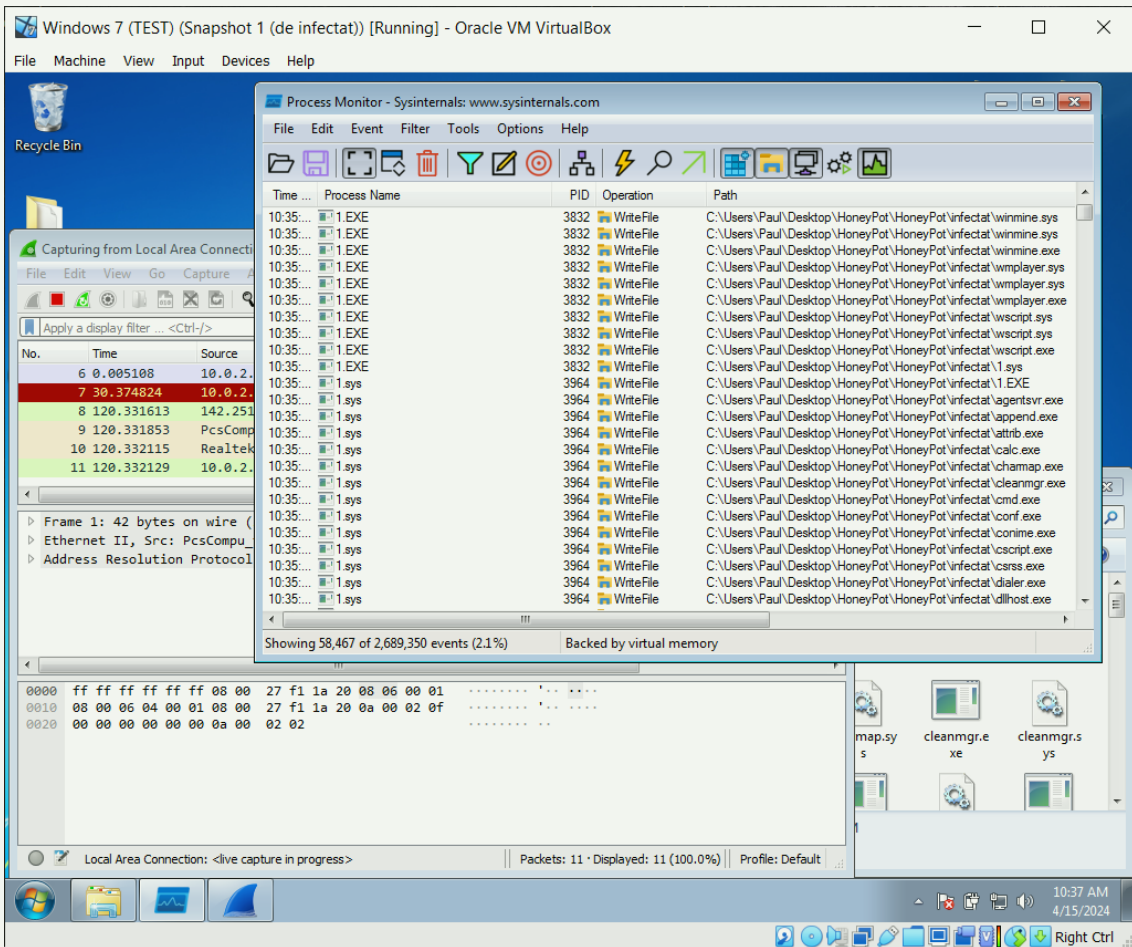
# ELIMINAREA ZGOMOTULUI PRODUS DE WIRESHARK



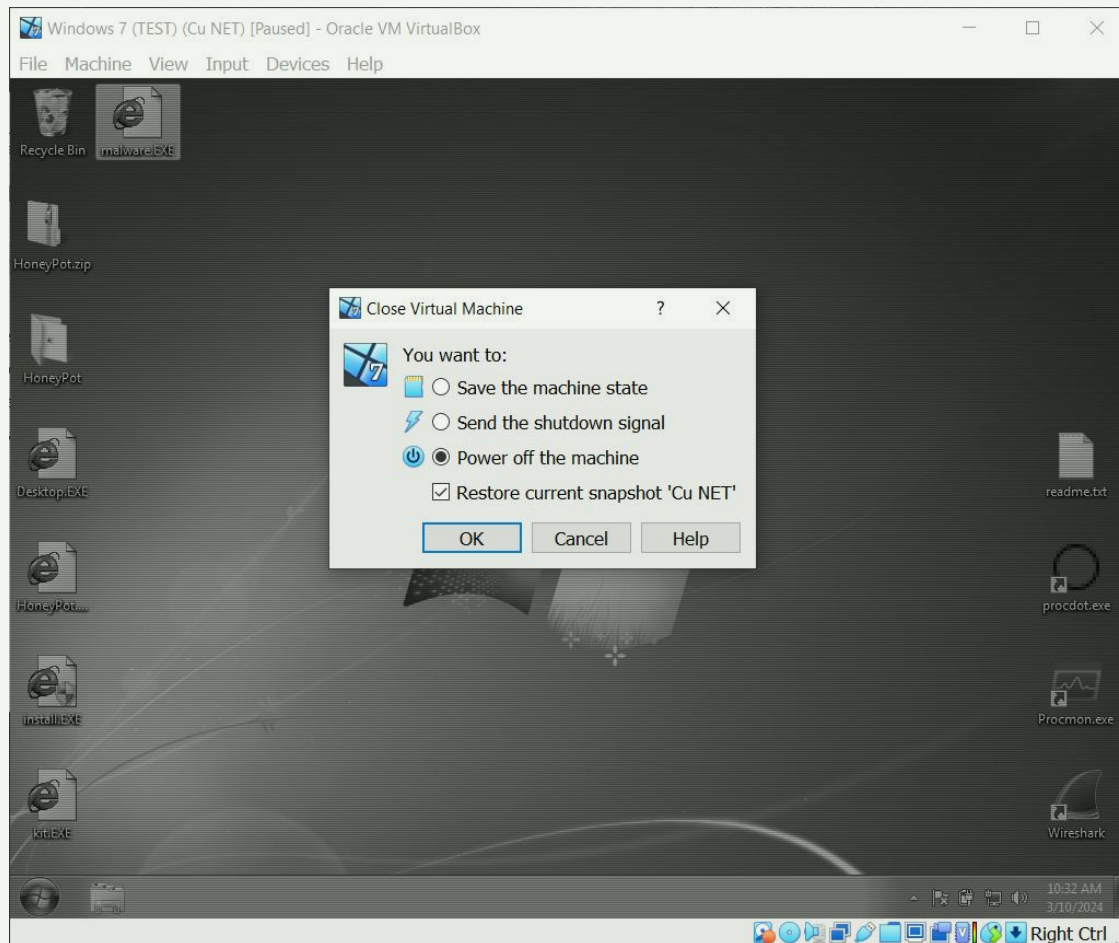
# DETONARE VIRUS & OBSERVATII



# INFECȚIE ÎN DESFĂȘURARE ...



# INCHIDERE PRIN RESTAURAREA MASINII VIRTUALE LA VECHEA STARE



- Inchiderea oricarui experiment prin restaurarea masinii virtuale la vechea stare!



**C.11.5**

# **VIZUALIZAȚI EVENIMENTELE ' POST DETONARE**





# BCOMPARE – ANALIZA HONEYPOT

## COMPARAREA CONȚINUTULUI A DOUĂ FIȘIERE

Observați diferența în cantitatea de informație dintre coada fișierului infectat și cel neinfectat.

The image displays two side-by-side screenshots of the BCompare application, comparing two files. The left window shows the infected file (agentsvr.exe) and the right window shows the non-infected file (agentsvr.exe). Red arrows point to specific areas of the files, and green callouts highlight the difference in file size and content.

On the left window (infected file), a red arrow points to the file size (275,968 bytes) and another points to the file content (Everything Else). A green callout labeled "Coadă fișier infectat" points to the file content.

On the right window (non-infected file), a red arrow points to the file size (256,512 bytes) and another points to the file content (Everything Else). A green callout labeled "Coadă fișier neinfectat" points to the file content.

A red arrow points from the bottom of the left window to the bottom of the right window, indicating the difference in file size.

# PROCESS MONITOR

## ANALIZA UNUI MALWARE ÎNTR-O MAȘINĂ VIRTUALĂ (VM)

### 1. Activitatea Proceselor

- **Crearea și terminarea proceselor.** Observați procesele create de malware. Anumite malware-uri pot lansa noi procese, fie pentru a executa sarcini specifice, fie pentru a se masca.
- **Relațiile între procese.** Identificați dacă malware-ul creează procese copil sau dacă interacționează cu alte procese existente.

### 2. Modificări în Sistemul de Fișiere

- **Crearea, modificarea și ștergerea fișierelor.** Fiți atenți la fișierele create sau modificate de malware, deoarece acestea pot indica activități precum extracția de date, instalarea de componente suplimentare sau încercări de persistență.
- **Accesări suspecte de fișiere.** Monitorizați accesul la fișiere de configurare sau sistem, care ar putea sugera încercări de modificare a setărilor sistemului.

### 3. Modificări în Registri

- **Crearea, modificarea și ștergerea cheilor de registru.** Malware-ul adesea modifică registry pentru a asigura persistența, a modifica setările sistemului sau a dezactiva funcții de securitate.
- **Acces la registre specifice.** Observați accesul la chei de registru ce controlează autorun, servicii, sau setări de rețea.

### 4. Activitatea de Rețea

- **Conexiuni de rețea.** Monitorizați orice încercare de conexiune la adrese IP sau domenii externe, ceea ce poate indica comunicații cu un server de comandă și control (C&C) sau încercări de exfiltrare a datelor.
- **Modificări ale setărilor de rețea.** Verificați dacă malware-ul modifică setările de rețea, cum ar fi proxy-uri sau DNS.

### 5. Hook-uri și DLL-uri Injectate

- **Injectarea de cod.** Multe tipuri de malware injectează cod în alte procese pentru a executa operațiuni în mod ascuns sau pentru a obține drepturi de acces mai mari.
- **Modificări ale DLL-urilor.** Atunci când un malware adaugă sau modifică DLL-uri, poate afecta funcționalitatea sistemului sau poate obține funcționalități suplimentare.

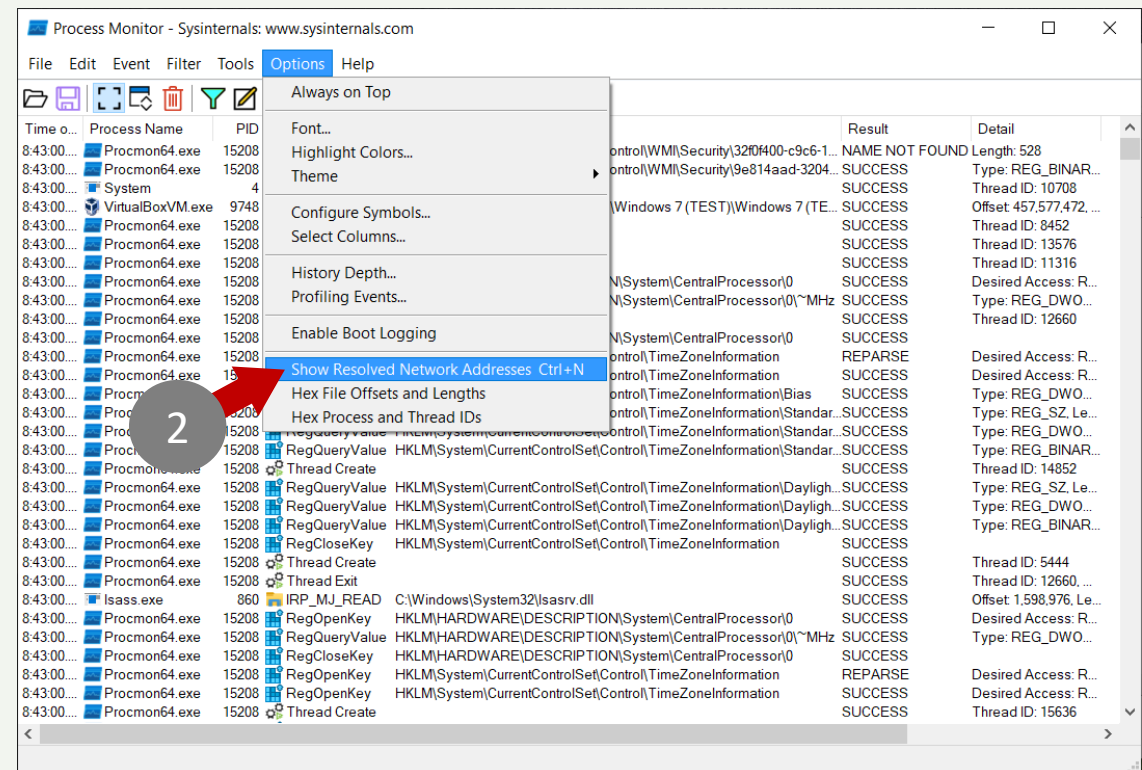
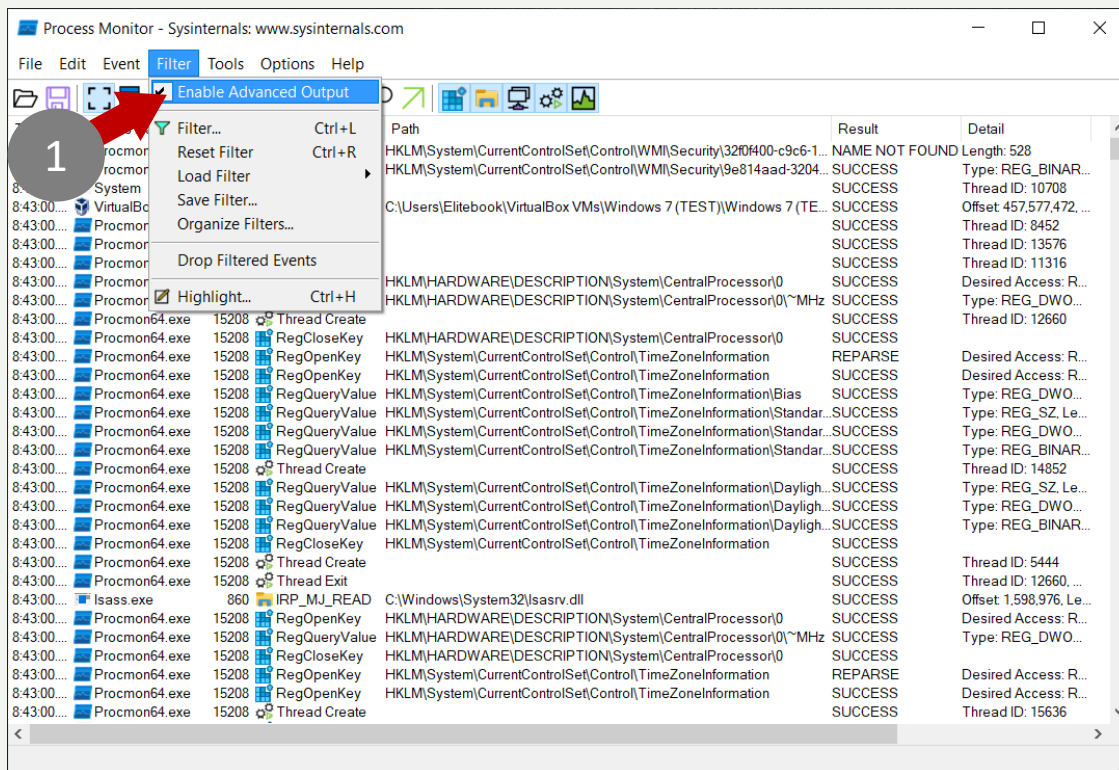
# SFATURI GENERALE

## PROCESS MONITOR

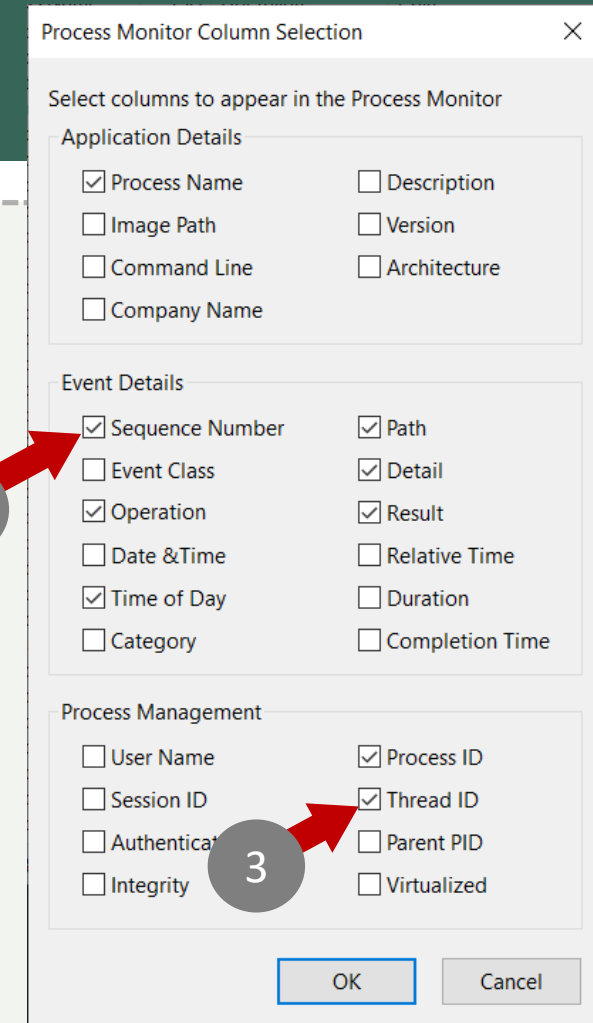
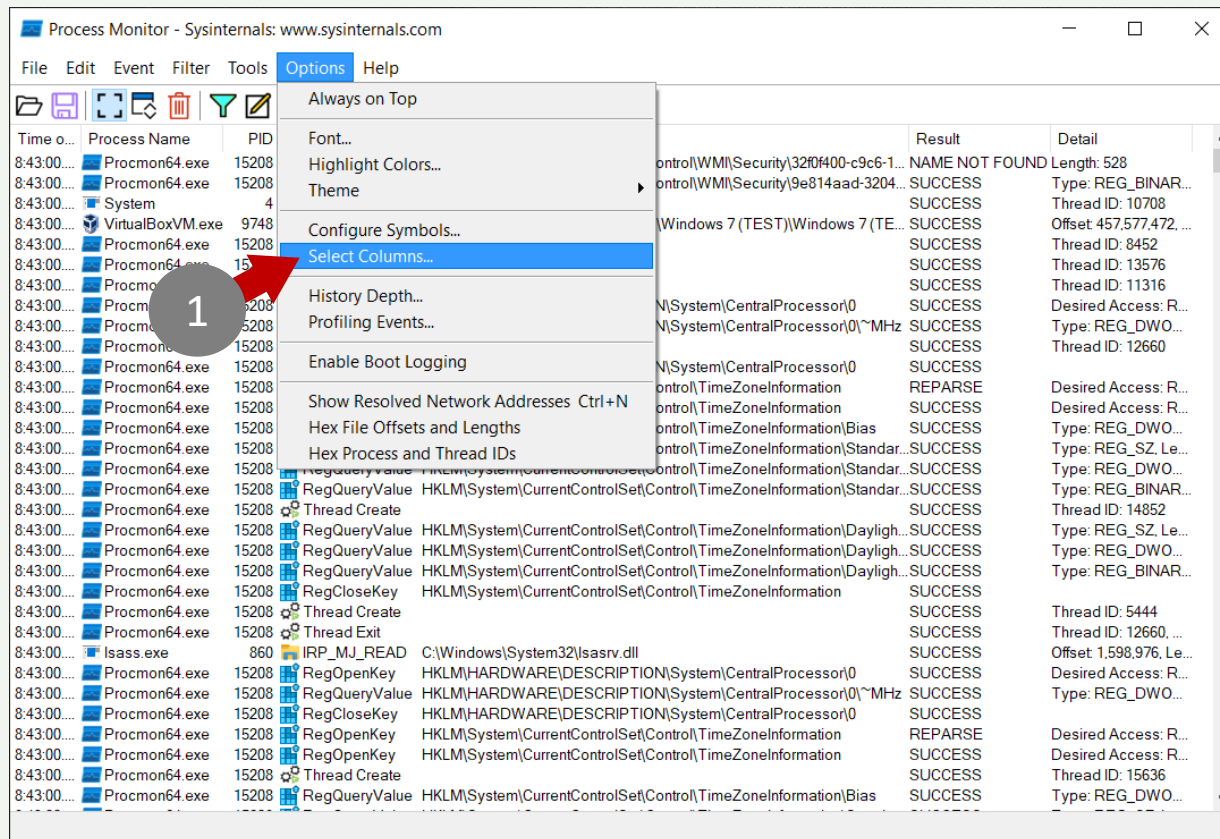
- **Filtrați și sortați datele.** Utilizați filtrele Process Monitor pentru a izola activitățile suspecte și a reduce zgomotul de fundal al activităților sistemului normal.
- **Exportați și salvați logurile.** Salvați logurile de monitorizare pentru analiză ulterioară și pentru a avea un punct de referință.
- **Utilizați snapshot-uri ale VM-ului.** Faceți snapshot-uri înainte și după detonarea malware-ului pentru a putea reveni rapid la o stare cunoscută și curată.
- **Folosiți alte unelte de analiză.** Complementați analiza cu alte unelte de securitate, cum ar fi antivirus, analizor de trafic de rețea, și unelte de analiză de malware pentru o imagine de ansamblu completă.

# PROCESS MONITOR

## SETUP PENTRU SALVAREA JURNALULUI (ORDINEA COLOANELOR)



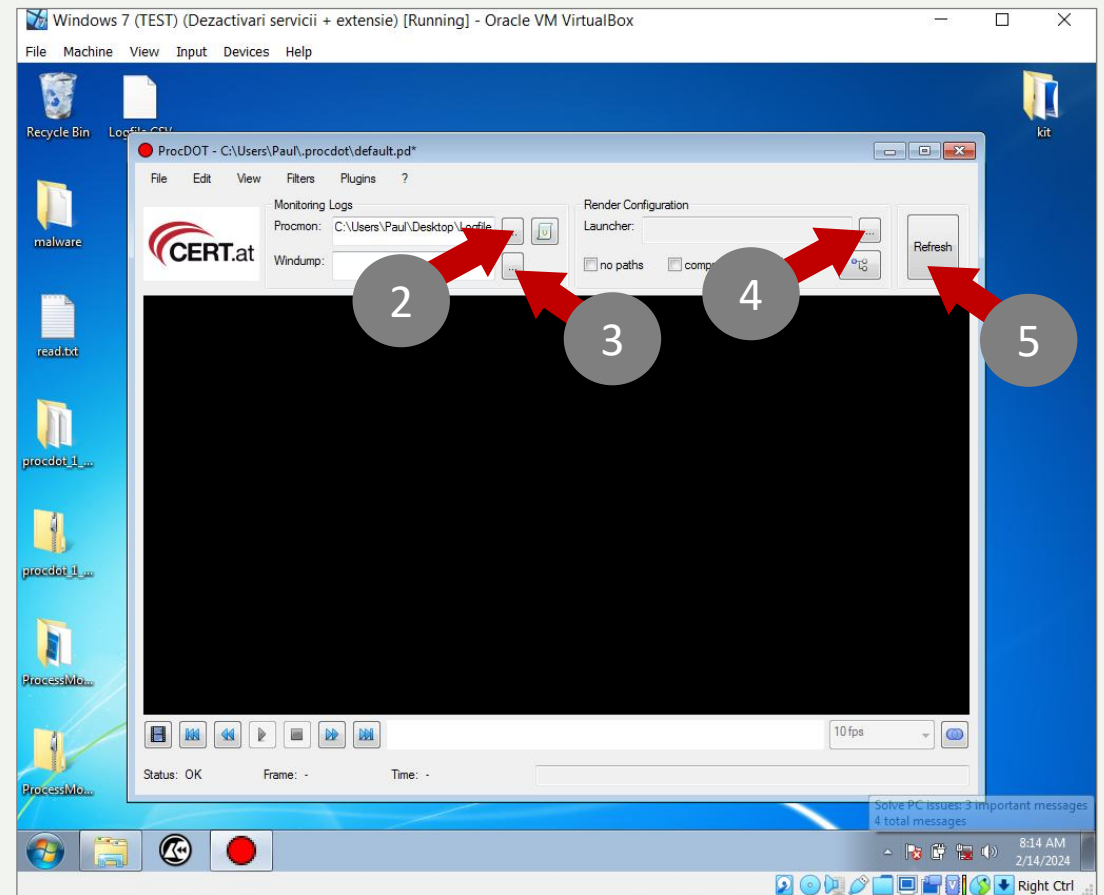
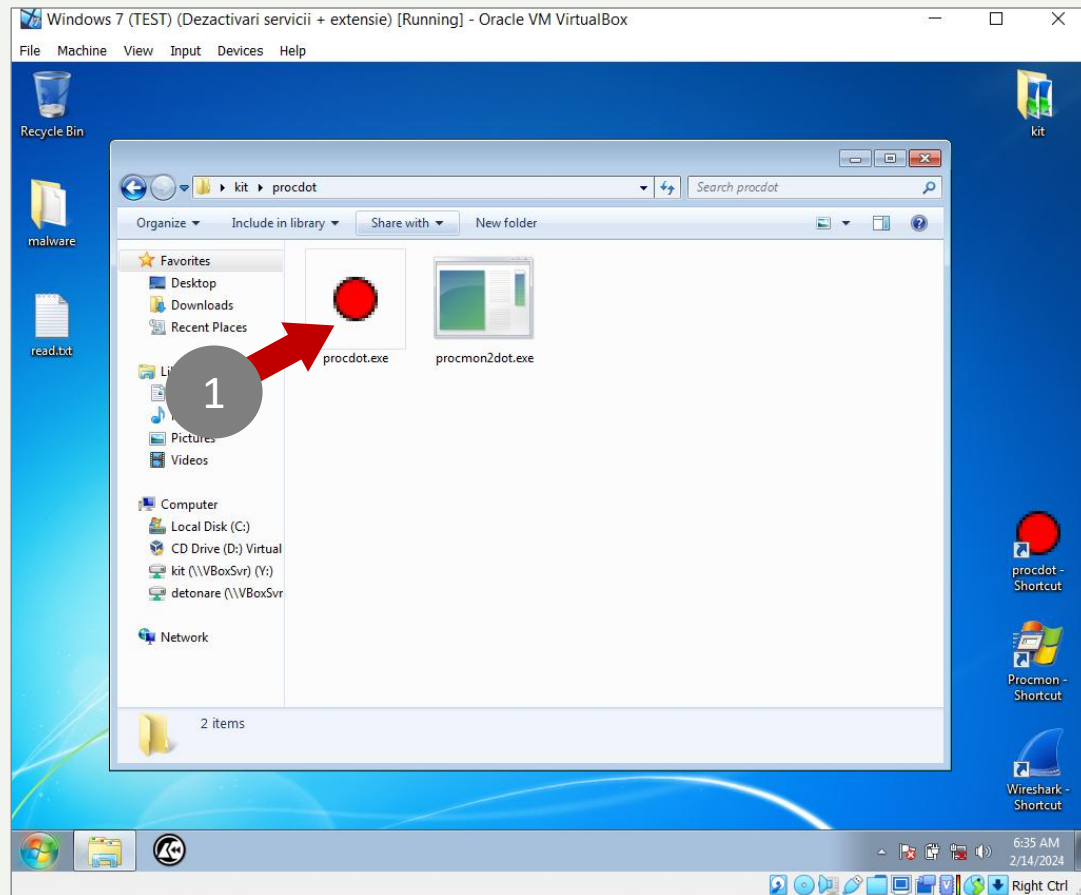
# PROCESS MONITOR SETUP





# PROCDOT

## RULAREA ȘI UTILIZAREA JURNALELOR PROCMON ȘI WIRESHARK



# PROC DOT

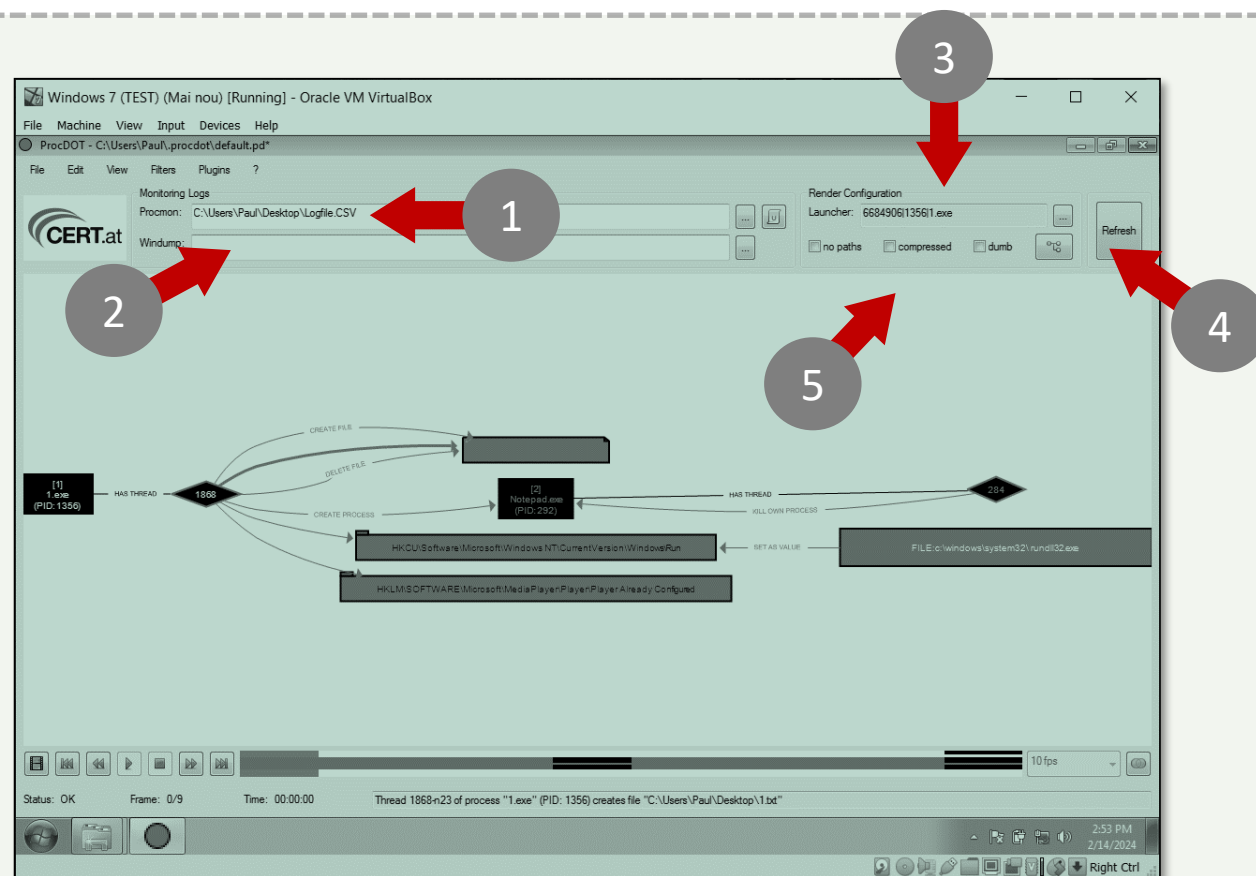
## REZULTAT FINAL

### Pornire rapidă în ProcDOT

=====

- 1) Selectați fișierele de jurnal (trist, dar adevărat, specificațiile pentru formatul de fișier nativ al Procmon (.PML) nu sunt disponibile (public). Prin urmare, trebuie să exportați fișierul dvs. .PML în .CSV, ceea ce se poate face cu ușurință prin „Salvare”., element de meniu din Procmon. Asigurați-vă că selectați „toate evenimentele”.)
- 2) Alegeți modul de reprezentare grafică (fără căi, comprimat)
- 3) Selectați primul proces relevant (malițios) (proces de lansare)
- 4) Faceți clic pe „Reîmprospătare”

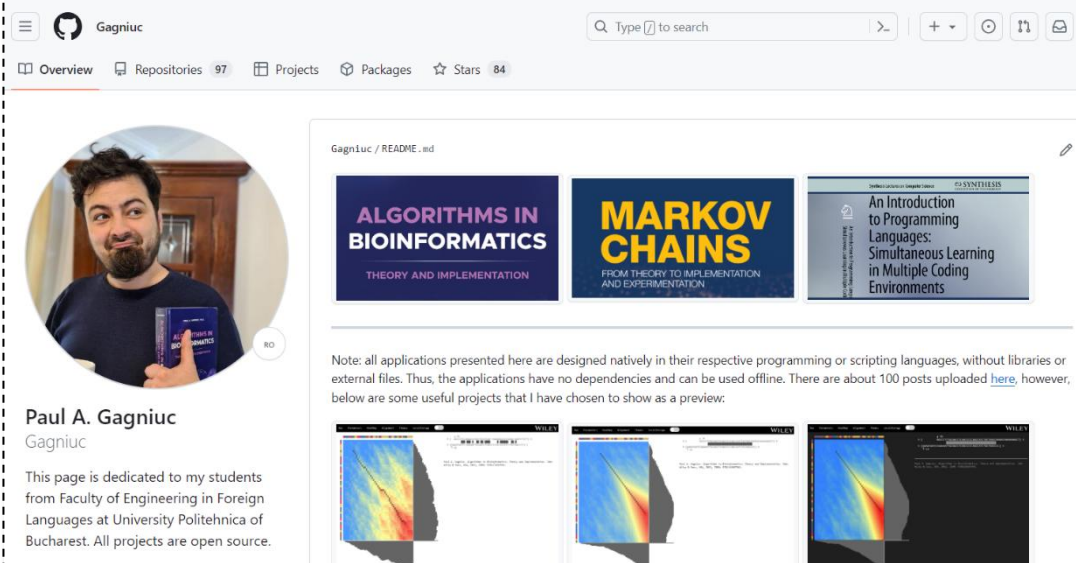
dot.exe îl găsiți în *graphviz*



# BIBLIOGRAFIE / RESURSE

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- Paul A. Gagniuc. *Coding Examples from Simple to Complex - Applications in MATLAB*, Springer, 2024, pp. 1-255.
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<https://github.com/gagniuc>



Gagniuc

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Gagniuc / README.md

**ALGORITHMS IN BIOINFORMATICS**  
THEORY AND IMPLEMENTATION

**MARKOV CHAINS**  
FROM THEORY TO IMPLEMENTATION AND EXPERIMENTATION

**An Introduction to Programming Languages: Simultaneous Learning in Multiple Coding Environments**

Note: all applications presented here are designed natively in their respective programming or scripting languages, without libraries or external files. Thus, the applications have no dependencies and can be used offline. There are about 100 posts uploaded [here](#), however, below are some useful projects that I have chosen to show as a preview.

Paul A. Gagniuc  
Gagniuc

This page is dedicated to my students from Faculty of Engineering in Foreign Languages at University Politehnica of Bucharest. All projects are open source.